

Formula Student Rules 2027

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Changelog

Rule	Version	Change
	1.0	Changed tempus to present tense across the document
	1.0	Unified wording across the document without changing the meaning
A 1.2.8	1.0	Added explicit bounds for points
A 3.3.1	1.0	Clarified rule
A 3.4.1	1.0	Unified wording for disqualification
A 3.5.3	1.0	Added explicit penalty for intent-violation
A 3.9	1.0	Added Code of Conduct
A 5.7	1.0	Shortened and clarified <u>SESA</u> section
A 6.5.2	1.0	Unified wording for disqualification
T 1.1.16	1.0	Updated definition of rollover protection envelope to unify text and illustration
T 1.2.1	1.0	Allowed UL94 VTM-0 for thin materials in addition to foams
T 2.1.3	1.0	Added new wording for rear tire keep-out-zone
T 2.1.4	1.0	Added variable keep-out-zone
T 2.6.4	1.0	Clarified rule about rim clearance
T 2.8.4	1.0	Added steering deformation limit and inspection method
T 2.8.9	1.0	Clarified rule
T 3.5.5	1.0	Clarified and expanded rule application and moved from T 3.16.1 to T 3.5
T 3.5.7	1.0	Shortened rule without losing definition
T 3.8.6	1.0	Clarified inspection requirement to ensure accessibility of tube dimensions
T 3.8.7	1.0	Expanded scope of the additional bracing requirement to items attached to main hoop and main hoop braces, rule moved to roll hoops section
T 3.12.3	1.0	Added minimum inclination requirement for recessed front hoop bracing load paths (e.g. damper recession) in monocoques
T 3.15.3	1.0	Extended the minimum clearance requirement to the side impact structures (12.5 mm); side impact structures (and bottom) may still be used for attachment locally
T 3.15.5	1.0	Added rule to limit hole size in (rear) impact structures whilst maintaining minimum required load paths as an alternative to penetration prevention
T 3.16.3	1.0	Clarified steel washers compliant with ISO 7093-1 or equivalent standards may be used as backing plates
T 3.16.5	1.0	Clarified rule by removing redundant sentence
T 3.16.6	1.0	Allowed alternative backing plate dimensions only for physically tested attachments (blind inserts)
T 3.17.2	1.0	Specified IA position in ready-to-race configuration
T 5.2.1	1.0	Replaced HANS by FHR and updated requirement on shoulder harness straps
T 5.5.4	1.0	Allowed variation on the shoulder harness mounting distance between belts in favor of FHR usage
T 5.7.2	1.0	Added allowed material for head restraint
T 6.1.10	1.0	Changed wording for pedal box and pedal box assembly
T 7.2.8	1.0	Removed requirement to mount catch cans to the chassis
T 7.3.1	1.0	Clarified rule
T 7.3.2	1.0	Added exception for unmodified OEM parts
T 8.2.1	1.0	Changed height restrictions
T 8.2.2	1.0	Changed width restrictions
T 9	1.0	Unified requirements regarding shielding for <u>CGS</u> and <u>HPHS</u>
T 9.3.3	1.0	Relaxed requirement regarding certification

Rule	Version	Change
T 10.1.5	1.0	Added rule for critical fasteners attached to the chassis
T 11.6.7	1.0	Added requirement to interrupt each sensor signal path for Technical Inspection.
T 11.7.2	1.0	Removed requirement to mount <u>LV</u> batteries to the chassis
T 13.2.1	1.0	Changed minimum tire clearance in jacked position
T 13.3.3	1.0	Unified Frontal Head Restraint references to use FHR with HANS just being an example
T 13.3.13	1.0	Refined driver communication equipment requirements
T 14.5.4	1.0	Clarified exception of T 11.2.1
T 14.9.3	1.0	Clarified ASSI visibility requirements
T 15.2.1	1.0	Clarified requirements on mechanical energy storage
T 15.2.2	1.0	Clarified which relay contact to use for EBS supply
T 15.2.2	1.0	Clarified that the <u>EBS</u> must also be switched by the <u>RES</u>
T 15.3.3	1.0	Clarified that brake actuation is included
T 15.3.4	1.0	Clarified redundancy requirement for brake pedal
EV 3.1.1	1.0	Changed wording from isolation to insulation
EV 4.5.3	1.0	Added relays to temperature rating requirement
EV 4.8.2	1.0	Changed description from ESO to trained person
EV 4.8.3	1.0	Changed wording from isolation to insulation
EV 5.1.3	1.0	Added electrical segment definition
EV 5.1.4	1.0	Added mechanical section definition
EV 5.2.1	1.0	Moved rule to cell subsection
EV 5.2.2	1.0	Moved from General Requirement to Cell subsection
EV 5.2.3	1.0	Defined operating limits for the cells
EV 5.3	1.0	Reorganized section
EV 5.3.1	1.0	Changed wording from isolation to insulation
EV 5.3.3	1.0	Added stack removal requirements
EV 5.3.4	1.0	Changed Segment to Section and removed the word “static”
EV 5.4	1.0	Reorganized section
EV 5.5	1.0	Reorganized section
EV 5.5.4	1.0	Changed segment to section
EV 5.5.7	1.0	Moved from EV5.4 to EV 5.5
EV 5.5.8	1.0	Clarified rule
EV 5.5.10	1.0	Clarified rule
EV 5.5.11	1.0	Changed segment to section and phrasing of vertical walls extending up to the lid
EV 5.5.12	1.0	Added that design must include all states of the cell
EV 5.5.13	1.0	Extended from tabs to entire high current path in the segment
EV 5.5.14	1.0	Removed exception for no calculations
EV 5.5.15	1.0	Added consideration of states of cell
EV 5.5.17	1.0	Clarified rule
EV 5.5.19	1.0	Simplified rule
EV 5.5.21	1.0	Clarified rule
EV 5.5.22	1.0	Moved rule to end of chapter
EV 5.8	1.0	Reworked cell temperature measurement requirements
EV 5.8.5	1.0	Added requirement for max distance for cell temp sensor from reference point
EV 5.8.6	1.0	Added requirement for temperature sensor contact
EV 5.8.7	1.0	Defined a reference point for temperature sensor placement

Rule	Version	Change
EV 5.8.8	1.0	Clarified rule
EV 5.8.14	1.0	Clarified rule
EV 5.8.15	1.0	Clarified rule
EV 7.1	1.0	Reorganized section
EV 7.1.5	1.0	Restructured requirements for charger
EV 7.1.5	1.0	Changed button diameter to 40mm
EV 7.1.5	1.0	Added AMS light to charger
EV 8.1.12	1.0	Relaxed hand cart size
IN 3.1	1.0	Removed rule that allowed stacks or modules to remain assembled to enable more flexibility during Technical Inspection
IN 3.1.1	1.0	Moved from EV section to IN and added segment
IN 4.1.1	1.0	Updated insulation test voltage to 1000 V for TS voltage above 500 V and converted requirements to tabular format
IN 5.1.1	1.0	Added requirement for pedal box accessibility
IN 11.1.1	1.0	Revised Brake Test wording
IN 12.1.8	1.0	Relaxed requirement for inspection mission during Post Inspection
S 2.2.4	1.0	Changed who is eligible to answer the judges' questions
S 2.2.5	1.0	Clarified rule
S 2.4.1	1.0	Clarified how Executive and Financial summary is scored
S 2.4.4	1.0	Clarified how supporting material is handled during the finals
S 2.4.6	1.0	Reduced maximum points for teams not participating in the finals
S 3	1.0	Removed BOM of entire vehicle but kept CCBOM for selected system(s)
S 3.4	1.0	Merged former BOM and CCBOM sections into one
S 3.4	1.0	Deleted rule about tires because it is no longer applicable
S 3.4.11	1.0	Removed redundant sentence
S 3.4.16	1.0	Added definition of "processes"
S 3.4.23	1.0	Clarified rule
D 9.4.1	1.0	Added clamping of EF_{team} to EF_{max}

Abbreviations

AIP	Anti Intrusion Plate	LVMS	Low Voltage Master Switch
AIR	Accumulator Isolation Relay	LVS	Low Voltage System
AMI	Autonomous Mission Indicator	OC	Off-Course
AMS	Accumulator Management System	PCB	Printed Circuit Board
APPS	Accelerator Pedal Position Sensor	R2D	Ready-to-drive
AS	Autonomous System	RES	Remote Emergency System
ASB	Autonomous System Brake	SCS	System Critical Signal
ASES	Accumulator Structural Equivalency Spreadsheet	SDC	Shutdown Circuit
ASF	Autonomous System Form	SE3D	Structural Equivalency 3D Model
ASMS	Autonomous System Master Switch	SES	Structural Equivalency Spreadsheet
ASR	Autonomous System Responsible	SESA	SES Approval
ASRQ	ASR Qualification	TPS	Throttle Position Sensor
ASSI	Autonomous System Status Indicator	TS	Tractive System
BOTS	Brake Over-Travel Switch	TSAC	Tractive System Accumulator Container
BPEFS	Business Plan Executive & Financial Summary	TSAL	Tractive System Active Light
BPP	Business Plan Presentation	TSMP	Tractive System Measuring Point
BSPD	Brake System Plausibility Device	TSMS	Tractive System Master Switch
CCBOM	Costed Carbonized Bill of Material	TVSD	Technical Vehicle System Documentation
CGS	Compressed Gas System	USS	Unsafe Stop
CO₂e	Carbon Dioxide Equivalents	VSV	Vehicle Status Video
CRD	Cost Report Documents		
CV	Internal Combustion Engine Vehicle		
DC	Driverless Cup		
DI	Direct Injection		
DNF	Did Not Finish		
DOO	Down or Out		
DQ	disqualified		
DV	Driverless		
EBS	Emergency Brake System		
EI	Flexural Rigidity		
ESF	Electrical System Form		
ESO	Electrical System Officer		
ESOQ	Electrical System Officer Qualification		
ETC	Electronic Throttle Control		
EV	Electric Vehicle		
FDM	Fused Deposition Modeling		
FHR	Frontal Head Restraint		
GWP	Global Warming Potential		
HPHS	High Pressure Hydraulic System		
HPI	High Pressure Injection		
HSC	Hybrid Storage Container		
HSF	Hybrid System Form		
HV	High Voltage		
HVD	High Voltage Disconnect		
HY	Combustion Hybrid Vehicle		
IA	Impact Attenuator		
IAD	Impact Attenuator Data		
IMD	Insulation Monitoring Device		
LCA	Life Cycle Assessment		
LPI	Low Pressure Injection		
LV	Low Voltage		

A Administrative Regulations

A 1 Competition Overview

A 1.1 Competition Objective

A 1.1.1 The competition challenges teams of university students to conceive, design, fabricate, develop and compete with small formula style race cars.

A 1.2 Competition Procedure

- A 1.2.1 The competition is split into the following classes:
- Internal Combustion Engine Vehicle (CV) including Combustion Hybrid Vehicle (HY)
 - Electric Vehicle (EV)
- A 1.2.2 All vehicles must meet the requirements defined in chapters T and either EV or CV , depending on their drivetrain type.
- A 1.2.3 The competition starts with Technical Inspection, described in chapter IN , to check the vehicle for safety and compliance with the rules.
- A 1.2.4 The competition is divided into a series of static and dynamic disciplines described in chapters S and D .
- A 1.2.5 The achievable points in each class are listed in Table 1.
- A 1.2.6 Vehicles of both classes can take part in an additional Driverless Cup (DC).
- A 1.2.7 The maximum achievable points for the DC are listed in Table 1.
- A 1.2.8 Each discipline's points and the overall points are bounded by $0 \leq P_{team} \leq P_{max}$, where P_{team} is the points achieved by the team including penalties and P_{max} is the maximum achievable points for the respective discipline or overall.

	<u>CV & EV</u>	<u>DC</u>
Static Disciplines:		
Business Plan Presentation	75 points	-
Cost and Manufacturing	100 points	-
Engineering Design	150 points	150 points
Dynamic Disciplines:		
Skidpad	50 points	-
Driverless (<u>DV</u>) Skidpad	75 points	75 points
Acceleration	50 points	-
Driverless (<u>DV</u>) Acceleration	75 points	75 points
Autocross	100 points	-
Driverless (<u>DV</u>) Autocross	-	100 points
Endurance	250 points	-
Efficiency	75 points	-
Trackdrive	-	200 points
Overall	1000 points	600 points

Table 1: Maximum points awarded

A 3 Rules of Conduct

A 1.2.9 The team with the most overall points wins the competition for its class or the DC, respectively.

A 1.3 Competition Information

A 1.3.1 The official language of the competition is English.

A 1.3.2 Every organizer of an event based on this document bears the sole responsibility.

A 1.3.3 The event specific rules and information are defined in the Event Handbook.

A 2 Vehicle Eligibility

A 2.1 Student Competition

A 2.1.1 Vehicles entered into the competition must be conceived, designed and maintained by the student team members without direct involvement from external professional engineers, racers, machinists or related professionals.

A 2.1.2 The student team may use any information from professionals or from academics as long as the information is given as a discussion of alternatives with their pros and cons.

A 2.1.3 Professionals must not make design decisions or drawings.

A 2.1.4 Students should perform fabrication tasks wherever possible.

A 2.2 First Year Vehicles

A 2.2.1 A vehicle may only be used for one year, counting from the first day on site of its first event.

A 2.2.2 To be classified as new, a vehicle must have a newly manufactured chassis.

A 2.2.3 In the following cases, the existing chassis may be used for one more consecutive year:

- Changing the powertrain from CV → EV or CV → HY or HY → EV
- Initial implementation of an Autonomous System, see [T 14](#)

The team must inform the officials ahead of time if they intend to use this exception.

A 3 Rules of Conduct

A 3.1 General Officials Authority

A 3.1.1 The officials reserve the right to revise the schedule of the event and/or interpret or modify the event or competition rules at any time and in any manner that is, in their sole judgment, required for safe and efficient operation.

A 3.1.2 All team members are required to cooperate with, and follow all instructions from the officials.

A 3.1.3 Official announcements are considered part of these rules.

A 3.1.4 All guidelines and clarifications posted in the “Rules and Important Documents” sections on the competition website for the current season including the Event Handbook are considered part of these rules.

A 3.2 Official Instructions

A 3.2.1 Failure of a team member to follow an instruction or command directed specifically to that team or team member results in 25 penalty points being deducted from the team’s overall score.

A 3 Rules of Conduct

A 3.3 Arguments with Officials

- A 3.3.1 Argument with, or disobedience to, any official results in penalty points for the team up to the team being disqualified from the event.

A 3.4 Unsportsmanlike Conduct

- A 3.4.1 In the event of unsportsmanlike conduct, 25 penalty points are deducted from the team's overall score. A second violation results in the team's disqualification from the event and its immediate expulsion from the event site.

A 3.5 Violations of Intent

- A 3.5.1 Violation of the intent of a rule is considered a violation of the rule itself.
- A 3.5.2 Any part, device or software fragment designed with the intent to violate a rule is considered a violation.
- A 3.5.3 Violations according to [A 3.5.2](#) result in a minimum of 100 penalty points being deducted from the team's overall score up to disqualification from the event.

A 3.6 Questions about the Rules

- A 3.6.1 Questions about the rules may be asked to the officials.
- A 3.6.2 The frequently asked questions (FAQ) section on the competition website must be checked before submitting a question.
- A 3.6.3 The officials only answer questions that are not already answered in the rules or FAQs or that require new or novel interpretation.
- A 3.6.4 Refer to the competition website for specific directions how to submit a rules question.

A 3.7 Protests

- A 3.7.1 A team may protest any rule interpretation, score or official action which they feel has caused some actual, non-trivial, harm to their team, or has had a substantive effect on their score.
- A 3.7.2 All protests must be submitted in writing as defined in the Event Handbook by the team captain within the announced protest period. In order to have a protest considered, a team must post a protest bond of 25 points which is forfeited if their protest is rejected.
- A 3.7.3 The decision of the officials regarding any protest is in a written form and is final.

A 3.8 Penalties

- A 3.8.1 Rule violations result in at least 20 penalty points, unless explicitly defined otherwise. The penalty points are deducted from the team's overall score.
- A 3.8.2 Overall penalties apply to CV, EV and DC score, except for penalties concerning disciplines which are not part of the respective class or cup.

A 3.9 Code of Conduct

- A 3.9.1 A Code of Conduct is published in the "Rules and Important Documents" section of the competition website. It forms part of these rules, see [A 3.1.4](#).
- A 3.9.2 The Code of Conduct applies to every person present at the event in any capacity, for the whole duration of the event and in every setting connected with it, including accommodation and campsites, transfers, social and evening activities, and official online channels.

A 4 General Requirements for Teams & Participants

- A 3.9.3 Acceptance of the Code of Conduct is a condition of team registration and of individual accreditation.
- A 3.9.4 Alleged breaches of the Code of Conduct are dealt with under the procedure set out in that document. The protest procedure, see [A 3.7.2](#), does not apply to them, and no protest bond is required. The objection procedure of the Code of Conduct applies instead.
- A 3.9.5 Team captains must ensure that every team member has read the Code of Conduct before arriving at the event. Dealing with a breach inside the team instead of reporting it is itself a breach.
- A 3.9.6 Organizers reserve the right to ban an individual, or take action against a team, in case of a serious breach of the Code of Conduct that occurs outside the event, especially if the safety of participants or the reputation of the competition, sponsors or other teams is compromised.

A 4 General Requirements for Teams & Participants

A 4.1 Teams per University

- A 4.1.1 A university can register one CV team and one EV team which both can take part in the DC.
- A 4.1.2 For the purposes of registering and competing, a university's CV team and EV team are considered to be separate and independent entities.
- A 4.1.3 Teams which are formed with members from two or more universities are treated as a single team.

A 4.2 Team Members and Participants

- A 4.2.1 Each participant must provide proof of valid private liability & individual health insurance for the event.
- A 4.2.2 All on-site participants must sign a liability waiver upon registering on site which can be found on the competition website.
- A 4.2.3 A team member may only be part of one team, work on one vehicle and take part in static and dynamic disciplines for only one team.
- A 4.2.4 Each team must have one team member identified as the team captain. The team captain is the main contact person for the officials during the registration process and the event.
- A 4.2.5 Persons who have previously attended any official event of any competition as a judge are not allowed to participate as team members.
- A 4.2.6 Team members must be enrolled as degree seeking undergraduate or graduate students in any university. Team members who have graduated within the seven month period before the event remain eligible to participate.
- A 4.2.7 Students seeking a PhD degree/PhD students or equivalent are not allowed to participate.
- A 4.2.8 Counting from the year of their first event, team members can only participate for four additional full years. First event is defined as the first registration of a team member for an event by the team captain, irrespective if the team or the team member actually physically participated.
- A 4.2.9 Team members must be at least 18 years of age.
- A 4.2.10 Team members who drive a competition vehicle at any time during an event must present a valid, government issued driver's license for passenger cars, containing a photograph.

A 5 Documentation & Deadlines

A 4.3 [EV ONLY] Electrical System Officer (ESO)

- A 4.3.1 Every participating team has to appoint one to four ESOs for the event.
- A 4.3.2 The ESOs are responsible for all electrical work carried out on the vehicle during the event.
- A 4.3.3 The ESOs are the only persons in the team who may declare the vehicle electrically safe, in order for work to be performed on any system of the vehicle by the team.
- A 4.3.4 The ESOs must be valid team members, i.e. they must have student status, see [A 4.2.6](#).
- A 4.3.5 The ESOs must be contactable by phone at all times during the event.
- A 4.3.6 An ESO must accompany the vehicle whenever it is operated or moved around at the event site.
- A 4.3.7 If only one ESO is named by the team, this ESO must not be a driver.
- A 4.3.8 An appropriate Electrical System Officer Qualification (ESOQ) requires the proof of a practical and theoretical training for working on the TS. Regarding content, the training should be on the basis of DGUV 209-093 stage 2E/3E¹ and held by an external expert.

A 4.4 Autonomous System Responsible (ASR)

- A 4.4.1 To operate the AS, the participating team must appoint at least one ASR for the event. This person is responsible for all autonomous operations of the vehicle during the event which includes any work on the AS as well as racing and testing.
- A 4.4.2 For vehicles with an electric drivetrain the ASR must fulfill [A 4.3](#) and therefore replaces the ESO. The team may register additional ESOs for Accumulator Inspection and work on TS only. The sum of registered ASRs and ESOs must not exceed four persons.
- A 4.4.3 The ASR is the only person in the team who is allowed to declare the AS safe, so that the vehicle may be operated in manual or autonomous mode.
- A 4.4.4 The ASR must be a valid team member, i.e. they must have student status, see [A 4.2.6](#).
- A 4.4.5 The ASR must accompany the vehicle whenever it is operated or moved around at the event site.
- A 4.4.6 If only one ASR is named by the team, this ASR must not be a driver.
- A 4.4.7 The ASR must be properly qualified to handle the AS and to understand and deal with problems and failures. As ASR Qualification (ASRQ), a bachelor degree in computer science, electrical engineering, mechatronics, automation engineering, robotics or similar, i.e. comparable study content or progress, is a sufficient qualification. The qualification certificate needs to be an official university document and contain information on completed courses.

A 5 Documentation & Deadlines

A 5.1 Required Documents and Forms

- A 5.1.1 Several documents and forms must be submitted by the action deadlines defined in the Event Handbook.

A 5.2 Submission

- A 5.2.1 Submitted documents may only be viewed by members of the submitting team, authorized judges and officials.

¹<https://fsg.one/4i5SP>

A 5 Documentation & Deadlines

- A 5.2.2 By submitting documents via the competition website, the team agrees that they may be reproduced and distributed by the officials, in both complete and edited versions, for educational purpose.
- A 5.2.3 Documents or videos that are largely incomplete or not readable/viewable are considered as not submitted.

A 5.3 Correction Requests

- A 5.3.1 If the officials request a correction for a document, the corrected version has to be submitted within 168 hours (7 days) following the request.
- A 5.3.2 Corrections requested before the initial deadline are treated as if they were requested at the initial deadline.
- A 5.3.3 Separate requests for different parts within one document or form are treated independently.
- A 5.3.4 If the corrected version of the document does not completely contain all requested corrections, the team is de-registered from the event.

A 5.4 Late Submission or Non-Submission

- A 5.4.1 For each submission later than the respective deadline 10 penalty points are deducted from the team's overall score.
- A 5.4.2 Teams that do not submit documents until 24 hours (1 day) after the respective deadline are de-registered from the event.

A 5.5 De-registration

- A 5.5.1 A team which is de-registered from the event has a single chance to apply for a place on the waiting list.
- A 5.5.2 To apply, the team must complete the following within 24 hours after the de-registration notification
- Submit an informal application to the officials
 - Correct the reason for de-registration (e.g. submission of a document)
- A 5.5.3 If the application is positively confirmed by the officials, the team:
- is placed at the end of the waiting list.
 - receives 20 penalty points.
- A 5.5.4 Applications are declined if the rework is inadequate or de-registration was caused by misbehavior of the team.

A 5.6 Vehicle Status Video (VSV)

- A 5.6.1 All teams must submit a video showing the vehicle driving in manual mode before the event. The video must be submitted before the deadline specified in the Event Handbook.
- A 5.6.2 [DC ONLY] The vehicle must instead drive in autonomous mode. No manual mode VSV is necessary.
- A 5.6.3 The video must show the following sequences in the described order:
- Standing still
 - 360° video around the vehicle (close up, may overlap with the following two bullet points)
 - [CV ONLY] Start engine / [EV ONLY] Activate TS
 - Straight driving, minimum distance is 30 m, minimum speed is 10 km/h for manual mode
 - Full stop (not needed for autonomous mode VSV)

A 6 General Rules

- 180° cornering
- Straight driving back to start point, minimum speed is 10 km/h for manual mode
- [DC ONLY] At the end of the VSV, the vehicle must be stopped by an emergency brake maneuver, see [T 15](#).
- Standing still, [CV ONLY] engine off / [EV ONLY] TS deactivated

A 5.6.4 The video must fulfill the following criteria:

- Continuous video from a third person view - no assembled sequences
- Vehicle must be clearly visible (light, video resolution, landscape mode, frames and frequency)
- Vehicle must run under its own power
- Driving must follow [A 6.3.4](#)
- Vehicle must be presented in ready-to-race condition including bodywork
- No crossing in front of the vehicle after [EV ONLY] TS activated / [CV ONLY] engine started
- [EV ONLY] TSAL, [DC ONLY] ASSI, and brake light, must be clearly visible in the video
- [EV ONLY] Ready-to-drive (R2D) sound must be audible in the video
- [DC ONLY] In addition to the third person view, an onboard view and a visualization of the vehicle's environment perception and path planning must be shown in split screen. All parts must be time synchronized.
- Must not exceed a length of 60 seconds.

A 5.6.5 If a team fails only the [DC ONLY] part, it is only de-registered from the DC.

A 5.6.6 A team which uploads a VSV with a previous year vehicle is de-registered from the event.

A 5.7 SES Approval (SESA)

A 5.7.1 A SESA is an additional, competition specific requirement. Teams using a monocoque must submit a SESA in addition to the SES. The SESA template is available on the competition website.

A 5.7.2 The SES must be checked and approved by a graduated engineer with appropriate expertise. Once the SES has been approved, the SESA must filled out, signed and submitted by the SESA deadline.

A 5.7.3 Any changes to the SES resulting from the approval process must be incorporated into the final SES and documented in a signed changelog submitted with the SESA.

A 5.7.4 Officials randomly compare the submitted SESA with the corresponding SES.

A 6 General Rules

A 6.1 Removing the Vehicle from the Site

A 6.1.1 Teams who remove their vehicle ([EV ONLY] or TS accumulator) from the event site after the event has begun are disqualified from the event.

A 6.2 Forfeit for Non-Appearance

A 6.2.1 It is the responsibility of each team to be in the right place at the right time.

A 6.2.2 If a team is not present and ready to compete at the scheduled time, they forfeit their attempt at that discipline.

A 6.3 Testing and Work Safety

A 6.3.1 Event organizers are not responsible for the use of the vehicles outside of their events.

A 6.3.2 The competition officials disassociate themselves from all activities of the teams besides their own event.

A 6 General Rules

- A 6.3.3 All teams are advised to follow common practices and common sense when working on the vehicle and when operating the vehicle, before, during and after an event.
- A 6.3.4 The following listed requirements are considered the minimum for a testing/operating environment to qualify as safe. Following these guidelines does not guarantee safety under all circumstances:
- Driver wearing full protection gear according to [T 13.3](#)
 - Working [AMS](#), [APPS](#), [ASSI](#), [BSPD](#), [EBS](#), [ETC](#) plausibility check, [IMD](#), [RES](#) and [TSAL](#) if applicable
 - Rules compliant chassis and mounted [IA](#)
 - No other passenger cars, trucks etc. being parked or driven on the same premises at the same time, unless the areas are clearly separated
 - No running under low visibility conditions
 - No running at speeds above typical competition speeds
 - No running in areas where crashing into obstacles at the height of the driver's head is possible, such that parts of the vehicle may pass below an obstacle, but the driver's head can be trapped between the obstacle and the main hoop for example
- A 6.3.5 Organizers reserve the right to disqualify a team registered for their event in case of unsafe driving behavior, especially if the reputation of the competition, sponsors and other teams is compromised. Examples of what could be seen as "unsafe driving behavior" are wheel-to-wheel racing or racing in unsuitable events, e.g. hill climb races or drag races, depending on the respective circumstances.

A 6.4 On-Site Working Safety

- A 6.4.1 Everyone in the dynamic area and everybody working on the vehicle must wear appropriate, closed-toed shoes.
- A 6.4.2 Appropriate personal protective equipment must be used.
- A 6.4.3 Hot work and machining operations that generate sparks or chips must only be carried out in the designated power tools area, as defined in the Event Handbook.
- A 6.4.4 Work that generates hazardous debris and dust, e.g. sanding or cutting of carbon fiber, must be performed in the designated power tools area. This excludes minor tasks, e.g. drilling a small hole, if safety precautions for the operator and nearby personnel are observed and all debris and dust are immediately cleaned up.
- A 6.4.5 Disposal procedures for hazardous waste, like oil and battery cells, are defined in the Event Handbook.
- A 6.4.6 When jacking up the vehicle a safe and stable support device rated for the load must be used.

A 6.5 Prohibited Substances

- A 6.5.1 Team members must not be under the influence of alcohol (0.0 ‰), cannabis, or any other recreational drug while on the event site.
- A 6.5.2 Violators are disqualified from the event and immediately expelled from the event site. Repeated offense results in the team's disqualification from the event and its immediate expulsion from the event site.
- A 6.5.3 Prescription medications and over-the-counter drugs used as directed for medical purposes are exempt from [A 6.5.1](#). However, team members must not be impaired in a manner that could compromise safety.

A 6.6 Vehicle Movement

- A 6.6.1 A Test Area for testing and tuning vehicles is available.
- A 6.6.2 Vehicles must not move under their own power anywhere other than on the Test Area or competition tracks.
- A 6.6.3 [EV ONLY] The detachable handle or key of the TSMS must be completely removed and kept by an ESO. The lockout/tagout function of the TSMS, see [EV 6.2.2](#), must be used.
- A 6.6.4 For vehicles with an AS, the detachable handle or key of the ASMS must be completely removed and kept by an ASR. The lockout/tagout function of the ASMS, see [T 14.5.7](#), must be used.
- A 6.6.5 Vehicles must be pushed at a normal walking pace by means of a “push bar”, see [T 13.1](#), and with a team member in the cockpit wearing the required driver equipment as defined in [T 13.3](#) except for helmet, arm restraints and balaclava.
- A 6.6.6 The team member in the cockpit must have full control of steering and braking.
- A 6.6.7 When the push bar is attached to the vehicle, the engine/TS must remain switched off.
- A 6.6.8 Vehicles with wings are required to have two team members walking on either side of the front wing whenever the vehicle is being pushed.

A 6.7 [CV ONLY] Engine Running

- A 6.7.1 Running engines is not allowed in the pits.
- A 6.7.2 Engine running and hybrid system activation is allowed in the Engine Test Area and in the dynamic area, when the following conditions are met:
- The vehicle must have passed Mechanical Inspection.
 - The vehicle must be jacked using the jacks, see [T 13.2](#).
 - A driver, wearing required driver equipment, see [T 13.3](#), must be seated in the cockpit.
 - A fire extinguisher must be immediately available.
 - Driven wheels can only remain fitted to the vehicle if they do not rotate.
 - No one is allowed under the vehicle while the engine is running.
- A 6.7.3 Cranking engines in the pits is allowed, when the following conditions are met:
- The vehicle has passed Mechanical Inspection.
 - The driven axles are securely jacked up.
 - Gearbox is in neutral.
 - All driven wheels are removed.
 - Connectors to all injectors and ignition coils are detached.
 - A fire extinguisher must be placed next to the engine.

A 6.8 [CV ONLY] Fueling

- A 6.8.1 Fueling may only take place at the official fuel station.
- A 6.8.2 Open fuel containers are not allowed at the event site.

A 6.9 [EV ONLY] Working on the Vehicle

- A 6.9.1 All activities require the TSAL to be green.
- A 6.9.2 A prominent manual sign indicating the “TSAL green” state must be present whenever the LVS is switched off and the requirements for an only green TSAL according to [EV 4.10](#) are met.
- A 6.9.3 All activities on the vehicle, except when working on an active TS, require the HVD to be removed.

A 6.10 [EV ONLY] Working on the Tractive System

- A 6.10.1 Activities on the TS, except for the accumulator, see [A 6.11](#), must take place in the pit.
- A 6.10.2 All activities require the attendance of the ESO.
- A 6.10.3 For activities on the inactive TS, the following procedure must be carried out:
- The vehicle must be barred from anyone not involved in the work, by using barrier tape.
 - The Tractive System Master Switch (TSMS) must be switched off.
 - It must be assured that the TS cannot be activated by, at a minimum, using the lockout/tagout of the TSMS.
 - A check for zero-potential must be carried out.
 - A sign that declares the vehicle is electrically safe must be installed. The name of the ESO who is supervising the activities must be noted on the sign. This ESO is the only person who may remove the sign and the barrier.
- A 6.10.4 In case of measurements on the active TS or an activation of the TS in the pit for testing purposes, the following steps must be followed:
- The vehicle must be barred from anyone not involved in the work, by using barrier tape.
 - The vehicle must be jacked up and the driven wheels removed.
 - One team member must be prepared to push a shutdown button at any time.
 - The TS must only be activated for as long as necessary.
 - Appropriate insulated tools and equipment must be used.
 - Safety glasses with side shields and compliant safety gloves must be worn by all participating team members when parts of the TS are exposed.
 - No other work on the vehicle is allowed when the TS is active.
- A 6.10.5 If the TSAL is flashing red or malfunctioning, the TS is considered as active.
- A 6.10.6 There must be at least one team member present, who is not directly involved in the work, but who could assist in case of an incident.

A 6.11 [EV ONLY] Working on Tractive System Accumulators

- A 6.11.1 Opening or working on TSACs is only allowed in the provided work places in the charging area, see [A 6.12](#).
- A 6.11.2 All activities require the attendance of an ESO.
- A 6.11.3 Whenever the TSACs are opened, the cell segments must be separated with the maintenance plugs, see [EV 5.4.7](#).
- A 6.11.4 Appropriate insulated tools and equipment must be used.
- A 6.11.5 Safety glasses with side shields and compliant safety gloves must be worn by all participating team members.
- A 6.11.6 There must be at least one team member present, who is not directly involved in the work conducted on the accumulator, but who could assist in case of an incident.
- A 6.11.7 Moving accumulator cells and/or accumulator segment(s) around at the event site is only allowed if they are inside a completely closed TSAC.
- A 6.11.8 All parts and modules of the TSAC that are not currently being worked on must be covered at least according to IPxxB while working on the accumulator container.

A 6.12 [EV ONLY] Charging

- A 6.12.1 There is a separated charging area on the event site. Charging TS accumulators is only allowed inside this area.

A 6 General Rules

- A 6.12.2 Accumulators must be removed from the vehicle and placed on the TSAC hand cart, see [EV 8.1](#), for charging.
- A 6.12.3 No grinding, drilling, etc. is allowed in the charging area.
- A 6.12.4 At least one team member who has knowledge of the charging process must stay with the accumulator(s) during charging.

T General Technical Requirements

T 1 Definitions

T 1.1 Chassis Definitions

- T 1.1.1 **Chassis** — the fabricated structural assembly that supports all functional vehicle systems. This assembly may be a single welded structure, multiple welded structures or a combination of composite and welded structures.
- T 1.1.2 **Bodywork** — the outermost surface of the chassis or any fairing parts and covers.
- T 1.1.3 **Cockpit** — the volume which accommodates the driver which is defined by the top of the vehicle, the floor closeout, the inner side of the bodywork, the front bulkhead and the firewall.
- T 1.1.4 **Chassis member** — a minimum representative single piece of uncut, continuous tubing or equivalent structure.
- T 1.1.5 **Front bulkhead** — a planar structure that defines the forward plane of the chassis and provides protection for the driver's feet (in front view, together with the AIP, covers the driver's feet).
- T 1.1.6 **Front bulkhead support** — a structure that defines the side of the chassis from the front bulkhead back to the top of the upper side impact structure and the bottom of the front hoop.
- T 1.1.7 **Front hoop** — a roll bar located above the driver's legs, in proximity to the steering wheel.
- T 1.1.8 **Impact Attenuator (IA)** — a deformable, energy absorbing device located forward of the front bulkhead.
- T 1.1.9 **Main hoop** — a roll bar located alongside or just behind the driver's torso.
- T 1.1.10 **Monocoque** — a chassis made of composite material.
- T 1.1.11 **Node-to-node triangulation** — an arrangement of chassis members projected onto a plane, where a co-planar load applied in any direction, at any node, results in only tensile or compressive forces in the chassis members as shown in Figure 1.

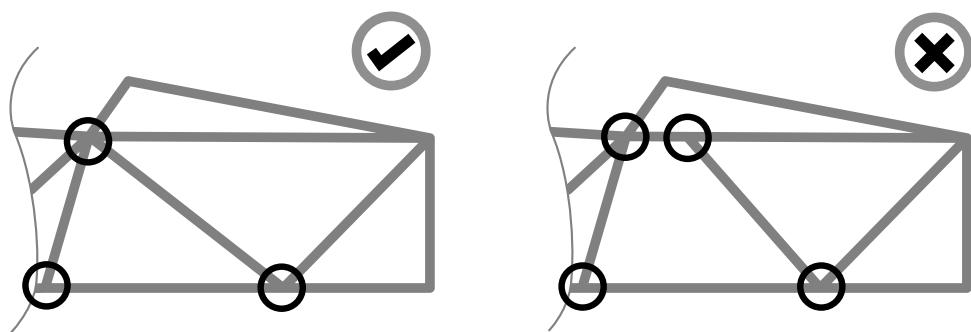


Figure 1: Node-to-node triangulation of chassis members (left correct and right incorrect).

- T 1.1.12 **Primary structure** — the primary structure is comprised of the following components:
- Main hoop
 - Front hoop
 - Roll hoop braces and supports
 - Impact structures as mentioned under [T 3.15](#)
 - Anti Intrusion Plate
 - Front bulkhead
 - Front bulkhead support system

T 1 Definitions

- All chassis members, guides and supports that transfer load from the driver's restraint system into the above mentioned components of the primary structure
- T 1.1.13 **Roll hoops** — both the front hoop and the main hoop are classified as “roll hoops”
- T 1.1.14 **Roll hoop bracing** — the structure from a roll hoop to the roll hoop bracing support.
- T 1.1.15 **Roll hoop bracing supports** — the structure from the lower end of the roll hoop bracing back to the roll hoop(s).
- T 1.1.16 **Rollover protection envelope** — the union of
- the primary structure and any additional structures fixed to the primary structure which meet the minimum specification defined in T 3.2 or equivalent
 - the plane from the top of the main hoop to the top of the front hoop
 - the plane from the top of the main hoop to the rearmost structure according to T 3.2 or equivalent
 - additional planes covering concave sections of structures according to T 3.2, e.g. to accommodate suspension or other components
- Example depicted in Figure 2.
- T 1.1.17 **Side impact structure** — the area of the side of the chassis between the front hoop and the main hoop and from the chassis floor to the height as required in T 3.15 above the lowest inside chassis point between front hoop and main hoop.
- T 1.1.18 **Surface envelope** — the surface envelope is the surface of the union of the rollover protection envelope, see T 1.1.16, and the volume defined by
- the top of the main hoop and the outside edges of the four tires
 - a plane from the top of the main hoop to the top of the front bulkhead
 - a plane from the top of the main hoop to the rearmost structure according to T 3.2 or equivalent
 - a projection in side view of the lower boundaries of the rollover protection envelope

Example depicted in Figure 3.

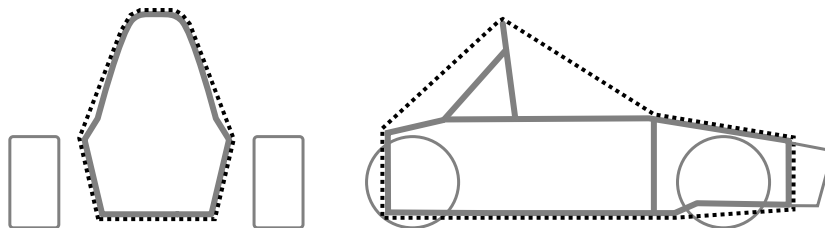


Figure 2: Rollover protection envelope T 1.1.16, thick lines are primary structure members according to T 3.2

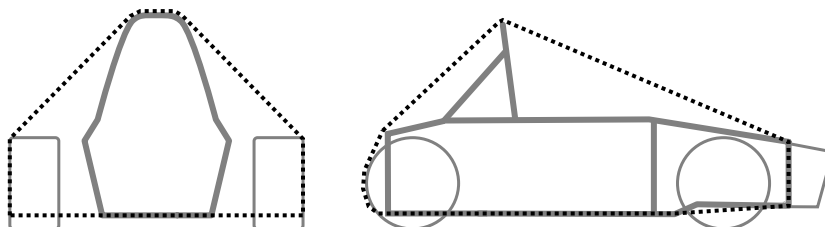


Figure 3: Surface Envelope T 1.1.18

T 1.2 Material Definitions

- T 1.2.1 **Fire Retardant** — a material meeting one of the following standards:
- UL94 V-0 for the minimum used material thickness

T 2 General Design Requirements

- FAR 25.853(a)(1)(i)
- UL94 HF-1 for foams; UL94 VTM-0 for materials that are not suitable for testing according to UL94 V-0 due to their thickness or flexibility.

Equivalent standards are only accepted, if the team shows equivalence and this is approved by the officials before the event. Compliance of alternative materials to above standards can be shown through testing with appropriate documentation in the SES/ASES and/or at Technical Inspection.

T 1.2.2 **Coolant** — a substance used for heat transfer by convection.

T 1.3 Electrical Definitions

T 1.3.1 **Direct Connection** — two devices or circuits are directly connected if the connection is not routed through any common PCB and does not include any devices or functionality other than overcurrent protection or connectors.

T 1.4 Driving Mode Definitions

T 1.4.1 **Manual Mode** — a vehicle is in manual mode when driven by a human driver. In this case the ASMS must be off (AS deactivated).

T 1.4.2 **Autonomous Mode** — a vehicle is in autonomous mode when the AS is activated. When a vehicle is in autonomous mode, there must be no person inside the vehicle.

T 2 General Design Requirements

T 2.1 Vehicle Configuration

T 2.1.1 The vehicle must be designed and fabricated in accordance with good engineering practices.

T 2.1.2 The vehicle must be open-wheeled, single seat and open cockpit (a formula style body) with four wheels that are not in a straight line.

T 2.1.3 Open wheel vehicles must satisfy the following, see also Figure 4:

- The wheel/tire assembly must be unobstructed when viewed from the side.
- No part of the vehicle may enter a keep-out-zone defined by two lines extending vertically from positions 75 mm in front of and 75 mm behind the outer diameter of the front and rear tires in the side view of the vehicle, with steering system straight ahead. This keep-out zone extends laterally outward from the inboard plane of the wheel/tire assembly indefinitely. The keep-out-zone at the rear tires extends up to 700 mm above the ground.
- An additional keep-out-zone extends from the top of the rear tire to 700 mm above the ground and 150 mm inboard from the inboard plane of the rear tire. It is bounded longitudinally by the 75 mm keep-out-zone of the rear tire.

T 2 General Design Requirements

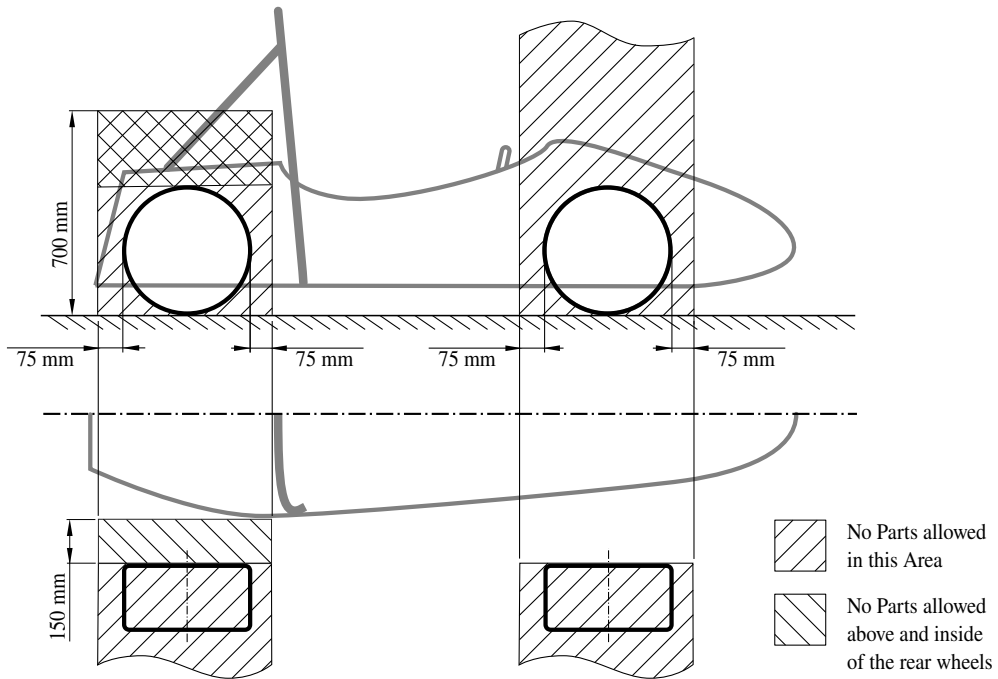


Figure 4: Keep-out-zones for the definition of an open-wheeled vehicle

T 2.1.4

No part of the vehicle forward of the vertical plane through the leading edge of the front tires must enter the variable keep-out zone, see Figure 5. It is defined by two independent, non-intersecting, cuboids of 75 mm × 250 mm × infinite length. One infinite-length face must rest on the ground, and the infinite-length edges must be parallel to the vehicle longitudinal axis. The cuboids must not extend outboard of the front tires.

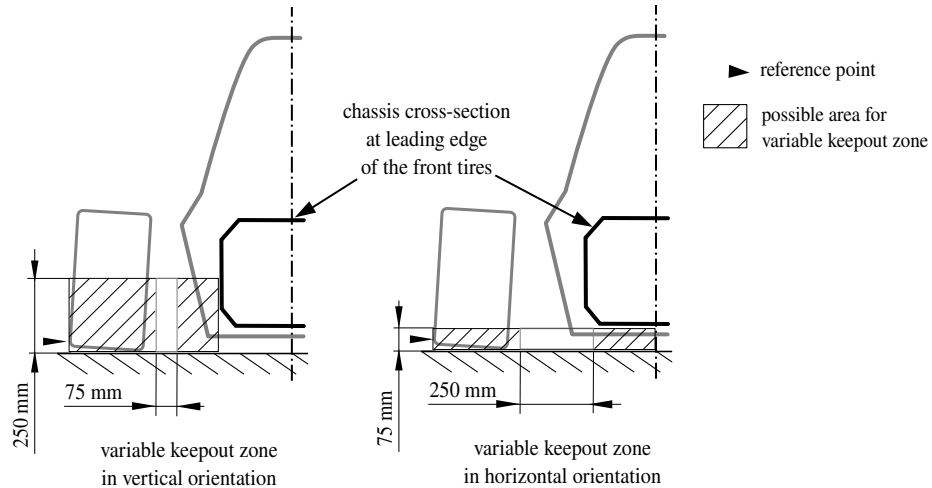


Figure 5: Variable keep-out-zone in front of the front tires

T 2.2 Ground Clearance

T 2.2.1

The minimum static ground clearance of any portion of the vehicle, other than the tires, including a driver, must be 30 mm. If an active suspension system is installed, the static ground clearance is measured in the lowest adjustable position.

T 2.2.2

Sliding skirts or other aerodynamic devices that by design, fabrication or as a consequence of moving, contact the track surface are prohibited.

T 2 General Design Requirements

T 2.3 Bodywork

- T 2.3.1 There must be no openings through the bodywork into the cockpit other than that required for the cockpit opening. Minimal openings around the front suspension and steering system components are allowed.
- T 2.3.2 In any side section view in front of the cockpit opening and outside the area defined in [T 8.2](#) all parts of the bodywork must have no external concave radii of curvatures. Any gaps between bodywork and other parts must be reduced to a minimum.
- T 2.3.3 Enclosed chassis structures and structures between the chassis and the ground must have two venting holes of at least 25 mm diameter in the lowest part of the structure to prevent accumulation of liquids. Additional holes are required when multiple local lowest parts exist in the structure.
- T 2.3.4 The bodywork in front of the front wheels with the tangent of surface above 45° relative to the forward direction must have a radius of at least 38 mm along the top, sides and bottom of all affected edges.

T 2.4 Minimum Edge Radii of Bodywork and Aerodynamic Devices

- T 2.4.1 For all edges that could come into contact with any standing pedestrian without reaching to the vehicle, the minimum radius of the bodywork and aerodynamic devices is 3 mm for all forward facing edges and 1 mm for all other edges.

T 2.5 Suspension

- T 2.5.1 The vehicle must be equipped with fully operational front and rear suspension systems including shock absorbers and a usable wheel travel of at least 50 mm and a minimum jounce of 25 mm with driver seated.
- T 2.5.2 All suspension mounting points must be visible at Technical Inspection, either by direct view or by removing any covers.

T 2.6 Wheels

- T 2.6.1 Any wheel mounting system that uses a single retaining nut must incorporate a device to prevent loosening of the nut and the wheel. A second nut ("jam nut") does not meet these requirements.
- T 2.6.2 Wheel lug bolts, drive pegs and studs must be made of steel or titanium. The team must be able to show good engineering practice and providing adequate strength by calculations. Wheel lug bolts, drive pegs and studs must not be hollow.
- T 2.6.3 Aluminium wheel nuts may be used, but they must be hard anodized and in pristine condition.
- T 2.6.4 The radial clearance between any non-rotating part and the rim must be at least 5 mm in static condition at any steering angle and any ride height.

T 2.7 Tires

- T 2.7.1 Vehicles must have two types of tires as follows:
- Dry tires - The tires on the vehicle when it is presented for Technical Inspection are defined as its "dry tires".
 - Wet tires - Wet tires may be any size or type of treaded or grooved tire provided:
 - The tread pattern or grooves were molded in by the tire manufacturer or were cut by the tire manufacturer or their appointed agent. Any grooves that have been cut must have documentary proof that it was done in accordance with these rules.

T 2 General Design Requirements

- There is a minimum tread depth of 2.4 mm.

- T 2.7.2 Tires on the same axle must have the same manufacturer, size and compound.
- T 2.7.3 Tire warmers are not allowed.
- T 2.7.4 Special agents that increase traction must not be added to the tires or track surface.

T 2.8 Steering System

- T 2.8.1 Steering systems using cables or belts for actuation are prohibited. This does not apply for autonomous steering actuators.
- T 2.8.2 The steering wheel must directly mechanically actuate the front wheels.
- T 2.8.3 The steering system must have positive steering stops that prevent the steering linkages from locking up. The stops must be placed on the rack and must prevent the tires and rims from contacting any other parts. Steering actuation must be possible during standstill.
- T 2.8.4 Allowable steering system free play is limited to a total of 7° measured at the steering wheel. The steering system must not exhibit excessive deformation under driver-applied loads. This may be checked by an official seated normally in the vehicle while applying normal steering force with the vehicle stationary.
- T 2.8.5 The steering wheel must be attached to the column with a quick disconnect. The driver must be able to operate the quick disconnect while in the normal driving position with gloves on.
- T 2.8.6 The steering wheel must be no more than 250 mm rearward of the front hoop. This distance is measured horizontally, on the vehicle centerline, from the rear surface of the front hoop to the forward most surface of the steering wheel with the steering system in any position.
- T 2.8.7 The steering wheel must have a continuous perimeter that is near circular or near oval. The outer perimeter profile may have some straight sections, but no concave sections.
- T 2.8.8 In any angular position, the top of the steering wheel must be no higher than the top-most surface of the front hoop.
- T 2.8.9 The steering rack must be mechanically attached to the primary structure and all stationary parts of the steering system must be within the rollover protection envelope, see [T 1.1.16](#).
- T 2.8.10 Joints between all components attaching the steering wheel to the steering rack must be mechanical and visible at Technical Inspection. Bonded joints are allowed in accordance with [T 3.2.8](#).
- T 2.8.11 Rear wheel steering, which can be electrically actuated, is allowed if mechanical stops limit the range of angular movement of the rear wheels to a maximum of 6°. This must be demonstrated with a driver in the vehicle and the team must provide the equipment for the steering angle range to be verified at Technical Inspection.

T 2.9 Wheelbase, Track and Rollover Stability

- T 2.9.1 The vehicle must have a wheelbase of at least 1525 mm.
- T 2.9.2 The smaller track of the vehicle (front or rear) must be no less than 75 % of the larger track.
- T 2.9.3 The track and center of gravity of the vehicle must combine to provide adequate rollover stability. This is tested by the Tilt Test [IN 7](#).

T 3 General Chassis Design

T 3.1 General Requirements

T 3.1.1 Among other requirements, the vehicle's structure must include:

- Two roll hoops that are braced
- A front bulkhead with support system and IA
- Side impact structures

T 3.2 Minimum Material Requirements

T 3.2.1 Table 2 shows the minimum requirements for the members of the primary structure if made from steel tubing.

Item or application	Minimum wall thickness	Minimum cross sectional area	Minimum area moment of inertia
Main and front hoops, shoulder harness mounting bar	2.0 mm	173 mm ²	11 320 mm ⁴
Impact structures, front bulkhead, roll hoop bracing, driver's restraint harness attachment (except as noted above)	1.2 mm	119 mm ²	8509 mm ⁴
Front bulkhead support, main hoop bracing supports	1.2 mm	91 mm ²	6695 mm ⁴

Table 2: Minimum Material Requirements

T 3.2.2 Steel tubing has to be made from unalloyed carbon steel with a maximum content of 0.3 % carbon, 1.7 % manganese and 0.6 % of any other element. All other steel grades are considered alternative materials and require additional testing and documentation, see [T 3.4](#).

T 3.2.3 Except for inspection holes, any holes drilled in any part which is a member of the primary structure must be considered in the SES.

T 3.2.4 The steel properties used for the calculations in the SES must be the following, or what is written in the material property documentation, whichever is lower:

Non-welded strength for continuous material calculations:

- Young's Modulus (E) = 200 GPa
- Yield Strength (S_y) = 305 MPa
- Ultimate Strength (S_u) = 365 MPa

Welded strength for discontinuous material such as joint calculations:

- Yield Strength (S_y) = 180 MPa
- Ultimate Strength (S_u) = 300 MPa

T 3.2.5 Any tubing with a wall thickness less than 1.2 mm or a minimum area moment of inertia less than 6695 mm⁴ is considered non-structural and is ignored when assessing compliance to any rule regarding the vehicle structure.

T 3.2.6 If a member of the primary structure (except for the roll hoops) is a bent tube or made from multiple tubes an additional tube must support it. This support tube must:

- have its attachment point at the position along the bend tube where it deviates farthest from a straight line connecting both ends.

T 3 General Chassis Design

- be of the same dimension as the supported tube(s).
- terminate at a node of the primary structure.
- be angled no more than 30° from the plane of the supported tube(s).

T 3.2.7 Any welded seams shape must not be mechanically altered in any way.

T 3.2.8 Where bonded joints are applicable and governed by the rules, a 50 % reduction must be applied to all nominal and tested adhesive values. Bonded joints in the primary structure must be documented in the SES.

T 3.3 Critical Component Mounting

T 3.3.1 Components, see [T 9.3.5](#), [CV 5.2.1](#) and [EV 5.5.8](#), themselves and their mountings must be able to withstand the following accelerations:

- 40 g in the longitudinal direction (forward/aft)
- 40 g in the lateral direction (left/right)
- 20 g in the vertical direction (up/down)

T 3.4 Alternative Materials

T 3.4.1 Alternative materials may be used for all parts of the primary structure and the TSAC with the following exceptions:

- The main hoop and the main hoop bracing must be steel
- The front hoop must be metal
- Any welded structures of the primary structure must be steel
- However, the front hoop may be an aluminium welded structure

T 3.4.2 If any other materials than steel tubing are used in the primary structure or the TSAC, physical testing is required to show equivalency to the minimum material properties for steel in [T 3.2](#).

T 3.4.3 If alloyed steel as defined by [T 3.2.2](#) is used, the team has to include tests and documentation in the SES to show structural equivalency. This may include, but is not limited to:

- Receipts and datasheets of the used tubing materials
- Documentation about welding processes and filler materials
- Documentation about heat treatments
- Tests showing adequate strength and elongation at break in the welded condition

T 3.4.4 Parts of the steering, brake and suspension system must not be manufactured using additive processes that produce anisotropic strength properties (e.g. consumer-grade FDM). Additively manufactured parts are permitted only if proven to have equivalent strength in all load directions.

T 3.5 Composite Structures

T 3.5.1 If composite structures are used in the primary structure and/or the TSAC:

- The Flexural Rigidity (EI) of that structure must be calculated with the tools and formulas in the SES and/or ASES.
- The EI must be calculated as the EI of a flat panel about its neutral axis.
- The structure used for flat panel calculations must have the same composition as the structure used in the primary structure or the TSAC.
- The actual geometry and curvature of the panel may only be considered if the flat panel EI equivalency exceeds 60 % of the minimum requirements ([T 3.2](#))
- Actual geometry calculations are not permitted for the side impact structure, front bulkhead and TSAC.

T 3.5.2 If composite materials are used in the primary structure or the TSAC, the SES and/or ASES must include:

- Material type(s)

T 3 General Chassis Design

- Cloth weights
 - Resin type
 - Fiber orientation
 - Number of layers
 - Core material
 - Lay-up technique
 - Adhesive film type(s)
 - 3-point-bend test and shear test data (the AIP is exempt from these tests, but must comply with [T 3.17.4](#))
- T 3.5.3 For any laminate in the primary structure and/or the TSAC, the maximum weight content of parallel fibers, relative to the weight of all fibers in the laminate, is 50 %. All fibers whose orientation falls within any 20° window $\pm 10^\circ$ are considered as parallel.
- T 3.5.4 If an asymmetrical lay-up is used in the primary structure, the thinner skin must have a thickness of at least 40 % of the thicker skin or 1 mm whichever is lower.
- T 3.5.5 When critical fasteners as defined in [T 10.1.1](#) are used to attach components to a sandwich structure, crushing of the sandwich core is not permitted.
- T 3.5.6 Wherever backing plates are required, they must be fully supported by the structure they are attached to.
- T 3.5.7 Backing plates must have a continuous near-circular or near-oval perimeter without concave sections or cut-outs other than bolt holes.
- T 3.5.8 Unidirectional fibers are not allowed in the outermost structural layers of a laminate used in the primary structure.

T 3.6 Laminate Testing

- T 3.6.1 If composite materials are used for any part of the primary structure (except for the AIP) or the TSAC the team must:
- build a representative test panel which must measure exactly 275 mm $\times$ 500 mm that has the same design, laminate and fabrication method as used for the respective part of the primary structure represented as a flat panel. The sides of the test panel must not be laminated (core material must be visible).
 - perform a 3-point bending test on this panel.
- The data from these tests and pictures of the test samples and test setup must be included in the SES. In the pictures, the following must be identifiable: distance between the two supports, dimensions of the load applicator and test sample marking as per [T 3.6.6](#). The test results must be used to derive strength and stiffness properties used in the SES formula for all laminate panels. Representative test panels for parts of the TSAC may use smaller dimensions, provided that the panel core thickness is 5 mm or smaller. This representative test panel must then measure 150 mm $\times$ 275 mm. In this case, the distance between the two test panel supports must be at least 200 mm and the load applicator must have a radius of at least 5 mm. [T 3.6.7](#) and [T 3.6.8](#) do not apply.
- T 3.6.2 If a panel represents side impact structure it must be proven that it has at least the same properties as two steel tubes meeting the requirements for side impact structure tubes for buckling modulus, yield strength and absorbed energy.
- T 3.6.3 A baseline 3-point bending test, performed with steel tube(s), must demonstrate a minimum rig compliance of 85 % and be documented in the SES.
- T 3.6.4 Composite structures with different core thicknesses but otherwise identical construction may use material properties derived from a single test panel, except for the front bulkhead and side impact structure. The panel with the thicker core must be tested and the structure

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using derived material properties must not use a core thickness of less than 66 % of the tested panel.

- T 3.6.5 When a laminate is not quasi-isotropic, i.e. has equal strength and stiffness in the 0°, 90° and ± 45° direction, the results from the 3 point bending test are assigned to the 0° lay-up direction. To show equivalence in the SES, the 0° lay-up direction must be used and oriented accordingly in the chassis.
- T 3.6.6 The test samples must be presented at Technical Inspection. All samples must be marked with the following non-removable (e.g.: permanent marker or engraving, but no sticker) information: laminated structure acronym and date of testing.
- T 3.6.7 The distance between the two test panel supports must be at least 400 mm.
- T 3.6.8 The load applicator used to test any panel or tube must be metallic and have a radius of 50 mm.
- T 3.6.9 The load applicator must overhang the test piece to prevent edge loading.
- T 3.6.10 There must be no material between the load applicator and the test piece.
- T 3.6.11 Perimeter shear tests must be completed which measure the force required to push or pull a 25 mm diameter flat punch through a flat laminate sample. The sample must be at least 100 mm × 100 mm. Core and skin thicknesses must be identical to those used in the actual primary structure and be manufactured using the same materials and processes. If an asymmetrical lay-up is used, the thinner skin must face the punch.
- T 3.6.12 The test fixture must support the entire sample, except for a 32 mm hole aligned co-axially with the punch. The sample must not be clamped to the fixture.

T 3.7 Structural Documentation

- T 3.7.1 All teams must submit a Structural Equivalency Spreadsheet (SES), Structural Equivalency 3D Model (SE3D) and [EV ONLY] Accumulator Structural Equivalency Spreadsheet (ASES).
- T 3.7.2 The SES and [EV ONLY] ASES can be downloaded from the competition website.
- T 3.7.3 The SE3D must contain a three dimensional CAD model of the chassis including all members of the primary structure and their mechanical attachment details.
- [MONOCOQUE ONLY] The SE3D must include the inner, outer skin and core thicknesses.
 - [MONOCOQUE ONLY] If the actual EI of a structure is used in the SES, this must be highlighted and included as a measurable instance.
 - [EV ONLY] The TSACs and their attachment must be included.
 - [CV ONLY] The air intake system, the fuel tank and filler neck must be included.
- T 3.7.4 Vehicles must be fabricated in accordance with the materials and processes described in the SES.
- T 3.7.5 Teams must bring a copy of the approved SES to Technical Inspection.

T 3.8 Roll Hoops

- T 3.8.1 Both roll hoops must be integrated into the primary structure using node-to-node triangulation or equivalent joining methods.
- T 3.8.2 The minimum radius of any bend, measured at the tube centerline, must be at least three times the tube outside diameter. Bends must be smooth and continuous with no evidence of crimping or wall failure. The minimum area moment of inertia, see T 3.2, must be maintained in all areas, including the bends of the manufactured tubes.

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- T 3.8.3 In a plane perpendicular to the longitudinal axis of the vehicle and through the lower end-points of the roll hoop, no part of the primary structure may lie below 30 mm of the endpoints of the roll hoop.
- T 3.8.4 Roll hoops attached to a composite primary structure must be mechanically attached at the top and bottom of both sides of that structure and at intermediate locations if needed to show equivalency. The lower roll hoop tubing attachment points must be within 50 mm of the endpoints of the roll hoop.
- T 3.8.5 Mounting plates welded to the roll hoops must be at least 2 mm thick steel or 3 mm thick aluminium, dependent of the roll hoop material.
- T 3.8.6 Both roll hoops must have one 4.5 mm inspection hole in a non-critical straight location. The outer surface of the tube must be unobstructed for at least 180° to allow measurement of the outer tube dimensions.
- T 3.8.7 If any item which extends outside of the rollover protection envelope is attached to the main hoop and/or main hoop braces, additional bracing is required to prevent bending loads in a rollover situation.

T 3.9 Main Hoop

- T 3.9.1 The main hoop must be constructed of a single piece of uncut, continuous, closed section steel tubing.
- T 3.9.2 In side view the portion of the main hoop which is above its upper attachment point to the side impact structure must be inclined less than 10° from vertical.
- T 3.9.3 In side view any bends in the main hoop above its upper attachment point to the primary structure must be braced to a node of the main hoop bracing support structure with tubing meeting the requirements of main hoop bracing.
- T 3.9.4 In side view any portion lower than the upper attachment point to the side impact structure must be inclined either forward or not more than 10° rearward.

T 3.10 Front Hoop

- T 3.10.1 The front hoop must be constructed of a continuous and closed section.
- T 3.10.2 If the front hoop is made from more than one piece it must be supported by node-to-node triangulation or an equivalent construction.
- T 3.10.3 In side view, no part of the front hoop can be inclined more than 20° from vertical.
- T 3.10.4 If the front hoop is a welded construction made from multiple aluminium profiles, the equivalent yield strength must be considered in the as-welded condition unless the team demonstrates and shows proof that it has been properly solution heat treated and artificially aged. The team must supply sufficient documentation proving the appropriate heat treatment process was performed.
- T 3.10.5 The front hoop requires six attachment points, two on each side connecting to the front bulk-head support structures and two connecting to the front hoop bracing, and must therefore show equivalency to 180 kN, as follows from [T 3.16.1](#) and [T 3.12.5](#).
- T 3.10.6 Fully laminating the front hoop to the monocoque is acceptable. Fully laminating means that the hoop has to be encapsulated with laminate around its whole circumference, see Figure 6. Equivalence to [T 3.8.4](#) must be shown in the [SES](#). The laminate encapsulating the front hoop must overlap by at least 25 mm on each side. It must have the same lay-up as the laminate that it is connecting to.

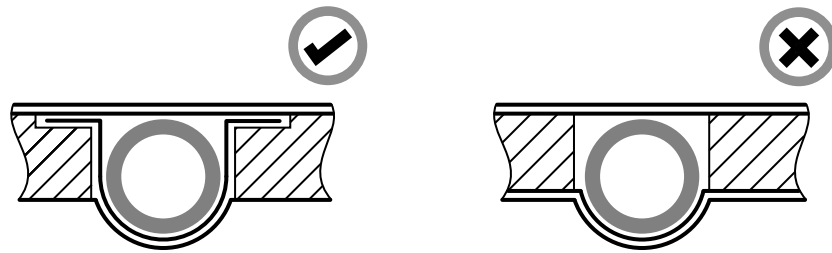


Figure 6: Front hoop laminating requirements

T 3.11 Main Hoop Bracing

- T 3.11.1 The main hoop must be supported to the front or the rear by bracing tubes on each side of the main hoop.
- T 3.11.2 In side view the main hoop and the main hoop braces must not lie on the same side of a vertical line coincident with the top of the main hoop.
- T 3.11.3 The main hoop braces must be attached to the main hoop no lower than 160 mm below the top-most surface of the main hoop. The included angle formed by the main hoop and the main hoop braces must be at least 30°.
- T 3.11.4 The main hoop braces must be straight.
- T 3.11.5 The lower ends of the main hoop braces must be supported back to the upper attachment point of the main hoop to the side impact structure and to the lower attachment point of the main hoop to the side impact structure by a node-to-node triangulated structure or equivalent composite structure.

T 3.12 Front Hoop Bracing

- T 3.12.1 The front hoop bracing attaches on each side of the front hoop as well as the structure forward of the driver's feet. A minimum of two tubes without any bends must be straight on a line in side view of the frame and must have a minimum distance of 100 mm between each other at the front hoop.
- T 3.12.2 The front hoop bracing structure must be attached no lower than 50 mm below the top-most surface of the front hoop, see Figure 7.
- T 3.12.3 In side view, any transition into a recessed section of a composite front hoop bracing structure with a depth greater than 50 mm must have a minimum inclination of 30° from vertical.
- T 3.12.4 If the front hoop is inclined more than 10° to the rear, additional braces extending rearwards are required.
- T 3.12.5 Composite front hoop bracing structures and their attachments cannot be counted towards the front bulkhead support structures and vice-versa for the structural equivalency documentation.
- T 3.12.6 Openings or reductions in effective panel height within a composite front hoop bracing or a composite front bulkhead support must not exceed a total area of 625 cm².

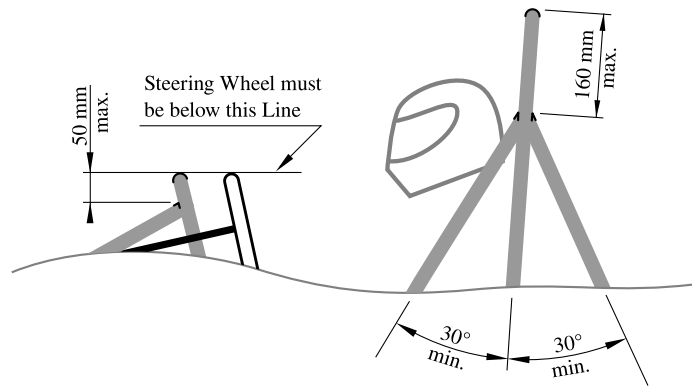


Figure 7: Front hoop bracing, main hoop bracing and steering wheel requirements

T 3.13 Front Bulkhead

- T 3.13.1 Any alternative material used for the front bulkhead must have a perimeter shear strength equivalent to a 1.5 mm thick steel plate.
- T 3.13.2 If the front bulkhead is part of a composite structure and is modeled as an “L” shape, the EI of the front bulkhead about the vertical and lateral axes must be equivalent to a steel tube meeting the requirements for the front bulkhead. The length of the section perpendicular to the bulkhead may be a maximum of 25 mm measured from the rearmost face of the bulkhead.

T 3.14 Front Bulkhead Support

- T 3.14.1 The front bulkhead must be supported back to the front hoop by a minimum of three tubes on each side; an upper member, a lower member and diagonal bracing to provide triangulation.
- The upper support member must be attached to the front bulkhead a maximum of 50 mm below the top-most surface of the front bulkhead, and attached to the front hoop a maximum of 50 mm below the upper side impact member. If the attachment point of the upper member is greater than 100 mm above the upper side impact member, node-to-node triangulated bracing is required to transfer load to the main hoop.
 - The lower support member must be attached to the base of the front bulkhead and the base of the front hoop.
 - The diagonal bracing must triangulate the upper and lower support members node-to-node.
- T 3.14.2 If the front bulkhead support is part of a composite structure, it must have equivalent EI to the sum of the EI of the six baseline steel tubes that it replaces and it must not be counted towards the composite front hoop bracing structures for the structural equivalency documentation, i.e. [T 3.12.5](#).
- T 3.14.3 The EI of the vertical side of the front bulkhead support structure must be equivalent to at least the EI of one baseline steel tube that it replaces.
- T 3.14.4 The perimeter shear strength of the monocoque laminate in the front bulkhead support structure must be at least 4 kN.

T 3.15 Impact Structures

- T 3.15.1 The side impact structure must consist of at least three steel tubes, see [T 3.2](#), on each side of the cockpit, see Figure 8.
- The upper member must connect the main hoop and the front hoop.
 - The upper member must be at a height between 240 mm and 320 mm above the lowest inside chassis point between the front and main hoop.
 - The lower member must connect the bottom of the main hoop and the bottom of the front hoop.

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- The diagonal member must triangulate the upper and lower member between the roll hoops node-to-node.
- T 3.15.2 Other impact structures, see [CV 1.3.2](#), [EV 4.4.2](#) and [EV 5.5.2](#), must be fully triangulated structures and consist of at least three steel tubes, see [T 3.2](#), on each side and rearward of the component that needs to be protected.
- If the component projects outwards from the roll hoops to the side, the front of the component must also be protected.
 - The upper member must not be higher than 320 mm above the lowest inside chassis point between the front and main hoop.
- T 3.15.3 No part of the TSAC, motor(s) or mountings may be within 25 mm of the rearmost impact structure. The TSAC must not attach directly to the rearmost impact structure; non-structural components, such as cooling ducts are permitted. A minimum 12.5 mm clearance from each side impact structure must be maintained to the TSAC and motor(s); local attachment to side impact structures and the bottom is allowed.
- T 3.15.4 If the impact structure is part of a composite structure, the following is required:
- The region of the structure up to a height of 320 mm above the lowest inside chassis point between the front and main hoop must have an EI equal to the three baseline steel tubes that it replaces. An example for side impact structure is given in Figure 9.
 - The vertical impact structure must have an EI equivalent to two baseline steel tubes and half the horizontal floor must have an EI equivalent to one baseline steel tube.
 - The vertical impact structure must have an absorbed energy equivalent to two baseline steel tubes, exceeding 65 J.
 - The perimeter shear strength must be at least 7.5 kN.
- T 3.15.5 The rear impact structure must be positioned such that its diagonal member or equivalent composite structure reacts the loads induced by the IA during vehicle impact, for any position of the minimum IA volume up to 350 mm above the ground.

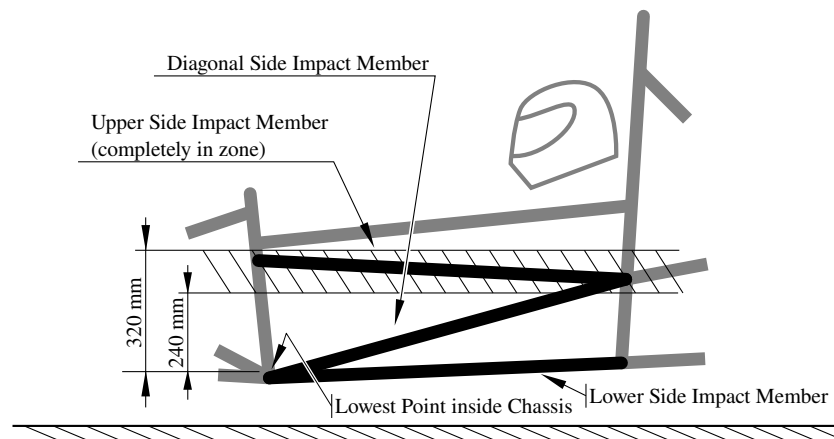


Figure 8: Side impact structure

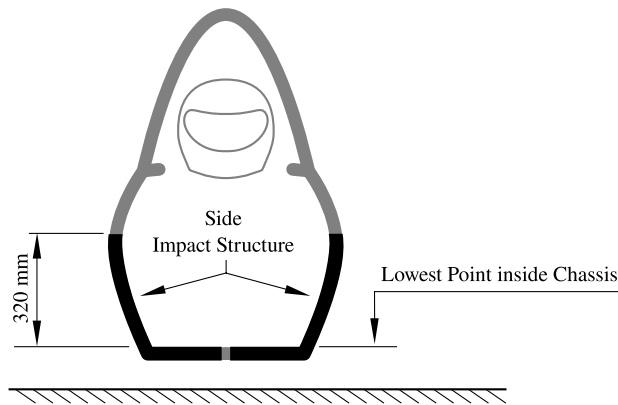


Figure 9: Side impact structure monocoque

T 3.16 Bolted Primary Structure Attachments

- T 3.16.1 If two parts of the primary structure are bolted together, each attachment point between the two parts must be able to carry a load of 30 kN in any direction.
- T 3.16.2 Data obtained from the laminate perimeter shear strength test must be used to prove that adequate shear area is provided.
- T 3.16.3 Each attachment point requires a minimum of two 8 mm metric grade 8.8 critical fasteners, see [T 10.1](#), and steel backing plates of minimum nominal thickness 2 mm and at least ISO 7093-1 washer size.
- T 3.16.4 For the attachment of front hoop bracing, main hoop bracing and main hoop bracing support to the primary structure the use of one 10 mm metric grade 8.8 bolt is sufficient, if the bolt is on the centerline of the tube, see Figure 10.

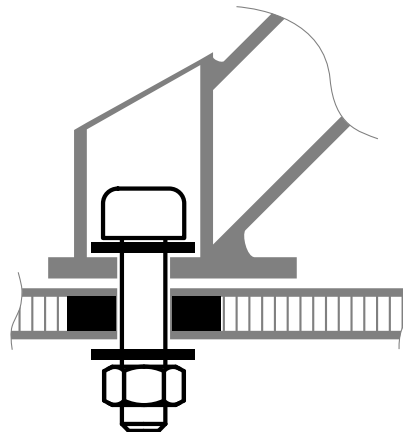


Figure 10: Bolted roll hoop bracing support

- T 3.16.5 For the [AIP](#) to front bulkhead attachment, and if two panels or plates of the primary structure are bolted together, for each 200 mm of reference perimeter a minimum of one 8 mm metric grade 8.8 bolt(s) must be used, rounded up to the next integer. The bolts must be evenly distributed over the circumference using good engineering practices. The reference perimeter is the outside perimeter of the attached part at the connection.
- T 3.16.6 For each bolt and its corresponding insert of [T 3.16.5](#) an exception in favor of blind inserts or alternative backing plate dimensions may be given only if the actual configuration is physically tested and documented in the [SES](#) and/or [ASES](#). Each bolt and its corresponding insert must withstand a force of 15 kN in both the axial pull-out and the weakest radial direction.

T 3.17 Impact Attenuator (IA)

T 3.17.1 Each vehicle must be equipped with an IA assembly consisting of an IA and AIP.

T 3.17.2 The IA must:

- be installed forward of the front bulkhead.
- be at least 100 mm high and 200 mm wide for a minimum distance of 200 mm forward of the front bulkhead.
- not be positioned more than 350 mm above the ground, for any portion of the required 100 mm × 200 mm × 200 mm volume, in any ready-to-race configuration.
- not be able to penetrate the front bulkhead in the event of an impact.
- be securely mounted directly to the AIP.
- not be part of the non-structural bodywork.
- be designed with a closed front section.
- not be wider or higher than the AIP.

T 3.17.3 The baseline design for the Anti Intrusion Plate (AIP) is a 1.5 mm solid steel or 4.0 mm solid aluminium plate.

- If the IA and AIP (IA assembly) are bolted to the front bulkhead, it must be the same size as the outside dimensions of the front bulkhead.
- If it is welded to the front bulkhead, it must extend at least to the centerline of the front bulkhead tubing in all directions.
- The AIP must not extend past the outside edges of the front bulkhead.

T 3.17.4 Alternative AIP designs are permissible if equivalency to T 3.17.3 is proven by physical testing as in T 3.19.2.

T 3.17.5 The IA may be attached to the AIP by a minimum of four 8 mm metric grade 8.8 bolts that are considered critical fasteners and must comply with T 10. Attachment(s) using adhesive must be able to carry a load of 60 kN in any direction.

T 3.17.6 The attachment of the IA assembly must be designed to provide an adequate load path for transverse and vertical loads in the event of off-center and off-axis impacts. Segmented foam attenuators must have the segments bonded together to prevent sliding or parallelogramming.

T 3.17.7 A team may use one of the “standard” FSAE IAs without physical testing, provided that all of the following is fulfilled:

- A 1.5 mm solid steel AIP is used, that is welded along its full perimeter to a steel bulkhead or a 4 mm solid aluminium AIP is used, that is bolted to any bulkhead with a minimum of eight 8 mm metric grade 8.8 bolts.
- The adhesive used to mount the “standard” IA to the AIP has a shear strength of at least 24 MPa.
- T 3.19.4 is fulfilled regarding aerodynamic device and sensor positions relative to the AIP.
- If the front bulkhead has an overall width greater than 400 mm and/or an overall height greater than 350 mm, a diagonal or X-bracing that is a front bulkhead support tube or an approved equivalent per T 3.2, or equivalent for monocoque bulkheads, must be included.
- If the “standard” honeycomb IA is used, the IA must be of pre-crushed type.

If one or more of the above conditions are not met, physical testing must be carried out to prove that the AIP does not permanently deflect more than 25 mm.

T 3.18 Impact Attenuator Data Requirement

T 3.18.1 All teams must submit an IA data report using the Impact Attenuator Data (IAD) template provided at the competition website.

T 3.19 Impact Attenuator Test Requirements

- T 3.19.1 The IA assembly, when mounted on the front of a vehicle with a total mass of 300 kg and impacting a solid, non-yielding impact barrier with a velocity of impact of 7 m/s, must meet the following requirements:
- Decelerate the vehicle at a rate not exceeding 20 g average and 40 g peak.
 - The energy absorbed in this event must meet or exceed 7350 J.
 - Teams using the standard IA are not required to submit test data with their IAD report, but all other requirements must be included.
- Equivalent (higher) test velocities are only allowed to accommodate for a lower total testing mass, as long as the energy absorbed is 7350 J or more. If these requirements cannot be met, a team must use the standard IA.
- T 3.19.2 During the IA test:
- The IA must be attached to the AIP using the intended vehicle attachment method.
 - The IA assembly must be attached to a test fixture that has geometry representative of the intended primary structure and equal or higher stiffness and strength. When no baseline design for the AIP is used, see T 3.17.3, the test fixture must be a copy of the intended primary structure (i.e. materials, lay-up, joining methods).
 - There must be at least 50 mm clearance rearwards of the AIP to the test fixture.
 - No part of the AIP may permanently deflect more than 25 mm beyond the position of the AIP before the test.
 - The representative test fixture of the intended primary structure may not be structurally compromised (i.e. no cracks) after the test.
- T 3.19.3 Teams using IAs (typically structural noses) directly attached to the front bulkhead, which shortcut the load path through the bulk of the AIP, must conduct an additional test. This test must prove that the AIP can withstand a load of 120 kN (300 kg multiplied by 40 g), where the load applicator matches the minimum IA dimensions.
- T 3.19.4 Vehicles with aerodynamic devices and/or sensors forward of the front bulkhead must not exceed the peak deceleration of T 3.19.1 for the combination of their IA assembly and the non-crushable object(s). One of the following three methods must be used to prove the design does not exceed 120 kN:
- Physical testing of the IA assembly including any non-crushable object(s) or geometrically representative dummies thereof forward of the AIP. When non-crushable object(s) are located in front view of the AIP, physical testing is required.
 - Combining the peak force from physical testing of the IA assembly with the failure load for the mounting of the non-crushable object(s), calculated from fastener shear and/or link buckling.
 - Combining the "standard" IA peak load of 95 kN with the failure load for the mounting of the non-crushable object(s), calculated from fastener shear and/or link buckling.
- T 3.19.5 Dynamic testing (sled, pendulum, drop tower, etc.) of the IA may only be conducted at a dedicated test facility. This facility may be part of the university, but must be supervised by professional staff. Teams are not allowed to design their own dynamic test apparatus.
- T 3.19.6 When using acceleration data from the dynamic test, the average deceleration must be calculated based on the raw unfiltered data. If peaks above the 40 g limit are present in the data, a 100 Hz, 3rd order, low pass Butterworth (–3 dB at 100 Hz) filter may be applied.

T 3.20 Non-Crushable Objects

- T 3.20.1 All non-crushable objects (e.g. pedals, master cylinders, hydraulic reservoirs) must be rearward of the rear most plane of the front bulkhead and at least 25 mm behind the AIP at any time, except for sensors, aerodynamic devices and their mountings.
- T 3.20.2 Sensor and aerodynamic devices may extend forward of the AIP, but their chassis attachment point must be located rearward of the AIP.

T 4 Cockpit

T 4.1 Cockpit Opening

- T 4.1.1 The size of the cockpit opening needs to be sufficient for the template shown on the left of Figure 11 to pass vertically from the cockpit opening to below the upper side impact member when held horizontally. The template may be moved fore and aft.
- T 4.1.2 If the side impact structure is not made of tubes, the template must pass until it is 320 mm above the lowest inside chassis point between the front and main hoop.
- T 4.1.3 The steering wheel, seat and all padding may be removed for the template to fit. Any other parts may only be removed if they are integrated with the steering wheel.

T 4.2 Cockpit Internal Cross Section

- T 4.2.1 The cockpit must provide a free internal cross section sufficient for the template shown on the right in Figure 11 to pass from the cockpit opening to a point 100 mm rearwards of the face of the rearmost pedal in an inoperative position. The template may be moved up and down. Adjustable pedals must be in their most forward position.
- T 4.2.2 The steering wheel and any padding that can be removed without the use of tools while the driver is seated may be removed for the template to fit.
- T 4.2.3 The driver's feet and legs must be completely contained within the primary structure when the driver is seated normally and the driver's feet are touching the pedals. In side and front views, any part of the driver's feet or legs must not extend above or outside of this structure.

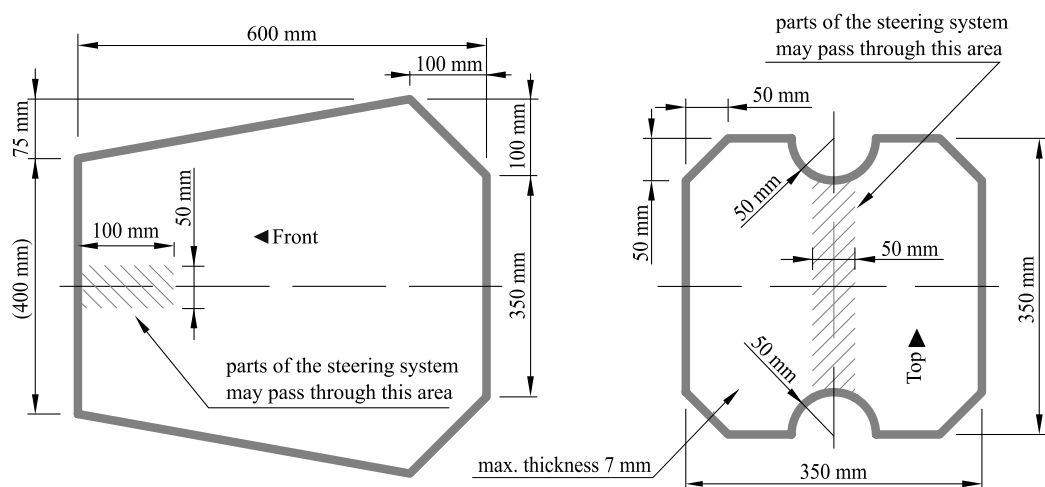


Figure 11: Cockpit opening template (left) and cockpit internal cross section template (right)

T 4.3 Percy (95th percentile male)

- T 4.3.1 When seated normally and restrained by the driver's restraint system, the helmet of a 95th percentile male and all of the team's drivers must, see Figure 12:

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- be a minimum of 50 mm away from the straight line drawn from the top of the main hoop to the top of the front hoop.
- be a minimum of 50 mm away from the straight line drawn from the top of the main hoop to the lower end of the main hoop bracing if the bracing extends rearwards.
- be no further rearwards than the rear surface of the main hoop if the main hoop bracing extends forwards.

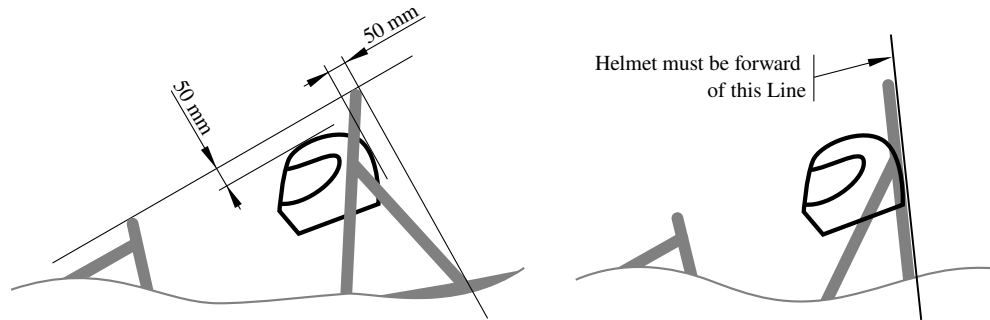


Figure 12: Minimum helmet clearance

- T 4.3.2 The 95th percentile male is represented by a two dimensional figure consisting of two circles of 200 mm diameter (one representing the hips and buttocks and one representing the shoulder region) and one circle of 300 mm (representing the head with helmet).
- T 4.3.3 The two 200 mm circles are connected by a straight line measuring 490 mm. The 300 mm circle is connected by a straight line measuring 280 mm with the upper 200 mm circle.
- T 4.3.4 The figure has to be positioned in the vehicle as follows, see Figure 13:
- The seat adjusted to the rearmost position
 - The pedals adjusted to the frontmost position
 - The bottom 200 mm circle placed on the seat bottom. The distance between the center of the circle and the rearmost actuation face of the pedals must be minimum 915 mm.
 - The middle circle positioned on the seat back
 - The upper 300 mm circle positioned 25 mm away from the head restraint.

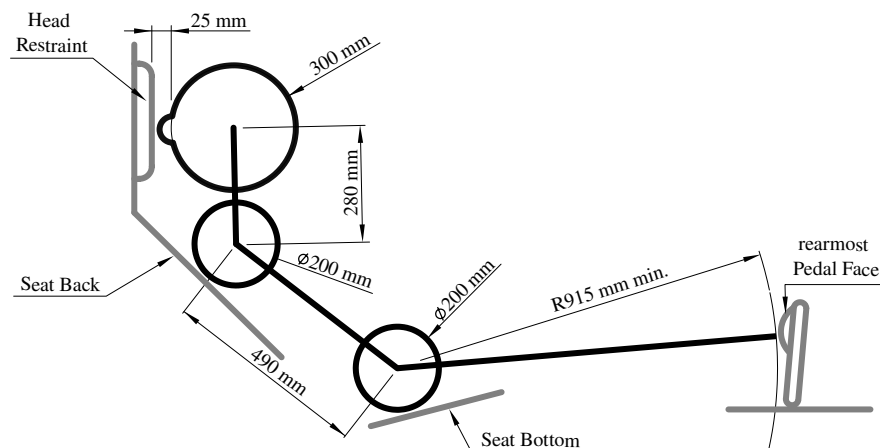


Figure 13: Percy placement

T 4.4 Side Tubes

- T 4.4.1 If there is any chassis member alongside the driver at the height of the neck of any of the drivers in the team, a metal tube or piece of sheet metal must be attached to the chassis to prevent the driver's shoulders from passing under that chassis member.

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T 4.5 Driver's Harness Attachment

- T 4.5.1 Any harness attachment to a monocoque must be using one 10 mm metric grade 8.8 bolt or two 8 mm metric grade 8.8 bolts (or bolts of an equivalent standard) and steel backing plates with a minimum thickness of 2 mm.
- T 4.5.2 Any harness that is fastened to the primary structure using brackets must use two 8 mm metric grade 8.8 or stronger fasteners.
- T 4.5.3 It must be proven that the attachments for shoulder and lap belts can support a load of 13 kN and the attachment points of the anti-submarine belts can support a load of 6.5 kN.
- T 4.5.4 If the lap belts and anti-submarine belts are attached less than 100 mm apart, these must support a total load of 19.5 kN.
- T 4.5.5 If the belts are attached to a laminated structure or the mounting brackets and tabs are not made from steel at least 1.6 mm thick, physical testing is required. The following requirements must be met:
- Load is applied to a test sample representing the tubular or laminated structure and must use the same brackets and tabs
 - Edges of the test fixture supporting the sample must be a minimum of 125 mm from the load application point.
 - The width of the shoulder harness test sample must not be any wider than the shoulder harness panel height used to show equivalency for the shoulder harness mounting bar.
 - Designs with attachments near a free edge must not support the free edge during the test.
 - Harness loads must be tested with the worst case for the range of the angles specified for the driver's harness.

T 4.6 Driver's Seat

- T 4.6.1 The lowest point of the driver's seat must in side view not extend below the upper face of the lowest side impact structure member or have a longitudinal tube (or tubes) that meets the material requirements for the side impact structure, see [T 3.2](#), passing underneath the lowest point of the seat.
- T 4.6.2 Adequate heat insulation must be provided to ensure that the driver is not able to contact any parts of the vehicle with a surface temperature above 60 °C. The insulation may be external to the cockpit or incorporated with the driver's seat or firewall. The design must address all three types of heat transfer with the following minimum requirements between the heat source and the part that the driver could contact:
- (a) Conduction insulation by:
 - (i) No direct contact, or
 - (ii) a heat resistant, conduction insulation material with a minimum thickness of 8 mm.
 - (b) Convection insulation by a minimum air gap of 25 mm.
 - (c) Radiation insulation by:
 - (i) A solid metal heat shield with a minimum thickness of 0.4 mm or
 - (ii) reflective foil or tape when combined with [T 4.6.2.a.ii](#).

T 4.7 Floor Closeout

- T 4.7.1 All vehicles must have a floor closeout made of one or more panels, which separate the driver from the ground.
- T 4.7.2 The closeout must extend from the front bulkhead to the firewall.

T 4 Cockpit

T 4.7.3 The panels must be made of a solid, non-brittle material.

T 4.7.4 If multiple panels are used, gaps between panels must not exceed 3 mm.

T 4.8 Firewall

T 4.8.1 A firewall must separate the cockpit from all components of the fuel supply system, hydraulic fluid (except brake system and dampers), flammable liquids, the LV battery and any TS component, see [EV 1.1.1](#).

T 4.8.2 The firewall must cover any straight line between the parts mentioned in [T 4.8.1](#) and any part of the tallest driver below a plane 100 mm above the bottom of the helmet.

T 4.8.3 The firewall must be a non-permeable surface made from a rigid, fire resistant material, see [T 1.2.1](#), which must be securely mounted to the vehicle's structure.

T 4.8.4 Any firewall must seal completely against the passage of fluids, especially at the sides and the floor of the cockpit.

T 4.8.5 Pass-throughs for wiring, cables, etc. are allowed if grommets, cable glands or connectors are used to seal the pass-through.

T 4.8.6 Multiple panels may be used to form the firewall but must overlap at least 5 mm and be sealed at the joints. Any sealing material must not be vital to the structural integrity of the firewall.

T 4.8.7 [EV ONLY] The TS firewall between driver and TS components must be composed of two layers:

- One solid layer, facing the TS side, must be made of aluminium with a thickness of at least 0.5 mm. This part of the TS firewall must be grounded according to [EV 3.1](#).
- The second layer, facing the driver, must be made of an electrically insulating and fire retardant material, see [T 1.2.1](#). The second layer must not be made of CFRP.
- The thickness of the second layer must be sufficient to prevent penetrating this layer with a 4 mm wide screwdriver and 250 N of force.

A sample of the TS firewall must be presented at Technical Inspection.

T 4.8.8 [EV ONLY] Conductive parts, except for the chassis and firewall mounting points, must not protrude through the TS firewall or must be properly insulated on the driver's side. The driver must not be able to touch uninsulated firewall mounting points while operating the vehicle.

T 4.8.9 [EV ONLY] TS parts outside of the rollover protection envelope, see [T 1.1.16](#) and [EV 4.4.3](#), do not need a firewall.

T 4.9 Accessibility of Controls

T 4.9.1 All vehicle controls must be operated from inside the cockpit without any part of the driver, e.g. hands, arms or elbows, being outside the vertical planes tangent to the outermost surface of the side impact structure.

T 4.10 Driver Visibility

T 4.10.1 The driver must have adequate visibility to the front and sides of the vehicle. Seated in a normal driving position, the driver must have a minimum field of vision of 100° to either side. The required visibility may be obtained by the driver turning their head and/or the use of mirrors.

T 4.10.2 If mirrors are required to meet [T 4.10.1](#), they must remain in place and be adjusted to enable the required visibility throughout all dynamic disciplines.

T 5 Driver Restraint System

T 4.11 Driver Egress

- T 4.11.1 All drivers must be able to exit to the side of the vehicle in less than 5 s with the driver in the fully seated position, hands in the driving position on the connected steering wheel (in all possible steering system positions) and wearing the required driver equipment as in [T 13.3](#). The egress time stops when the driver has both feet on the ground.

T 5 Driver Restraint System

T 5.1 Definitions

- T 5.1.1 **6-point system** — consists of a two-piece lap belt (minimum width 50 mm), two shoulder straps (minimum width 75 mm) and two leg or anti-submarine straps (minimum width 50 mm).
- T 5.1.2 **7-point system** — same as the 6-point system except it has three anti-submarine straps.
- T 5.1.3 **upright driving position** — position with a seat back angled at 30° or less from the vertical as measured along the line joining the two 200 mm circles of the 95th percentile male template as defined in [T 4.3](#) and positioned per [T 4.3.4](#).
- T 5.1.4 **reclined driving position** — position with a seat back angled at more than 30° from the vertical as measured along the line joining the two 200 mm circles of the 95th percentile male template as defined in [T 4.3](#) and positioned per [T 4.3.4](#)

T 5.2 Belts - General

- T 5.2.1 All drivers must use a 6-point or 7-point restraint harness meeting the following specifications:
- All driver restraint systems must meet SFI Specification 16.1, SFI Specification 16.5, SFI Specification 16.6 or FIA specification 8853/2016.
 - The belts must bear the appropriate dated labels.
 - The material of all straps must be in perfect condition.
 - There must be a single metal-to-metal latch type quick release for all straps.
 - All lap belts must incorporate a tilt lock adjuster (“quick adjuster”). A tilt lock adjuster in each portion of the lap belt is highly recommended. Lap belts with “pull-up” adjusters are recommended over “pull-down” adjusters.
 - Vehicles with a “reclined driving position” must have either anti-submarine belts with tilt lock adjusters (“quick adjusters”) or have two sets of anti-submarine belts installed.
 - The shoulder harness must be the “over-the-shoulder type”. Only separate shoulder straps are allowed (i.e. “Y”-type shoulder straps are not allowed). The “H”-type configuration is allowed.
 - The shoulder harness straps must be threaded through the three bar adjusters in accordance with the manufacturer’s instructions.
 - When a Frontal Head Restraint (FHR) device is used by the driver, minimum 50 mm wide shoulder harnesses are allowed.
- T 5.2.2 SFI spec harnesses must be replaced following December 31st of the 2nd year after the date of manufacture as indicated by the label. FIA spec harnesses must be replaced following December 31st of the year marked on the label.
- T 5.2.3 The restraint system must be worn tightly at all times.
- ### T 5.3 Belt, Strap and Harness Installation - General
- T 5.3.1 The lap belt, shoulder harness and anti-submarine strap(s) must be securely mounted to the primary structure. This structure and any guide or support for the belts must meet the minimum requirements of [T 3.2](#).

T 5 Driver Restraint System

- T 5.3.2 Harnesses, belts and straps must not pass through a firewall, i.e. all harness attachment points must be on the driver's side of any firewall.
- T 5.3.3 The attachment of the driver's restraint system requires an approved SES per T 3.7. The lap belts and anti submarine belts must not be routed over the sides of the seat. Where the belts or harness pass through a hole in the seat, the seat must be rolled or grommited to prevent chafing of the belts.

T 5.4 Lap Belt Mounting

- T 5.4.1 The lap belt must pass around the pelvic area below the anterior superior iliac spines (the hip bones).
- T 5.4.2 The lap belts must come through the seat at the bottom of the sides of the seat to maximize the wrap of the pelvic surface and continue in a straight line to the anchorage point.
- T 5.4.3 In side view, the lap belt must be capable of pivoting freely by using either a shouldered bolt or an eye bolt attachment.
- T 5.4.4 With an "upright driving position", in side view the lap belt must be at an angle of between 45° and 65° to the horizontal.
- T 5.4.5 With a "reclined driving position", in side view the lap belt must be between an angle of 60° and 80° to the horizontal.
- T 5.4.6 In "upright driving position" the centerline of the lap belt at the seat bottom should be between 0 mm to 76 mm forward of the seat back to seat bottom junction as in Figure 14. In "reclined driving position" it may also be rearward of the junction point.

T 5.5 Shoulder Harness

- T 5.5.1 The shoulder harness must be mounted behind the driver to a structure that meets the requirements of the primary structure. However, it cannot be mounted to the main hoop bracing or attendant structure without additional bracing to prevent loads being transferred into the main hoop bracing.
- T 5.5.2 If the harness is mounted to a tube that is not straight, the joints between this tube and the structure to which it is mounted must be reinforced in side view by triangulation tubes to prevent torsional rotation of the harness mounting tube. Supporting calculations are required. Analysis method: Use 7 kN load per attachment and the range of angles in T 5.5.5, calculate that the bent shoulder harness bar triangulation stresses are less than as welded yield strength T 3.2.4 for combined bending and shear and does not fail in column buckling. If the team chooses not to perform the strength analysis T 3.2.6 applies.
- T 5.5.3 The strength of any shoulder harness bar and bracing tubes must be proven in the relevant tab of the team's SES submission.
- T 5.5.4 The shoulder harness mounting points must be between 180 mm and 230 mm apart, measured center to center. When a FHR is used, the shoulder harness mounting must comply with the FHR manufacturer's instructions and FIA guidance.
- T 5.5.5 From the driver's shoulders rearwards to the mounting point or structural guide, the shoulder harness must be between 10° above the horizontal and 20° below the horizontal as in Figure 14.

T 5 Driver Restraint System

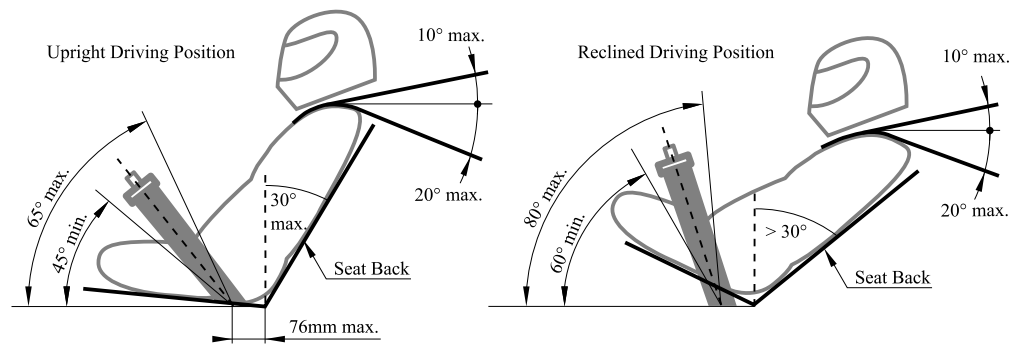


Figure 14: Lap belt and shoulder harness mounting

T 5.6 Anti-Submarine Belt Mounting

T 5.6.1 The anti-submarine belts of a 6 point harness must be mounted in one of the following setups:

- With the belts going vertically down from the groin, or angled up to 20° rearwards. The anchorage points must be approximately 100 mm apart.
- With the anchorage points on the primary structure at or near the lap belt anchorages, the driver sitting on the anti-submarine belts and the belts coming up around the groin to the release buckle.

T 5.7 Head Restraint

T 5.7.1 A head restraint must be provided on the vehicle to limit the rearward motion of the driver's head.

T 5.7.2 The head restraint must:

- be vertical or near vertical in side view.
- be padded with an energy absorbing material with a minimum thickness of 40 mm that meets either the SFI 45.2 standard, or is listed in the FIA technical list n°17 as a type A or B material for single seater cars.
- include a rectangular area of at least a minimum width and height of 150 mm.
- be located so that for each driver:
 - the restraint is no more than 25 mm away from the back of the driver's helmet, with the driver in their normal driving position.
 - the contact point of the back of the driver's helmet on the head restraint is no less than 50 mm from any edge of the head restraint.

T 5.7.3 The head restraint and its mounting must withstand a force of 890 N applied in the rearward direction at any point on its surface.

T 5.8 Roll Bar Padding

T 5.8.1 Any portion of the roll bar, roll bar bracing or chassis which might be contacted by the driver's helmet must be covered with a minimum thickness of 12 mm of padding which meets SFI spec 45.1 or FIA 8857-2001.

T 5.9 Driver's Leg Protection

T 5.9.1 All moving suspension and steering system components and other sharp edges inside the cockpit between the front hoop and a vertical plane 100 mm rearward of the pedals, must be shielded with solid material.

T 5.9.2 Covers over suspension and steering system components must be removable to allow inspection of the mounting points.

T 6 Brake System

T 6.1 Brake System - General

- T 6.1.1 The vehicle must be equipped with a hydraulic brake system that acts on all four wheels and is operated by a single control.
- T 6.1.2 The brake system must have two independent hydraulic circuits such that in the case of a leak or failure at any point in the system, effective braking power is maintained on at least two wheels. Each hydraulic circuit must have its own fluid reserve, either by the use of separate reservoirs or by the use of a dammed reservoir.
- T 6.1.3 The brake system must be sealed to prevent leakage.
- T 6.1.4 While parts of the brake system are disconnected, e.g. by a quick-release decoupling
- [EV ONLY] the vehicle must not be able to enter R2D mode, see [EV 4.11.7](#).
 - the ASB must not pass its initial checkup sequence, see [T 15.3.1](#), in autonomous mode.
- T 6.1.5 A single brake acting on a limited-slip differential is acceptable.
- T 6.1.6 “Brake-by-wire” systems are prohibited in manual mode.
- T 6.1.7 Unarmored plastic brake lines are prohibited.
- T 6.1.8 The brake system must be protected from failure of the drivetrain, see [T 7.3.2](#), from touching any movable part and from minor collisions.
- T 6.1.9 Any part of the brake system must be within the surface envelope, see [T 1.1.18](#).
- T 6.1.10 The brake pedal, including its load-transferring face and mounting, must withstand a 2 kN force without failure of the brake system or pedal box. The whole pedal box assembly must be securely mounted to the primary structure without excessive deformation under driver-applied loads. This may be tested by pressing the pedal with the maximum force an official can exert when seated normally.
- T 6.1.11 The brake pedal, including pedal face, must be fabricated from steel or aluminium or machined from steel, aluminium or titanium.
- T 6.1.12 Adjustable pedals must have a positive stop to prevent loosening from the chassis.
- T 6.1.13 The first 90 % of the brake pedal travel may be used to regenerate brake energy without actuating the hydraulic brake system. The remaining brake pedal travel must directly actuate the hydraulic brake system, but brake energy regeneration may remain active.

T 6.2 Brake Over-Travel Switch (BOTS)

- T 6.2.1 A BOTS must be installed on the vehicle as part of the SDC, as in [EV 6](#) or [CV 4.1](#). This switch must be installed so that in the event of a failure in at least one of the brake circuits the brake pedal over-travel results in the SDC being opened. This must function for all possible brake pedal and brake balance settings without damaging any part of the vehicle.
- T 6.2.2 Repeated actuation of the switch must not close the SDC, and it must be designed so that the driver cannot reset it.
- T 6.2.3 The BOTS must be a push-pull, push-rotate or flip type mechanical switch. It may consist of a series connection of switches.

T 6.3 Brake Light

- T 6.3.1 The vehicle must be equipped with one brake light that is illuminated if and only if

T 7 Powertrain

- the hydraulic brake system is actuated
- [EV ONLY] or the electric brake system is actuated, see [EV 2.2.3](#)

T 6.3.2 The brake light must

- be a red light.
- be clearly visible from the rear.
- meet requirements according to [T 11.10](#).

T 6.3.3 In side view the brake light must be oriented vertically and mounted between the wheel centerline and driver's shoulder level. Viewed from the back it must be positioned approximately at the vehicle's centerline.

T 7 Powertrain

T 7.1 Transmission and Drive

T 7.1.1 Movement of the vehicle without a person in the vehicle and with the master switch(es) in the off position must be possible.

T 7.2 Coolant Fluid and System Sealing

T 7.2.1 [CV ONLY] Water-cooled internal combustion engines must only use plain water as the coolant.

T 7.2.2 Other cooling systems may only use plain water, air or oil as the coolant, see [T 1.2.2](#).

T 7.2.3 Cooling systems using plain water (except outboard wheel motors and their cooling hoses) must have a heat resistant (Permanently rated for at least 100 °C), rigid and securely mounted cover which covers any part of the tallest driver below a plane 100 mm above the bottom of the helmet.

T 7.2.4 Any cooling or lubrication system must be sealed to prevent leakage.

T 7.2.5 Separate catch cans must be employed to retain fluids other than plain water from any vents of the cooling system or combustion engine lubrication system. Each catch can must have a minimum volume of 10 % of the fluid being contained or 900 mL whichever is greater.

T 7.2.6 Any vent for systems containing plain water must have a catch can with a minimum volume of 10 % of the fluid being contained or 100 mL, whichever is greater.

T 7.2.7 All parts of the engine, motor(s) cooling and lubrication system, including their mountings, must be rated for at least 100 °C or the temperatures the respective fluid may reach, whichever is higher.

T 7.2.8 Catch cans must be securely mounted and located rearwards of the firewall below the driver's shoulder level.

T 7.2.9 Any catch can must vent through a hose with a minimum internal diameter of 3 mm down to the bottom level of the chassis and must exit outside the bodywork.

T 7.3 Drivetrain Shields and Guards

T 7.3.1 The lowest point of any lubrication system can only be lower than the line between the lowest point of the main hoop and the lowest triangulated chassis member behind the lubrication system if it is protected from hitting the ground by a structure mounted directly to the chassis.

T 7.3.2 Exposed rotating final drivetrain parts, such as gears, clutches, chains and belts must be fitted with scatter shields. Scatter shields and their mountings must:

- Be constructed of non-perforated 2 mm steel or 3 mm aluminium alloy 6061-T6.
- Be attached with 6 mm metric grade 8.8 or stronger and must comply with [T 10.1](#).

T 8 Aerodynamic Devices

The mounting of exposed rotating final drivetrain parts, except unmodified OEM parts must follow [T 10.1](#).

T 7.3.3 In addition to [T 7.3.2](#), scatter shields must:

- Cover chains and belts from the drive sprocket to the driven sprocket/chain wheel/belt or pulley.
- Start and end parallel to the lowest point of the driven sprocket/chain wheel/belt or pulley.
- Be at least three times the width of the chain or belt and be centered at the centerline of the chain or belt

T 7.3.4 The tractive electric motor(s) must have a housing or separate scatter shield from non-perforated 2 mm aluminium alloy 6061-T6 or equivalent. The scatter shield may be split into two equal sections, each 1 mm thick.

T 7.3.5 Finger guards are required to cover any parts that spin while the vehicle is stationary. Finger guards may be made of lighter material, sufficient to resist finger forces. Mesh or perforated material may be used but must prevent the passage of a 12 mm diameter object through the guard.

T 8 Aerodynamic Devices

T 8.1 Definition Aerodynamic Device

T 8.1.1 A specifically designed structure mounted on the vehicle to guide the airflow around the vehicle, increasing the downforce acting on the vehicle and/or lowering its drag. The mounting of this structure is not regarded as an aerodynamic device, unless it is intentionally designed to be one.

T 8.2 Restrictions for Aerodynamic Devices

T 8.2.1 Height restrictions:

- All aerodynamic devices forward of a vertical plane through the leading edge of the front tires must be lower than 350 mm from the ground.
- All aerodynamic devices rearward of a vertical plane through the rearmost portion of the front face of the driver head restraint support, excluding any padding, set to its most rearward position must be lower than 1100 mm from the ground.
- All other aerodynamic devices must be lower than the virtual horizontal plane defined by the topmost points of the two tires on the left or right side of the vehicle, whichever side is closer to the device.

T 8.2.2 Width restrictions:

- All aerodynamic devices lower than the topmost points of the tires and further rearward than a vertical plane through the leading edge of the front tire, must not be wider than a vertical plane touching the most outboard point of the front and rear wheel/tire.
- All aerodynamic devices between the top edge of the rear tires and 700 mm above the ground, must not be wider than 150 mm inboard of the most inboard point of each rear tire.
- All aerodynamic devices between 700 mm and 1100 mm above the ground, must not extend outboard of the most outboard point of the rear wheel/tire.

T 8.2.3 Length restrictions:

- All aerodynamic devices must not extend further rearward than 250 mm from the rearmost part of the rear tires.
- All aerodynamic devices must not extend further forward than 700 mm from the fronts of the front tires.

T 9 Compressed Gas Systems and High Pressure Hydraulic Systems

- T 8.2.4 All restrictions must be fulfilled with the wheels pointing straight and with any suspension setup with or without a driver seated in the vehicle.

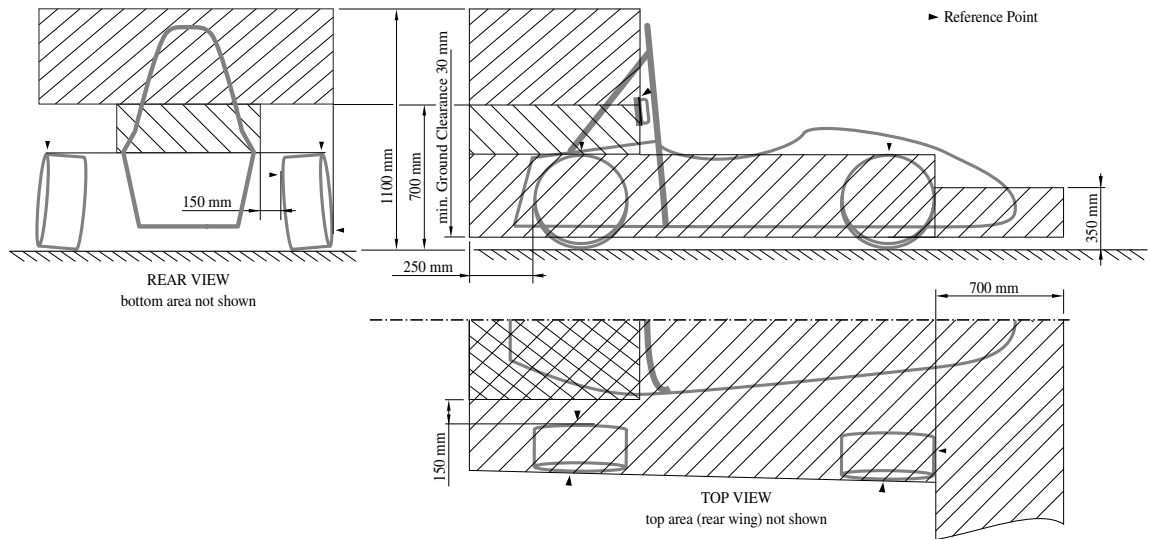


Figure 15: Maximum dimensions and positioning of aerodynamic devices. The positioning space is further restricted, see T 2.1.

T 8.3 Aerodynamic Devices Stability and Strength

- T 8.3.1 Any aerodynamic device must be able to withstand a force of 200 N distributed over a minimum surface of 225 cm² and not deflect more than 10 mm in the load carrying direction.
- T 8.3.2 Any aerodynamic device must be able to withstand a force of 50 N applied in any direction at any point and not deflect more than 25 mm.

T 9 Compressed Gas Systems and High Pressure Hydraulic Systems

T 9.1 Definitions

- T 9.1.1 **Compressed Gas System (CGS)** — any system that uses a compressed gas as an actuating medium, except for gas springs.
- T 9.1.2 **High Pressure Hydraulic System (HPHS)** — any system that uses a non-compressible fluid that is pressurized at 2100 kPa or higher as an actuating medium. Any part of the brake system, see T 6, that is not part of the ASB, see T 15, is not considered HPHS.

T 9.2 General requirements

- T 9.2.1 All parts of any CGS or HPHS must be designed for the maximum possible operating pressure.
- T 9.2.2 All parts of any CGS or HPHS and their mountings must be located within the rollover protection envelope, see T 1.1.16, and must be protected from collision or damage.
- T 9.2.3 The driver and anyone standing outside the vehicle must be shielded from:
- All gas cylinders/tanks and their pressure regulators of any CGS.
 - All parts of any HPHS.

The shielding must be steel or aluminium with a minimum thickness of 1 mm.

T 9.3 Compressed Gas Systems

- T 9.3.1 The working gas of any CGS must be air or pure nitrogen (N_2).
- T 9.3.2 The maximum possible operating pressure inside any CGS must not exceed 10 bar. Gas cylinders/tanks may exceed this limit, if a (series connection of) pressure regulator(s) which limit(s) the output pressure to a maximum of 10 bar is mounted directly onto them.
- T 9.3.3 If exceeding the maximum possible operating pressure limit according to [T 9.3.2](#) gas cylinders/tanks and the pressure regulators mounted directly onto them must be of proprietary manufacture and labeled with π , DOT or TC certification. The purchase date of any gas cylinder/tank must be within the last five years.
- T 9.3.4 The maximum possible operating pressure of any CGS must be limited by means of a pressure relief valve. The relief pressure threshold of the valve must be non-adjustable.
- T 9.3.5 Gas cylinders/tanks must be mounted according to [T 3.3.1](#) and their axis must not point at the driver.
- T 9.3.6 Gas cylinders/tanks must be insulated from any heat sources.

T 10 Fasteners

T 10.1 Critical Fasteners

- T 10.1.1 Critical fasteners are defined as bolts, nuts, and other fasteners utilized in the primary structure, the (autonomous) steering, (autonomous) brake, driver's harness, suspension systems and those specifically designated as critical fasteners in the respective rule.
- T 10.1.2 All threaded critical fasteners must be steel, at least 4 mm metric grade 8.8 (unmodified OEM parts 3 mm metric grade 8.8), equivalent size or of that specified in the referencing rule, whichever is larger. The strength classification of critical fasteners must be identifiable.
- T 10.1.3 All threaded critical fasteners must be of the type hexagon bolts (ISO 4017, ISO 4014 or an equivalent standard) or socket head cap screws (ISO 4762, DIN 7984, ISO 7379 or an equivalent standard) including their fine-pitch thread versions.
- T 10.1.4 Any bolted joint in the primary structure and mounting of the TSAC to the chassis, using either tabs or brackets, must have an edge distance ratio "e/D" of 1.5 or greater. "D" equals the hole diameter and "e" equals the distance from the hole centerline to the nearest free edge of the tab or bracket. Any tabs attaching suspension members to the primary structure are not required to meet this rule.
- T 10.1.5 All threaded critical fasteners attached to the chassis must not be lower than the rollover protection envelope, see [T 1.1.16](#).
- T 10.1.6 For steering and suspension systems, alternative fasteners are allowed if equivalency to [T 10.1.2](#) and [T 10.1.3](#) can be shown.

T 10.2 Securing Fasteners

- T 10.2.1 All critical fasteners must be secured from unintentional loosening by the use of positive locking mechanisms.
- T 10.2.2 The following methods are accepted as positive locking mechanisms:
- Correctly installed safety wiring.
 - Cotter pins.
 - Nylon lock nuts (ISO 7040, ISO 10512, EN 1663 or an equivalent standard) for locations where no temperature rating above 80 °C is required.

T 11 Electrical Components

- Prevailing torque lock nuts (DIN 980, ISO 7042 or an equivalent standard, and jet nuts or K-nuts).
- Locking plates.
- Tab washers.

Any locking mechanism based on pre-tensioning or an adhesive is not considered a positive locking mechanism.

- T 10.2.3 Snap or retaining rings according to DIN 471, DIN 472, or equivalent standard are allowed in unmodified OEM applications or for securing bearings or springs or brake disc floaters given that they do not bear any loads under normal driving conditions. The groove must be in pristine condition and manufactured according to the standard of the snap or retaining ring.
- T 10.2.4 A minimum of two full threads must project from any lock nut.
- T 10.2.5 All spherical rod ends and spherical bearings on the steering or suspension system must be in double shear or captured by having a screw/bolt head or washer with an outer diameter that is larger than the spherical bearing housing inner diameter.
- T 10.2.6 Adjustable tie-rod ends must be constrained with a jam nut to prevent loosening.
- T 10.2.7 Other positive locking mechanisms than specified can only be accepted for brake disc floaters, given they are unmodified OEM applications and installed according to the manufacturer's specifications.

T 11 Electrical Components

T 11.1 Low Voltage System (LVS)

- T 11.1.1 The Low Voltage System (LVS) is defined as
- [CV ONLY] all electrical circuits of the vehicle.
 - [EV ONLY] every electrical part that is not part of the TS, see [EV 1.1.1](#)
- T 11.1.2 The maximum allowed voltage that may occur between any two electrical connections in the LVS is 60 V DC or 50 V AC RMS.
- T 11.1.3 All LVS parts must be adequately insulated.
- T 11.1.4 [CV ONLY] The following systems are excluded from the LVS voltage limit, see [T 11.1.2](#):
- High voltage systems for ignition
 - High voltage systems for injectors
 - Voltages internal to unmodified OEM charging systems designed for < 60 V DC output.
- T 11.1.5 [CV ONLY] The maximum allowed voltage for motor controller/inverters internal low power control signals is 75 V DC
- T 11.1.6 [EV ONLY] The LVS must not use orange wiring or conduit.
- T 11.1.7 [EV ONLY] The LVS must be grounded to the chassis.
- T 11.1.8 Batteries to operate a real-time clock as well as battery-operated driver communication and onboard cameras are not part of the LVS.

T 11.2 Master Switches

- T 11.2.1 Master switches, see [T 11.3](#), [EV 6.2](#), and [T 14.5](#), must be a mechanical switch of the rotary type, with a red, removable handle. The handle must have a width of at least 50 mm and must only be removable in electrically open position. They must be direct acting, i.e. they must not act through a relay or logic.
- T 11.2.2 Master switches must be located on the right side of the vehicle, in proximity to the main hoop, at the 95th percentile male driver's shoulder height, as defined in [T 4.3](#), and be easily

T 11 Electrical Components

actuated from outside the vehicle. The center of any master switch must not be mounted lower than the vertical distance of the template's, see [T 4.3](#), middle circle center to the ground surface multiplied by 0.8.

T 11.2.3 The "ON" position of the switch must be in the horizontal position and must be marked accordingly. The "OFF" position of the master switch must also be clearly marked.

T 11.2.4 Master switches must be securely mounted and must not be removed during maintenance.

T 11.2.5 Master switches must be mounted next to each other.

T 11.3 Low Voltage Master Switch (LVMS)

T 11.3.1 An LVMS according to [T 11.2](#) must completely disable

- [EV ONLY] power to the LVS
- [CV ONLY] power from the LV battery and the alternator to the LVS

T 11.3.2 The LVMS must be mounted in the middle of a completely red circular area of ≥ 50 mm diameter placed on a high contrast background.

T 11.3.3 The LVMS must be marked with "LV" and a symbol showing a red spark in a white edged blue triangle.

T 11.4 Shutdown Buttons

T 11.4.1 A system of three shutdown buttons must be installed on the vehicle.

T 11.4.2 Each shutdown button must be a push-pull or push-rotate mechanical emergency switch where pushing the button opens the SDC, see [EV 6.1](#) and [CV 4.1](#).

T 11.4.3 One button must be located on each side of the vehicle behind the driver's compartment at approximately the level of the driver's head. The minimum allowed diameter of the shutdown buttons on both sides of the vehicle is 40 mm. The buttons must be easily reachable from outside the vehicle.

T 11.4.4 One shutdown button serves as a cockpit-mounted shutdown button and must:

- have a minimum diameter of 24 mm.
- be located in easy reach of a belted-in driver.
- be alongside of the steering wheel and unobstructed by the steering wheel or any other part of the vehicle.

T 11.4.5 The international electrical symbol consisting of a red spark on a white-edged blue triangle must be affixed in close proximity to each shutdown button.

T 11.4.6 Shutdown buttons must be securely mounted and must not be removed during maintenance.

T 11.4.7 Shutdown buttons must be red.

T 11.5 Inertia Switch

T 11.5.1 An inertia switch must be part of the SDC, see [CV 4.1](#) and [EV 6.1](#), such that an impact results in the SDC being opened. The inertia switch must latch until manually reset.

T 11.5.2 The device must trigger due to a longitudinal impact load which decelerates the vehicle at between ≤ 8 g and ≤ 11 g depending on the duration of the deceleration

T 11.5.3 The device must not include any semiconductor components.

T 11.5.4 The device must be securely mounted and installed according to manufacturer specification. It must be removable to test functionality.

T 11.6 Brake System Plausibility Device (BSPD)

- T 11.6.1 A standalone non-programmable circuit, the BSPD, must open the SDC, see [EV 6.1](#) and [CV 4.1](#), when hard braking occurs, whilst
- [EV ONLY] ≥ 5 kW power is delivered to the motors.
 - [CV ONLY] the throttle position is more than 25 % over idle position.
- The SDC must remain open until power cycling the LVMS or the BSPD may reset itself if the opening condition is no longer present for more than 10 s.
- T 11.6.2 The action of opening the SDC must occur if the implausibility is persistent for more than 500 ms.
- T 11.6.3 The BSPD must be directly supplied, see [T 1.3.1](#), from the LVMS, see [T 11.3](#).
- T 11.6.4 Standalone is defined as there is no additional functionality implemented on all required PCBs. The interfaces must be reduced to the minimum necessary signals, i.e. power supply, required sensors and the SDC. Supply and sensor signals must not be routed through any other devices before entering the BSPD. If other systems use the sensors in parallel, the team must show at Technical Inspection they do not interfere with the BSPD.
- T 11.6.5 To detect hard braking, a brake system pressure sensor must be used. The threshold must be chosen such that there are no locked wheels and the brake pressure is ≤ 30 bar.
- T 11.6.6 [EV ONLY] To measure power delivery, a DC circuit current sensor only must be used. The threshold must be chosen to an equivalent of ≤ 5 kW for maximum TS voltage.
- T 11.6.7 It must be possible to separately interrupt each sensor signal path for Technical Inspection.
- T 11.6.8 All necessary signals are SCS, see [T 11.9](#).
- T 11.6.9 [EV ONLY] The BSPD including all required sensors must not be installed inside the TSAC.

T 11.7 Low Voltage Batteries

- T 11.7.1 LV batteries are all batteries connected to the LVS.
- T 11.7.2 LV batteries must be securely mounted and located within the rollover protection envelope, see [T 1.1.16](#).
- T 11.7.3 Any wet-cell battery located in the cockpit must be enclosed in a non-conductive, waterproof (according to IPX7 or higher, IEC 60529) and acid resistant container.
- T 11.7.4 LV batteries must have a rigid and sturdy casing.
- T 11.7.5 Completely closed LV battery cases must have an overpressure relief. Venting gases must be separated from the driver by a firewall.
- T 11.7.6 LV batteries must be protected from short circuits, not more than 100 mm from ungrounded terminals.
- T 11.7.7 Battery packs based on lithium chemistry other than lithium iron phosphate (LiFePO_4) and all hybrid system energy storages regardless of chemistry type:
- must have a fire retardant casing, see [T 1.2.1](#).
 - must include overcurrent protection that trips at or below the maximum specified discharge current of the cells.
 - must include overtemperature protection of at least 30 % of the cells, meeting [EV 5.8.5](#), that trips when any cell leaves the allowed temperature range according to the manufacturer's datasheet, but not more than 60 °C, for more than 1 s and disconnects the battery.

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- must include voltage protection of all cells that trips when any cell leaves the allowed voltage range according to the manufacturer's datasheet for more than 500 ms and disconnects the battery.
- It must be possible to display all cell voltages and measured temperatures, e.g. by connecting a laptop.
- must meet [EV 5.8.14](#).
- Signals needed to fulfill these requirements are [SCS](#), see [T 11.9](#).

T 11.8 Accelerator Pedal Position Sensor (APPS)

- T 11.8.1 [T 11.8](#) only applies for [EVs](#) or [CVs](#) using [ETC](#).
- T 11.8.2 The [APPS](#) must be actuated by a foot pedal.
- T 11.8.3 Pedal travel is defined as percentage of travel from fully released position to a fully applied position where 0 % is fully released and 100 % is fully applied.
- T 11.8.4 The foot pedal must return to the 0 % position when not actuated. The foot pedal must have a positive stop preventing the mounted sensors from being damaged or overstressed. Two springs must be used to return the foot pedal to the 0 % position and each spring must work when the other is disconnected. Springs in the [APPS](#) are not accepted as return springs.
- T 11.8.5 At least two separate sensors must be used as [APPSs](#). The sensors may share the housing.
- T 11.8.6 If analog sensors are used, they must have different, non-intersecting transfer functions. A short circuit between the signal lines must always result in an implausibility according to [T 11.8.9](#).
- T 11.8.7 The [APPS](#) signals are [SCSs](#), see [T 11.9](#).
- T 11.8.8 If an implausibility occurs between the values of the [APPSs](#) and persists for more than 100 ms
- [EV ONLY] The power to the motor(s) must be immediately shut down completely. It is not necessary to completely deactivate the [TS](#), the motor controller(s) shutting down the power to the motor(s) is sufficient.
 - [CV ONLY] The power to the electronic throttle must be immediately shut down.
- T 11.8.9 Implausibility is defined as a deviation of more than ten percentage points pedal travel between any of the used [APPSs](#) or any failure according to [T 11.9](#).
- T 11.8.10 If three sensors are used, then in the case of an [APPS](#) implausibility, any two sensors that are plausible may be used to define the torque target and the 3rd [APPS](#) may be ignored.
- T 11.8.11 It must be possible to separately disconnect each [APPS](#) signal wire to check all functionalities.
- T 11.8.12 A fully released accelerator pedal in manual mode must result in:
- [EV ONLY] a wheel torque of ≤ 0 N m.
 - [CV ONLY] an idle position or lower throttle set-point. This may only be exceeded during a gearshift for a maximum of 500 ms.

T 11.9 System Critical Signal (SCS)

- T 11.9.1 [SCS](#) are defined as all electrical signals which
- influence actions on the [SDC](#), see [CV 4.1](#) and [EV 6.1](#).
 - influence the wheel torque.
 - [EV ONLY] influence indicators according to [EV 5.8.12](#), [EV 4.10](#) or [EV 6.3.8](#).
 - influence indicators according to [T 14.10.4](#).
- T 11.9.2 Any of the following [SCS](#) single failures must result in a safe state of all connected systems:
- (a) Failures of signals transmitted by cable:
- Open circuit

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- Short circuit to ground
- (b) Failures of analog sensor signals transmitted by cable:
 - Short circuit to supply voltage
- (c) Failures of sensor signals used in programmable devices:
 - Implausibility due to out of range signals, e.g. mechanically impossible angle of an angle sensor.
- (d) Failures of digitally transmitted signals by cable or wireless:
 - Data corruption (e.g. checked by a checksum)
 - Loss and delay of messages (e.g. checked by transmission time outs)

Signals might be a member of multiple signal classes, e.g. analog signals transmitted by cable might be a member of [T 11.9.2.a](#), [T 11.9.2.b](#) and [T 11.9.2.c](#).

- T 11.9.3 If a signal failure is correctable, e.g. due to redundancy or worst case values, the safe state must be entered as soon as an additional non correctable failure occurs.
- T 11.9.4 The maximum allowed delay of messages according to [T 11.9.2.d](#) must be chosen depending on the impact of delayed messages to the connected system, but must not exceed 500 ms.
- T 11.9.5 Safe state is defined depending on the signals as follows:
- Signals only influencing indicators – Indicating a failure of its own function or of the connected system
 - LV battery signals – At least one pole is electrically disconnected from the rest of the vehicle
 - ASSI, see [T 14.9](#) – indicating “AS Off”
 - [EV ONLY] For all other signals – opened SDC and opened AIRs
 - [CV ONLY] For all other signals – opened SDC and stopped engine
- T 11.9.6 Indicators according to [T 11.9.1](#) with safe state “illuminated” (e.g. absence of failures is not actively indicated) must be illuminated for 1 s to 3 s for visible check after power cycling the LVMS.

T 11.10 System status light

- T 11.10.1 Any system status light(s), see [T 6.3](#) and [T 14.9](#), must meet the following requirements:
- Black background.
 - Rectangular, triangular or near round shape.
 - Minimum illuminated surface of 15 cm² with even luminous intensity.
 - Clearly visible in very bright sunlight.
 - If LED lights are used without a diffuser, they must not be more than 20 mm apart.
 - If a single line of LED lights is used, the minimum length is 150 mm.

T 11.11 Fans and Turbines

- T 11.11.1 The maximum combined total rated power of any active devices designed to move air is 500 W, this includes cooling fans but does not apply to [CV 1.8](#).

T 11.12 Sensors, Cameras & Electrical Components Mounting

- T 11.12.1 All sensors, cameras and components must be securely mounted.
- T 11.12.2 Sensors, cameras and components must not come into contact with the driver’s helmet under any circumstances.
- T 11.12.3 All sensors, cameras and components (including housing, mounting and covers) must be positioned within the surface envelope, see [T 1.1.18](#), or within the box defined in [T 8.2](#).
- T 11.12.4 Passive antennas which do not incorporate active electronic parts, e.g. amplifiers with the longest side < 100 mm may additionally to [T 11.12.3](#) protrude from the surface envelope, see [T 1.1.18](#), but not their mounting.

T 11.13 Legal & Work Safety

T 11.13.1 All sensors must fulfill the local legislative specifications (i.e. eye-protection classification for laser sensors, power limitation for radar sensors, etc.) in the country of competition.

T 12 Vehicle Identification

T 12.1 Vehicle Number

T 12.1.1 Each vehicle is assigned a number at the time of its entry into an event.

T 12.1.2 Vehicle numbers must appear on the vehicle at the front and both sides as follows:

- Height: At least 150 mm high;
- Font: Roman Sans-Serif characters. Italic, outline, serif, or shadow numbers are prohibited.
- Stroke width and spacing between numbers: At least 20 mm.
- Color: Either white numbers on a black background or black numbers on a white background.
- Background shape: The number background must be one of the following: round, oval, square or rectangular. There must be at least 25 mm between the edge of the numbers and the edge of the background.
- Clear: The numbers must not be obscured by parts of the vehicle.
- Alignment: In front view the number at the front and in side view the numbers on the sides must be horizontally aligned.

T 12.2 Hybrid Vehicle Identification

T 12.2.1 Combustion Hybrid Vehicles have to be identified by an additional marking next to all three vehicle numbers with the capital letters "HY" in Roman Sans-Serif characters at least 75 mm high in white on a red background.

T 12.3 University Name

T 12.3.1 The university name must be written fully. Only the following abbreviations in the university name are allowed, if the city name is written fully:

- University → Uni
- Technical University → TU
- University of Applied Sciences → UAS
- Berufsakademie → BA
- If the university officially uses an abbreviation in their proper name, this abbreviation is allowed.

T 12.3.2 The university name must be displayed and written in Roman Sans-Serif characters that are at least 50 mm tall, in front of the main hoop on both sides of the vehicle. The top of each character must not be more than 50 mm below the top of the bodywork.

T 12.3.3 The characters must be clearly visible at a distance, unobstructed in side view and placed on a high contrast background.

T 12.4 Timing Equipment

T 12.4.1 The event organizers provide timing equipment for all vehicles.

T 13 Vehicle and Driver Equipment

T 13.1 Push Bar

- T 13.1.1 Each team must have a removable device (called the push bar) that attaches to the rear of the vehicle and allows two people to push and pull the vehicle while standing erect behind the vehicle.
- T 13.1.2 The push bar must have a red color.
- T 13.1.3 The university name must be written on the push bar. The characters must be clearly visible and placed on a high contrast background.
- T 13.1.4 The push bar must be capable of slowing and stopping the forward motion of the vehicle and pulling it rearwards.
- T 13.1.5 [CV ONLY] An approved fire extinguisher, see [T 13.4.1](#), must be mounted to the push bar such that it is quickly accessible.
- T 13.1.6 [EV ONLY] Two pairs of high-voltage insulating gloves and a multimeter with two 4 mm banana plug test leads rated for 600 V CAT III or better must be attached to the push bar. The HV gloves must be protected by a case or similar means from mechanical damage, humidity and sunlight. It must be possible to open the case without using tools.

T 13.2 Jacks

- T 13.2.1 Each team must have one or two removable devices (jacks) that hold the vehicle, so that all driven wheels are at least 75 mm off the ground.
- T 13.2.2 Lifting the vehicle and positioning of the device(s) must be done by the team in a safe way.
- T 13.2.3 In lifted position:
- The vehicle must be adequately supported and stable.
 - It must be safe for a driver to enter and exit the vehicle, without additional devices.
 - The device(s) must not extend out of the vehicle's projected surface area.
- T 13.2.4 On both sides of the vehicle the device's pickup points must be indicated by orange triangles.
- T 13.2.5 The university name must be written on the jacks. The characters must be clearly visible and placed on a high contrast background.

T 13.3 Driver Equipment

- T 13.3.1 The equipment specified below must be worn by the driver anytime while in the cockpit with the engine running or with the TS active for electric vehicles and anytime between starting a dynamic discipline and either finishing or abandoning a dynamic discipline. Removal of any driver equipment during the discipline results in a DQ.
- T 13.3.2 A well-fitting, unmodified, closed face helmet that meets one of the following certifications and is labeled as such:
- Snell SA2020, EA2016, SA2025 or newer
 - SFI 31.1/2015, 31.1/2020 or newer
 - FIA 8860-2010, FIA 8860-2018, FIA 8859-2015, 8859-2024 or newer
- Open-faced helmets and off-road helmets (helmets without integrated eye shields) are not approved. All helmets to be used in the event must be presented during Technical Inspection where approved helmets are stickered.

T 13 Vehicle and Driver Equipment

- T 13.3.3 If FHR systems (e.g. HANS) are used, they must be certified to one of the following standards and be labeled as such:
- FIA 8858-2010
 - FIA 8860-2004
 - SFI 38.1
- T 13.3.4 A balaclava which covers the driver's head, hair and neck, made from acceptable fire resistant material as defined in [T 13.3.14](#), or a full helmet skirt of acceptable fire resistant material.
- T 13.3.5 A fire resistant one piece suit, made from a minimum of two layers that covers the body from the neck down to the ankles and the wrists. The suit must be certified to one of the following standards and be labeled as such:
- SFI 3.2A/5 (or higher)
 - SFI 3.4/5 (or higher)
 - FIA Standard 8856-2000
 - FIA Standard 8856-2018
- T 13.3.6 Fire resistant underwear (long pants and long sleeve t-shirt). This fire resistant underwear must be made from acceptable fire resistant material as listed in [T 13.3.14](#) and must cover the driver's body completely from neck down to ankles and wrists.
- T 13.3.7 Fire resistant socks made from acceptable fire resistant material as defined in [T 13.3.14](#), that cover the bare skin between the driver's suit and the boots or shoes.
- T 13.3.8 Fire resistant shoes made from acceptable fire resistant material as defined in [T 13.3.14](#). The shoes must be certified to the standard and labeled as such:
- SFI Spec 3.3
 - FIA Standard 8856-2000
 - FIA Standard 8856-2018
- T 13.3.9 Fire resistant gloves made from acceptable fire resistant material as defined in [T 13.3.14](#). Gloves of all leather construction or fire resistant gloves constructed using leather palms with no insulating fire resisting material underneath are not allowed.
- T 13.3.10 Fire resistant clothing must not be older than 10 years, recognizable since no FIA hologram label present.
- T 13.3.11 Arm restraints are required and must be worn such that the driver can release them and exit the vehicle unassisted regardless of the vehicle's position. Arm restraints must be commercially manufactured according to SFI Standard 3.3 or equivalent.
- T 13.3.12 All driver equipment covered in [T 13.3](#) must be in good condition. Specifically, it must not have any tears, rips, open seams, areas of significant wear or abrasion or stains which might compromise fire resistant performance. Teams must be able to show that the clothing still meets the requirements of [T 13.3](#) if driver's clothing is embroidered. This requires, but is not limited to, that only the outer shell is perforated. The officials reserve the right to impound all non-approved driver equipment until the end of the event.
- T 13.3.13 Driver communication equipment must be securely mounted to the vehicle, except for the microphone and speakers worn by the driver. Wired driver communication equipment, must automatically disconnect during driver egress. The connector must be placed behind the driver's shoulder and the cable length reduced to the minimum.
- T 13.3.14 For the purpose of this section some, but not all, of the approved fire resistant materials are: Nomex, Aramid, Belcotex and Indura.
- T 13.3.15 T-shirts, socks or other undergarments made from nylon or any other synthetic material which melts when exposed to high heat are prohibited.

T 14 Autonomous System (AS)

T 13.4 Fire Extinguishers

T 13.4.1 Each team must have at least two foam type fire extinguishers with valid inspection tag.

T 13.4.2 Fire extinguishers must have a rating of at least:

- USA, Canada and Brazil: 10BC or 1A 10BC
- Europe: 34B or 5A 34B
- Australia: 20BE or 1A 10BE

T 13.4.3 Except for the initial inspection, one extinguisher must readily be available in the team's paddock area.

T 14 Autonomous System (AS)

T 14.1 Definitions

T 14.1.1 Each vehicle must implement a full AS according to T 14, to run in autonomous mode.

T 14.1.2 [CV ONLY] The following definitions apply to maintain the same wording as for EVs:

- R2D – engine is running and a gear is engaged.
- TS active – engine is running but gearbox is in neutral, also assumed for TS not active.
- TS activation button – engine start button
- AIR - fuel pump relay, see Figure 20

T 14.2 Data Logger

T 14.2.1 The officials provide a standardized data logger that must be installed during the event. Further specifications for the data logger and required hardware and software interfaces can be found in the Event Handbook.

T 14.2.2 The team needs to provide two sets of signals to the data logger:

- Basic set of signals as defined in the Event Handbook
- Vehicle-individual set of signals that is monitored by the ASB to ensure redundancy and fault detection

T 14.2.3 [CV ONLY] The mounting of the data logger must be sealed during Technical Inspection.

T 14.3 Remote Emergency System (RES)

T 14.3.1 Every vehicle must be equipped with a standard RES specified in the Event Handbook. The system consists of two parts, the remote control and the vehicle module.

T 14.3.2 The RES must be purchased by the team.

T 14.3.3 The RES has two functions:

- When the remote emergency stop button is pressed, it must open the SDC defined in T 14.4.
- When the "Go" button is pressed, the preselected autonomous mission is started.

T 14.3.4 The RES vehicle module must be directly, see T 1.3.1, integrated in the vehicle's SDC with one of its relays hard-wired in series to the shutdown buttons.

T 14.3.5 The RES relay, which is integrated into the SDC, must be bypassed by a normally closed relay, when driving manually. The relay must:

- be directly supplied, see T 1.3.1, by the ASMS, see T 14.5.
- have a safety certified forcibly guided or a mirrored normally open contact which is directly connected, see T 1.3.1, in series to the ASMS.

T 14 Autonomous System (AS)

T 14.3.6 The antenna of the RES must be mounted unobstructed and without interfering parts in proximity, e.g. other antennas, etc.

T 14.4 Shutdown Circuit

- T 14.4.1 The SDC may only be closed by the AS, if the following conditions are fulfilled:
- Manual Driving: Manual Mission is selected, the AS has checked that ASB is deactivated, i.e. no autonomous brake actuation possible.
 - Autonomous Driving: Autonomous Mission is selected, ASMS is switched on and sufficient brake pressure is built up, i.e. brakes are closed.

T 14.5 Autonomous System Master Switch (ASMS)

- T 14.5.1 Each vehicle must be equipped with an ASMS according to T 11.2.
- T 14.5.2 The ASMS must be mounted in the middle of a completely blue circular area of ≥ 50 mm diameter placed on a high contrast background.
- T 14.5.3 The ASMS must be marked with "AS".
- T 14.5.4 The power supply of all steering and brake actuators must be switched by the LVMS and the ASMS. Other than the direct acting requirement stated in T 11.2.1, a relay with its coil supplied by the ASMS may be used in parallel to the ASMS.
- T 14.5.5 When the ASMS is in "Off" position, the following must be fulfilled:
- No steering, braking and propulsion actuation can be performed by request of the autonomous system.
 - The sensors and the processing units can stay operational.
 - The vehicle must be able to be pushed as specified in A 6.6.
 - It must be possible to operate the vehicle manually as a normal CV or EV.
- T 14.5.6 While the ASMS is switched to the "On" position
- no person may be (seated) inside the vehicle.
 - no other manual steps other than activating the TS may be performed at the vehicle.
- T 14.5.7 The ASMS must be fitted with a "lockout/tagout" capability to prevent accidental activation of the AS. The ASR must ensure that the ASMS is locked in the off position whenever the vehicle is outside the dynamic area or driven in manual mode.

T 14.6 Steering Actuation

- T 14.6.1 Steering system actuation, i.e. movement, must only happen if the vehicle is in R2D mode or during "AS Emergency", see T 14.8.3.
- T 14.6.2 Manual steering must be possible without manual release steps, e.g. operating manual valves and/or (dis-)connecting mechanical elements, while the ASMS is switched "Off".

T 14.7 Actuator Decoupling and Sensor Repositioning

- T 14.7.1 Actuators may be disconnected if the decoupling mechanism cannot block manual operation in any position.
- T 14.7.2 Sensors according to T 11.12 may be repositioned to multiple fixed positions if full rules compliance is ensured in all fixed positions.
- T 14.7.3 The mechanisms used for actuator decoupling, see T 14.7.1, or sensor repositioning, see T 14.7.2, must ensure that
- no parts except covers are removed from the vehicle.

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- covers are only removed from the vehicle temporarily during the process.
- all parts remain in physical contact to the vehicle without using a tether line or similar, i.e. fasteners may be loosened but must not be fully undone.
- they are securely locked in all (fixed) positions.

T 14.8 Autonomous System Status Definitions

- T 14.8.1 The EBS is considered to be “activated”, if the power supply path defined in T 15.2.2 is cut after passing the initial checkup sequence, see T 15.3.1. Brakes may only be released by operating the deactivation point(s) defined in T 15.1.7.
- T 14.8.2 The EBS is no longer considered to be activated, if all brakes have been released using the deactivation point(s) defined in T 15.1.7.
- T 14.8.3 The status of the AS must be determined according to the flowchart in Figure 16.

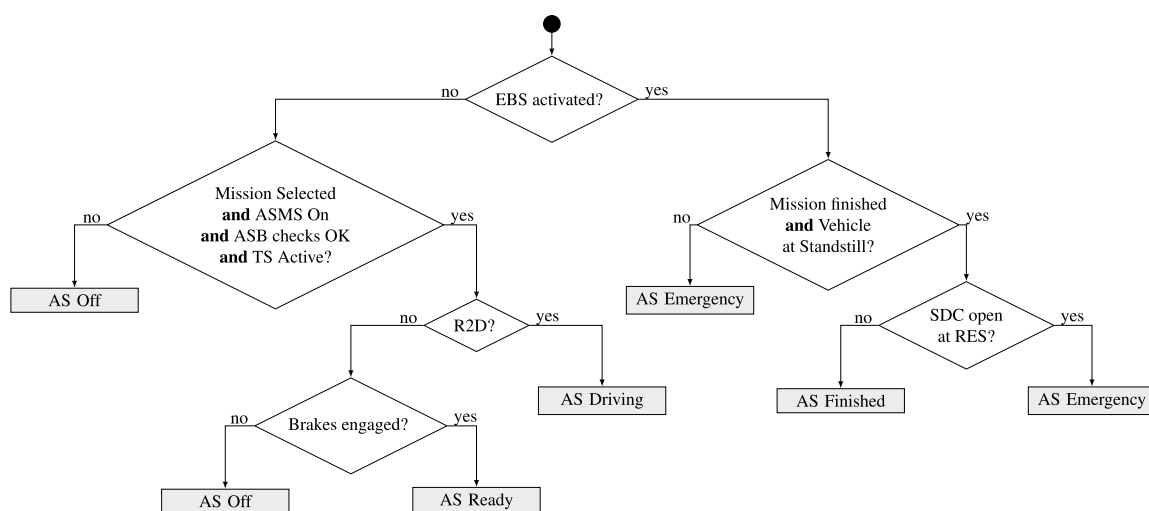


Figure 16: AS Status Flowchart

- T 14.8.4 In autonomous mode, R2D mode may only be entered by the “Go” signal from the RES, after the system has remained in “AS Ready” for at least 5 s.
- T 14.8.5 In autonomous mode, the vehicle must not start moving, until the system has remained in “AS Driving” for at least 3 s.

T 14.9 Autonomous System Status Indicators (ASSIs)

- T 14.9.1 The vehicle must include three ASSIs that must indicate the status of the AS, as defined in T 14.8, correlating to illumination as shown:

AS Off	AS Ready	AS Driving	AS Emergency	AS Finished
off	yellow continuous	yellow flashing	blue flashing	blue continuous

During “AS Driving” and “AS Emergency” the ASSIs must be flashing continuously with a frequency between 2 Hz and 5 Hz and a duty cycle of 50 %. The ASSIs must not perform any other functions.

- T 14.9.2 One ASSI must be located on each side of the vehicle behind the driver’s compartment, in a region 160 mm below the top of the main hoop and 600 mm above the ground. The third ASSI must be located at the rear of the vehicle, on the vehicle centerline, in a region 160 mm below the top of the main hoop and 100 mm above the brake light.
- T 14.9.3 The entire illuminated surface of at least one ASSI must be visible:
- from any angle of the vehicle from a point 1.60 m vertically from ground level, within 3 m horizontal radius from the top of the main hoop.

T 15 Autonomous System Brake (ASB)

- in direct sunlight.

T 14.9.4 Each ASSI must meet the requirements according to [T 11.10](#).

T 14.9.5 The status “AS Emergency” has to be indicated by an intermittent sound with the following parameters:

- on-/off-frequency: 1 Hz to 5 Hz
- duty cycle 50 %
- sound level between 80 dB(A) and 90 dB(A), fast weighting in a radius of 2 m around the vehicle.
- duration between 8 s and 10 s after entering “AS Emergency”

T 14.10 Autonomous Missions

T 14.10.1 The AS must at least implement the following missions:

- Acceleration
- Skidpad
- [DC ONLY] Autocross
- [DC ONLY] Trackdrive
- EBS Test
- Inspection
- Manual Driving

T 14.10.2 The Inspection Mission is defined by slowly spinning the drivetrain and actuating the steering system with a sine wave while the vehicle is jacked up and all wheels are removed. After 25 s to 30 s the AS must transition to “AS Finished”.

T 14.10.3 It must be possible to select any mission without the use of an external device.

T 14.10.4 The selected mission must be indicated by the Autonomous Mission Indicator (AMI).

T 14.10.5 The AMI must be easily readable by any person standing next to the ASMS and can either be part of the dashboard or located next to the ASMS.

T 14.11 Autonomous System Form (ASF)

T 14.11.1 Before the event, all teams must submit a clearly structured documentation of their entire AS, including ASB, called ASF.

T 15 Autonomous System Brake (ASB)

T 15.1 Technical Requirements

T 15.1.1 To run in autonomous mode, the vehicle must be equipped with an ASB that features an EBS, see [T 15.2](#), as part of it.

T 15.1.2 All parts of the ASB and their mountings must be located within the rollover protection envelope, see [T 1.1.16](#), with the following exceptions:

- brake system according to [T 6](#)
- deactivation point(s) according to [T 15.1.8](#)

T 15.1.3 The TS is not considered to be a brake system.

T 15.1.4 Manual braking must always be possible. In case of manual and autonomous braking simultaneously, always the highest of both pressures must be applied to the brakes.

T 15.1.5 Master brake cylinders must not be connected in series.

T 15.1.6 The ASB may be part of the hydraulic brake system. For all components inside the vehicle's brake circuit [T 6](#) is applied. On all remaining pneumatic and hydraulic components [T 9](#) applies.

T 15 Autonomous System Brake (ASB)

T 15.1.7 The ASB must be designed so that it can be easily deactivated by two deactivation points.

T 15.1.8 Any deactivation point of the ASB must:

- work without the aid of electrical power
- be in proximity to each other
- be mounted within the surface envelope, see [T 1.1.18](#)
- either be mounted in proximity to the ASMS or on the top side of the vehicle between front bulkhead and front hoop close to the vehicle's centerline
- be protected against unintended actuation, e.g. by being hit by a cone, while driving
- be operable by maximum two simple push/pull and/or turning actions, the order and direction of these actions must be shown next to the deactivation points.
- be marked with "Brake release"
- have a red handle

T 15.2 Emergency Brake System (EBS)

T 15.2.1 The EBS must only use passive systems which are primarily designed and manufactured for the purpose of mechanically storing energy. Electrical power-loss at EBS must lead to a direct emergency brake maneuver with the performance specified in [T 15.4](#).

T 15.2.2 The EBS must be supplied by

- LVMS
- ASMS
- the forcibly guided or mirrored normally open contact of the relay according to [T 14.3.5](#)
- RES
- a relay which is supplied by the SDC, i.e. parallel to the AIRs, but must not be delayed.

T 15.3 Functional Safety

T 15.3.1 An initial check has to be performed to ensure that the ASB is able to

- operate as specified in [T 15.3.2](#) and [T 15.3.4](#)
- build up brake pressure as expected

before AS transitions to "AS Ready".

T 15.3.2 After the initial check the ASB and its SCS must be continuously monitored for failures.

T 15.3.3 The functionality required to achieve [T 15.3.1](#) including brake actuation and [T 15.3.2](#) must be implemented on a single PCB. Sensor signals may be routed and preprocessed through additional PCBs.

T 15.3.4 The vehicle must automatically transition to the safe state, if:

- the functionality according to [T 15.2.1](#) cannot be ensured.
- an (additional) single point of failure would lead to total loss of brake capability.

If the ASB actuates the brake system, see [T 6](#), exclusively through the brake pedal, see [T 6.1.10](#), and equivalent to the actuation by the driver's foot the transition to the safe state does not need to be ensured in the event of a failure of the brake pedal itself.

T 15.3.5 The safe state is the vehicle at a standstill, brakes engaged to prevent the vehicle from rolling, and an open SDC.

T 15.3.6 To get to the safe state, the vehicle must perform an autonomous brake maneuver described in section [T 15.4](#) and [IN 11.2](#).

T 15.4 Emergency Brake System Performance

T 15.4.1 The system reaction time, i.e. the time between opening of the SDC and the start of the deceleration, must not exceed 200 ms.

T 15 Autonomous System Brake (ASB)

- T 15.4.2 The average deceleration must be greater than 10 m/s^2 under dry track conditions.
- T 15.4.3 In case of a single failure the ASB should be designed to achieve at least half of the performance specified in [T 15.4.2](#).
- T 15.4.4 Whilst decelerating, the vehicle must remain in a stable driving condition.

CV Internal Combustion Engine Vehicles (CVs)

CV 1 Internal Combustion Engine Powertrains

CV 1.1 Engine Limitation

CV 1.1.1 The engine(s) used to power the vehicle must be piston engine(s) using a four-stroke primary heat cycle with a displacement not exceeding 710 cm³ per cycle. Hybrid powertrains are allowed with electric motors that are connected to the engine, gearbox, wheels, exhaust turbines or superchargers. Hybrid powertrains must use a purely electrical energy storage. Starter motors that cannot provide propulsion when the engine is running are not considered hybrid.

CV 1.2 Starter

CV 1.2.1 Each vehicle must be equipped with an on-board starter, which must be used to start the vehicle.

CV 1.2.2 For autonomous operation, the vehicle must be equipped with an additional engine start button next to the LVMS, see [T 11.3](#), that can be easily actuated from outside the vehicle. Using the external engine start button, the engine may only start if

- the ASMS, see [T 14.5](#), is switched on and
- the gearbox is in neutral.

CV 1.2.3 If [CV 1.2.2](#) requires an additional engine start button, there must be a green light next to it, that indicates that the gearbox is in neutral. It must be marked with the letter "N". This letter must have a minimum height of 25 mm.

CV 1.2.4 The AS must not be able to (re-)start the engine.

CV 1.3 Air Intake System

CV 1.3.1 All parts of the engine air and fuel control systems (including the throttle and the complete air intake system, including the air filter and any air boxes) must lie within the surface envelope, see [T 1.1.18](#).

CV 1.3.2 Any portion of the air intake system that is less than 350 mm above the ground must be protected from impacts, see [T 3.15](#). Impact protection must follow [T 3.16](#) when having bolted attachments.

CV 1.3.3 The intake manifold must be securely mounted to the engine block or cylinder head with brackets and mechanical fasteners that are considered critical fasteners and must comply with [T 10](#). This mounting must ensure that the intake manifold including the injectors is securely mounted. Rubber bushings or hoses are not considered a secure mounting.

CV 1.3.4 Intake systems with significant mass or cantilever from the cylinder head must be supported to prevent stress to the intake system resulting from engine movement and chassis torsion.

CV 1.4 Throttle

CV 1.4.1 The vehicle must be equipped with a throttle body. The throttle body may be of any size or design.

CV 1 Internal Combustion Engine Powertrains

- CV 1.4.2 The throttle must be actuated mechanically by a foot pedal, i.e. via a cable or a rod system, see [CV 1.5](#), or by an ETC system, see [CV 1.6](#).
- CV 1.4.3 Throttle position is defined as percentage of travel from fully closed to fully open where 0 % is fully closed and 100 % is fully open. The idle position is the average position of the throttle body while the engine is idling.
- CV 1.4.4 The throttle system mechanism must be protected from debris ingress to prevent jamming.

CV 1.5 Mechanical Throttle Actuation

- CV 1.5.1 [CV 1.5](#) can only be used if no AS is used.
- CV 1.5.2 The throttle actuation system must use at least two return springs located at the throttle body, so that the failure of any one of the two springs does not prevent the throttle returning to the idle position.
- CV 1.5.3 Each return spring must be capable of returning the throttle to the idle position with the other disconnected.
- CV 1.5.4 Springs in the TPS are not allowed as return springs.
- CV 1.5.5 Throttle cables must be located at least 50 mm from any exhaust system component and out of the exhaust stream.
- CV 1.5.6 Throttle cables or rods must have smooth operation and must not have the possibility of binding or sticking. They must be protected from being bent or kinked by the driver's foot during operation or when entering the vehicle.
- CV 1.5.7 A positive pedal stop must be incorporated on the accelerator pedal to prevent over-stressing the throttle cable or actuation system.

CV 1.6 Electronic Throttle Control (ETC)

- CV 1.6.1 [CV 1.6](#) only applies if ETC is used.
- CV 1.6.2 The team must be able to demonstrate the functionality of all safety features and error detections of the ETC system at Technical Inspection, see [IN](#).
- CV 1.6.3 The ETC system must be equipped with at least the following sensors:
- Accelerator Pedal Position Sensors (APPSs) as defined in [T 11.8](#).
 - Two Throttle Position Sensors (TPSs) to measure the throttle position.
- CV 1.6.4 All ETC signals are SCS, see [T 11.9](#).
- CV 1.6.5 When power is removed, the electronic throttle must immediately close at least to idle position $\pm 5\%$. An interval of one second is allowed for the throttle to close to idle, failure to achieve this within the required interval must result in immediate disabling of power to ignition, fuel injectors and fuel pump. This action must remain active until the TPS signals indicate the throttle has returned to idle position $\pm 5\%$ for at least one second.
- CV 1.6.6 If plausibility does not occur between the values of at least two TPSs and this persists for more than 100 ms, the power to the electronic throttle must be immediately shut down. Plausibility is defined as a deviation of less than ten percentage points between the sensor values as defined in [CV 1.4.3](#) and no detected failures as defined in [T 11.9](#). AS must check this signal consistency on a low level itself.
- CV 1.6.7 The electronic throttle must use at least two sources of energy capable of returning the throttle to the closed position. One of the sources may be the device that normally actuates

CV 1 Internal Combustion Engine Powertrains

the throttle, e.g. a DC motor, but the other device(s) must be a return spring that can return the throttle to the idle position in the event of a loss of actuator power.

CV 1.6.8 Springs in the TPSs are not acceptable as return springs.

CV 1.6.9 The power to the electronic throttle must be immediately shut down, as defined in [CV 1.6.5](#), if the throttle position differs by more than 10 % from the expected target TPS position for more than 500 ms.

CV 1.7 Intake System Restrictor

CV 1.7.1 In order to limit the power capability from the engine(s), a single circular restrictor must be placed in the intake system and all engine(s) airflow must pass through this restrictor. The only allowed sequences of components are the following:

- For naturally aspirated engines, the sequence must be: throttle body, restrictor, and engine, see Figure 17
- For turbocharged or supercharged engines, the sequence must be: restrictor, compressor, throttle body, engine, see Figure 18

CV 1.7.2 The maximum restrictor diameters which must be respected at all times during the event are:

- Gasoline fueled vehicles - 20 mm
- E 85 fueled vehicles - 19 mm

CV 1.7.3 The restrictor must be located to facilitate measurement during the Technical Inspection process.

CV 1.7.4 The circular restricting cross section must not be movable or flexible in any way, e.g. the restrictor must not be part of the movable portion of a barrel throttle body.

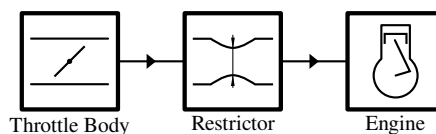


Figure 17: Intake configuration for naturally aspirated engines.

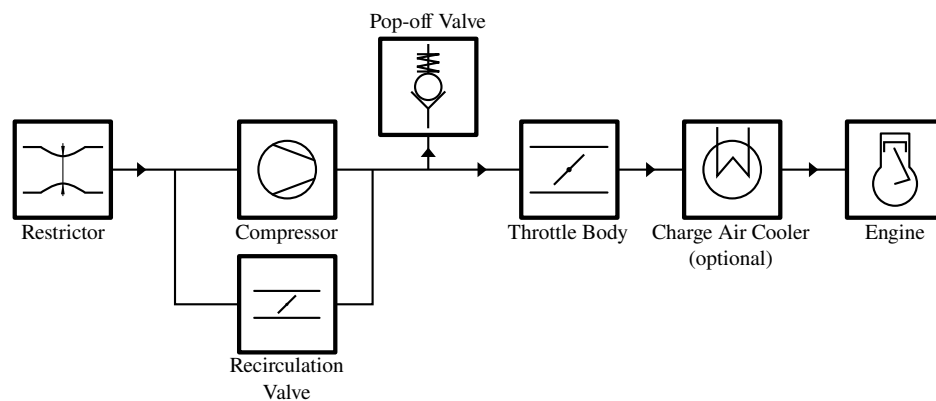


Figure 18: Intake configuration for turbocharged or supercharged engines.

CV 1.8 Turbochargers and Superchargers

CV 1.8.1 The intake air may be cooled with an intercooler. Only ambient air may be used to remove heat from the intercooler system. Air-to-air and water-to-air intercoolers are allowed. The coolant of a water-to-air intercooler system must be plain water without any additives.

CV 1.8.2 If pop-off valves, recirculation valves, or heat exchangers (intercoolers) are used, they may only be positioned in the intake system as shown in Figure 18.

CV 1.8.3 Plenums anywhere upstream of the throttle body are prohibited. A "plenum" is any tank or volume that is a significant enlargement of the normal intake runner system.

CV 2 Fuel and Fuel System

- CV 1.8.4 The maximum allowable internal diameter of the intake runner system between the restrictor and throttle body is 60 mm diameter, or the equivalent area of 2827 mm² if non-circular.

CV 1.9 Crankcase / Engine Lubrication Venting

- CV 1.9.1 Any crankcase or engine lubrication vent lines routed to the intake system must be connected upstream of the intake system restrictor.
- CV 1.9.2 Crankcase breathers that pass through the oil catch tank(s) to exhaust systems, or vacuum devices that connect directly to the exhaust system, are prohibited.

CV 2 Fuel and Fuel System

CV 2.1 Fuel

- CV 2.1.1 The available fuel types are unleaded gasoline 98RON and E 85.
- CV 2.1.2 The vehicles must be operated with the fuel provided at the event.
- CV 2.1.3 No agents other than fuel and air may be induced into the combustion chamber.
- CV 2.1.4 The temperature of fuel introduced into the fuel system must not be changed with the intent to improve calculated efficiency.

CV 2.2 Fuel System Location Requirements

- CV 2.2.1 The fuel tank must be located within the rollover protection envelope, see [T 1.1.16](#), except the fuel filler neck if it is 350 mm above the ground.
- CV 2.2.2 All parts of the fuel storage and supply system must lie within the surface envelope, see [T 1.1.18](#).
- CV 2.2.3 All parts of the fuel storage and supply system must be adequately protected against any heat sources and located at least 50 mm from any exhaust system component.
- CV 2.2.4 All parts of the fuel system which can come in contact with the fuel must be rated for permanent contact with fuel.

CV 2.3 Fuel Tank

- CV 2.3.1 The fuel tank is defined as the part of the fuel containment device that is in contact with the fuel. It may be made of a rigid material or a flexible material.
- CV 2.3.2 The fuel tank must be securely mounted to the vehicle structure with mountings that allow some flexibility such that chassis flex cannot unintentionally load the fuel tank.
- CV 2.3.3 The fuel tank must not touch any part of the vehicle other than its mounting and parts of the fuel system at any time.
- CV 2.3.4 Any fuel tank that is made from a flexible material, for example a bladder fuel cell or a bag tank, must be enclosed within a rigid fuel tank container which is securely mounted to the vehicle structure. Fuel tank containers (containing a bladder fuel cell or bag tank) may be load carrying.
- CV 2.3.5 The fuel system must have a provision for emptying the fuel tank if required.
- CV 2.3.6 The fuel tank, by design, must not have a variable capacity.

CV 2.4 Fuel Lines for Low Pressure Systems

CV 2.4.1 Fuel lines between fuel tank and fuel rail and return lines must:

- have reinforced rubber fuel lines with an abrasion protection with a fuel hose clamp which has a full 360° wrap, a nut and bolt system for tightening and rolled edges to prevent the clamp cutting into the hose, or
- have metal braided hoses with crimped-on or reusable, threaded fittings.
- be rated for temperatures of at least 120 °C.

CV 2.4.2 The use of unmodified OEM fuel lines and connectors, including those manufactured from plastic, is acceptable.

CV 2.4.3 For reinforced rubber fuel lines with hose & clamp style connections beaded or barbed hose fittings must be used.

CV 2.4.4 Fuel lines must be securely mounted.

CV 2.4.5 All fuel lines must be shielded from possible rotating equipment failure or collision damage.

CV 2.4.6 All fuel lines must be fitted in such a way that any leakage cannot result in the accumulation of fuel in the cockpit.

CV 2.5 Fuel Injection System Requirements

CV 2.5.1 The following requirements apply to LPI fuel systems:

- The fuel lines must comply with [CV 2.4](#).
- The fuel rail must be securely mounted to the engine cylinder block, cylinder head, or intake manifold with mechanical fasteners. The threaded fasteners used to secure the fuel rail are considered critical fasteners and must comply with [T 10](#).
- The use of fuel rails made from plastic, carbon fiber or rapid prototyping flammable materials is prohibited. However, the use of unmodified OEM Fuel Rails manufactured from these materials is acceptable.

CV 2.5.2 The following requirements apply to HPI and DI fuel systems:

- All high pressure fuel lines must be stainless steel rigid line or Aeroquip FC807 smooth bore PTFE hose with stainless steel reinforcement and visible Nomex tracer yarn. Use of elastomeric seals is prohibited. Lines must be securely mounted every 100 mm by mechanical fasteners to structural engine components.
- The fuel rail must be securely mounted to the engine cylinder head with mechanical fasteners. The fastening method must be sufficient to hold the fuel rail in place with the maximum regulated pressure acting on the injector internals and neglecting any assistance from in-cylinder pressure acting on the injector tip. The threaded fasteners used to secure the fuel rail are considered critical fasteners and must comply with [T 10](#).
- The fuel pump must be securely mounted to structural engine components.
- A fuel pressure regulator must be fitted between the high and low pressure sides of the fuel system in parallel with the DI boost pump. The external regulator must be used even if the DI boost pump comes equipped with an internal regulator.
- Before the Tilt Test specified in [IN 7](#), engines fitted with mechanically actuated fuel pumps must be run to fill and pressure the system downstream of the high pressure pump.

CV 2.6 Fuel Tank Filler Neck and Sight Tube

CV 2.6.1 The fuel tank must have a filler neck which:

- has at least an inner diameter of 35 mm at any point between the fuel tank and the top of the fuel filler cap.

CV 2 Fuel and Fuel System

- is accompanied by a clear fuel resistant sight tube above the top of the fuel tank with a length of at least 125 mm vertical height for reading the fuel level, see Figure 19.
- is made of material that is permanently rated for temperatures of at least 120 °C.

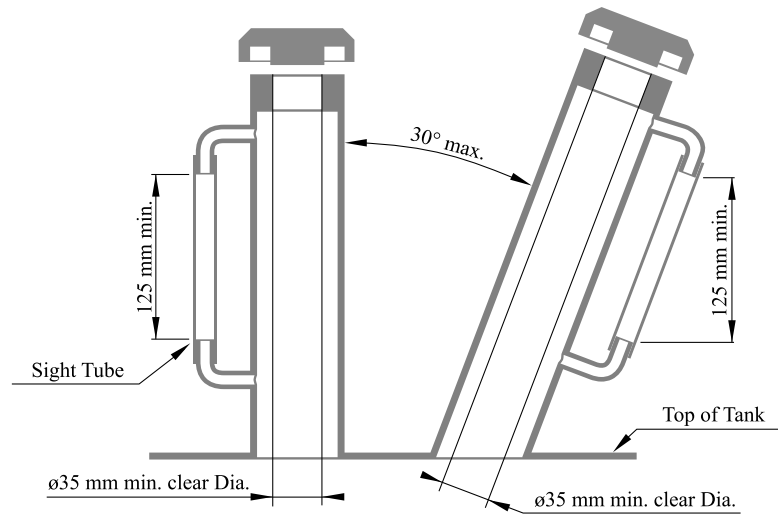


Figure 19: Minimum requirements fuel tank filler neck.

- CV 2.6.2 A clear filler neck tube may be used as a sight tube.
- CV 2.6.3 Above the lowest point of the sight tube, the filler neck must not be angled more than 30° from the vertical.
- CV 2.6.4 A permanent, non-movable, clear and easily visible fuel level line must be located between 12 mm and 25 mm below the top of the visible portion of the sight tube. This line is used as the fill line for the Tilt Test, see [IN 7.1](#), and before and after Endurance to measure the amount of fuel used during Endurance.
- CV 2.6.5 The filler neck opening must be directly accessible without removing any parts of the vehicle except for the fuel filler cap.
- CV 2.6.6 The filler neck must have a fuel filler cap that can withstand severe vibrations or high pressures such as could occur during a vehicle rollover event.

CV 2.7 Tank Filling Requirement

- CV 2.7.1 The fuel tank must be capable of being filled to capacity without manipulating the tank or the vehicle in any manner. The fuel system must be designed in a way that during refueling of the vehicle on a level surface, the formation of air cavities or other effects that cause the fuel level observed at the sight tube to drop after movement or operation of the vehicle (other than due to consumption) is prevented.
- CV 2.7.2 The fuel system must be designed such that the spillage during refueling cannot contact the driver position, exhaust system, hot engine parts, or the ignition system.

CV 2.8 Venting Systems

- CV 2.8.1 The fuel tank venting systems must be designed such that fuel cannot spill during hard cornering or acceleration.
- CV 2.8.2 All fuel vent lines must be equipped with a check valve to prevent fuel leakage when the tank is inverted. All fuel vent lines must exit outside the bodywork.

CV 3 Exhaust System and Noise Control

CV 3.1 Exhaust System General

- CV 3.1.1 The exhaust outlet must be routed to the side or rear of the vehicle and so that the driver is not subjected to fumes at any speed considering the draft of the vehicle.
- CV 3.1.2 The exhaust outlet(s) must not extend more than 450 mm behind the centerline of the rear axle and must be no more than 600 mm above the ground.
- CV 3.1.3 Any exhaust components (headers, mufflers, etc.) that protrude from the side of the chassis in front of the rear axle must be shielded to prevent contact by persons approaching the vehicle or a driver exiting the vehicle. The temperature of the outer surface must not be harmful to a person touching it.
- CV 3.1.4 The application of fibrous/absorbent material, e.g. "headerwrap", to the outside of an exhaust manifold or exhaust system is prohibited.

CV 3.2 Maximum Sound Level

- CV 3.2.1 The maximum sound level test speed for a given engine is the engine speed that corresponds to an average piston speed of 15.25 m/s. The calculated speed is rounded to the nearest 500 rpm. The maximum allowed sound level up to this calculated speed is 110 dB(C), fast weighting.
- CV 3.2.2 The idle test speed for a given engine is up to the team and determined by their calibrated idle speed. If the idle speed varies then the vehicle is tested across the range of idle speeds determined by the team. At idle the maximum allowed sound level is 103 dB(C), fast weighting.

CV 4 Shutdown System

CV 4.1 Shutdown Circuit

CV 4.1.1

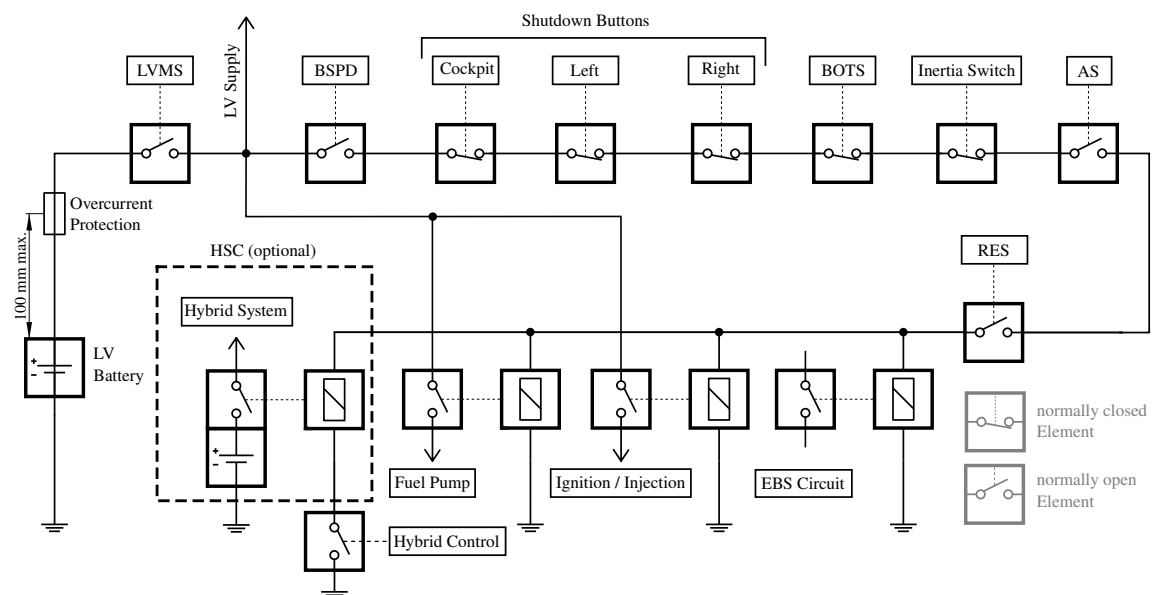


Figure 20: Explanatory example schematic of the required Shutdown Circuit (SDC)

The Shutdown Circuit (SDC) directly controls all electric power to the ignition, fuel injectors and all fuel pumps. It must act through a minimum of two mechanical relays. One relay for

CV 5 Hybrid System

the fuel pump and at least one relay for injection and ignition. An explanatory schematic of the required SDC is shown in Figure 20.

- CV 4.1.2 The SDC is defined as a series connection of at least the LVMS, see [T 11.3](#), the BSPD, see [T 11.6](#), three shutdown buttons, see [T 11.4](#), the BOTS, see [T 6.2](#) and the inertia switch, see [T 11.5](#).
- CV 4.1.3 All circuits that are part of the SDC must be designed in a way, that in the de-energized/ disconnected state they open the SDC.
- CV 4.1.4 [HY ONLY] The HSC AIR as per [CV 5.2.2](#) must be part of the SDC in such a way that one side of the relay coil is directly incorporated into the SDC and the other side is controlled by the hybrid control system.

CV 5 Hybrid System

CV 5.1 Hybrid System General

- CV 5.1.1 **Hybrid System** — the hybrid storage container, motors and every part that is electrically connected to them.
- CV 5.1.2 The hybrid system must be a LVS, [T 11.1](#) and [T 11.7](#) are applied for all hybrid system components.
- CV 5.1.3 All electrical parts of the hybrid system except for ground terminals must be covered at least according to IPxxB when energized.
- CV 5.1.4 **Hybrid Storage Container (HSC)** — the electric energy storage system, including the AIR and overcurrent protection, that is used in the hybrid system.
- CV 5.1.5 Moving energy into the HSC from a different electric storage system is prohibited during any dynamic discipline.
- CV 5.1.6 A firewall, see [T 4.8](#), must be present between the HSC and the fuel tank.
- CV 5.1.7 The HSC must be positioned according to [T 11.7.2](#), all other hybrid system components must be positioned within the surface envelope, see [T 1.1.18](#).
- CV 5.1.8 The high current path, see [EV 1.2.2](#), of the hybrid system must meet [EV 4.5.15](#).
- CV 5.1.9 Motors must meet [EV 2.1](#).
- CV 5.1.10 The hybrid system may only be activated when the combustion engine is running or during engine start.

CV 5.2 Hybrid Storage Container

- CV 5.2.1 The HSC must be attached to the primary structure, see [T 1.1.12](#), according to [T 3.3.1](#).
- CV 5.2.2 A disconnection mechanism, designed as an AIR must be integrated inside of the HSC, disconnecting the positive pole of the HSC. The AIR must be compliant with [EV 5.6.3](#).
- CV 5.2.3 The maximum combined total weight of all elements in the hybrid system that store the electric energy, e.g. battery cells or supercapacitors, including all casings and tabs that are integral to them, is 3 kg.
- CV 5.2.4 Holes, both internal and external, in the HSC, are only allowed for the wiring harness, ventilation, cooling, or fasteners. The total cutout area must be below 25 % of the area of the respective single wall.
- CV 5.2.5 The HSC must be removable to be inspected at the Mechanical Inspection and it must be possible to easily check the weight limit.

CV 5.3 Hybrid System Form (HSF)

- CV 5.3.1 An HSF has to be submitted using the HSF template.
- CV 5.3.2 The HSF template is available on the competition website.
- CV 5.3.3 If no HSF is submitted, the team must not use the hybrid system at the event. [A 5.4.2](#) is not applied for the HSF.

EV Electric Vehicles (EVs)

EV 1 Definitions

EV 1.1 Tractive System

EV 1.1.1 **Tractive System (TS)** — every part that is electrically connected to the motors and TS accumulators. The LVS may be supplied by the TS if a galvanic isolation between both systems is ensured.

EV 1.1.2 **TS enclosures** — every housing or enclosure that contains parts of the TS.

EV 1.2 Electrical

EV 1.2.1 **Galvanic Isolation** — two electric circuits are defined as galvanically isolated if all of the following conditions are true:

- The resistance between both circuits is $\geq 500 \Omega/V$, related to the maximum TS voltage of the vehicle, at a test voltage of maximum TS voltage or 250 V, whichever is higher.
- The isolation test voltage RMS, AC for 1 min, between both circuits is higher than three times the maximum TS voltage or 750 V, whichever is higher.
- The working voltage of the isolation barrier, if specified in the datasheet, is higher than the maximum TS voltage.

Capacitors that bridge galvanic isolation must be class-Y capacitors.

EV 1.2.2 **High Current Path** — any path of a TS circuitry that, during normal operation, carries more than 1 A.

EV 2 Electric Powertrain

EV 2.1 Motors

EV 2.1.1 Only electric motors are allowed.

EV 2.1.2 Motor attachments must follow [T 10](#).

EV 2.1.3 Motor casings must follow [T 7.3](#).

EV 2.1.4 The motor(s) must be connected to the TS accumulator through a motor controller.

EV 2.2 Power Limitation

EV 2.2.1 The TS power at the outlet of the TSAC must not exceed 80 kW.

EV 2.2.2 The TS current at the outlet of the TSAC must not exceed 500 A.

EV 2.2.3 Regenerating energy is allowed and unrestricted with regard to power.

EV 2.2.4 Wheels must not be spun in reverse.

EV 3 General Requirements

EV 3.1 Grounding

EV 3.1.1 TS enclosures, see [EV 1.1.2](#), must consist of either

- a grounded solid layer made of at least 0.5 mm thick electrically conductive material, aluminium or better, see [EV 3.1.3](#) or

EV 4 Tractive System

- be made out of electrically insulating materials providing an insulation resistance of at least 2 MΩ, measured at a voltage of 500 V, between the inside and the outside of the enclosure. The TS enclosure must be rigid and must prevent possible mechanical penetrations. Protruding electrically conductive parts, such as fasteners or connectors, must follow [EV 3.1.2](#)

The TSAC might use at least 0.9 mm thick steel layer as the grounded layer.

- EV 3.1.2 Electrically conductive seat, driver harness, and firewall mounting points as well as the aluminium layer of the TS firewall, see [T 4.8](#), and the LVS ground measuring point, see [EV 4.7.8](#), must be grounded, see [EV 3.1.3](#).
- EV 3.1.3 An electrically conductive part is grounded if its resistance to LVS ground is below 100 mΩ, measured with a current of 1 A, and the grounding wire is able to continuously carry at least 10 % of the TS accumulator main fuse current rating.
- EV 3.1.4 Parts of the vehicle which are or may become electrically conductive within 100 mm of any TS component must have a resistance below 100 Ω to LVS ground.
- EV 3.1.5 The rotating part of the wheels does not need to be grounded.

EV 3.2 Overcurrent Protection

- EV 3.2.1 All electric systems must have appropriate overcurrent protection.
- EV 3.2.2 The continuous current rating of the overcurrent protection must not be greater than the continuous current rating of any electric component, for example, wire, busbar, or other conductors that it protects. I.e. if multiple pins of a connector are used to carry currents in parallel, each pin must be appropriately protected.
- EV 3.2.3 All used fuses must have an interrupt current rating that is higher than the theoretical short circuit current of the system that it protects.
- EV 3.2.4 All overcurrent protection devices must be rated for the highest voltage in the systems they protect. All devices used must be rated for DC.
- EV 3.2.5 All overcurrent protection devices that are part of the TS must not rely on programmable logic. The overcurrent protection function of motor controllers/inverters for the motor outputs may rely on programmable logic.
- EV 3.2.6 The overcurrent protection must be designed for the expected surrounding temperature range but at least for 0 °C to 85 °C
- EV 3.2.7 The TS high current path, see [EV 1.2.2](#), through the TS accumulator(s) must be fused.

EV 4 Tractive System

EV 4.1 General Requirements

- EV 4.1.1 The maximum allowed voltage that may occur between any two electric connections is 600 V DC and for motor controller and AMS internal low power control signals 630 V DC.
- EV 4.1.2 All components in the TS must be rated for the maximum TS voltage. The TS area of a PCB, see [EV 4.3.6](#), is considered as one component. Every input connected to the TS must be rated to the maximum TS voltage.
- EV 4.1.3 All components must be rated for the maximum possible temperature that may occur during usage.

EV 4.2 Tractive System Enclosures

- EV 4.2.1 TS enclosures, see [EV 1.1.2](#), must be labeled with reasonably sized stickers according to “ISO 7010-W012” (triangle with a black lightning bolt on a yellow background). The sticker must also contain the text “High Voltage” if the voltage is more than 60 VDC or 50 VAC RMS.

EV 4.3 Separation of Traction System and Grounded Low Voltage System

- EV 4.3.1 The entire TS and LVS must be galvanically isolated, see [EV 1.2.1](#) and [IN 4.1.1](#).
- EV 4.3.2 All connections from a TS component to external devices, such as laptops must include galvanic isolation, see [EV 1.2.1](#).
- EV 4.3.3 TS and LVS circuits must be physically segregated such that they are not running through the same conduit or connector, except for interlock circuit connections.
- EV 4.3.4 Where both TS and LVS are present within an TS enclosure, they must be separated by barriers made of moisture-resistant insulating materials or maintain 20 mm spacing through air, or over a surface.
- EV 4.3.5 Components and cables capable of movement must be positively restrained to maintain spacing.
- EV 4.3.6 If TS and LVS are on the same PCB, they must be on separate well-defined areas of the board, meeting the spacing requirements in Table 3, each area clearly marked with “TS” or “LV”. The outline of the area required for spacing must be marked. Grooves and cut-outs must have a minimum width of 1.5 mm to influence the creepage path. “Conformal coating” refers to a coating insulator on a PCB. Solder resist is not a coating.

Voltage	Clearance Distance	Creepage Distance	
		General	conformal coating
0 VDC to 50 VDC	1.0 mm	4 mm	1.0 mm
50 VDC to 150 VDC	1.0 mm	5 mm	1.0 mm
150 VDC to 300 VDC	1.5 mm	10 mm	2.0 mm
300 VDC to 600 VDC	3.0 mm	20 mm	4.0 mm

Table 3: Spacing required between TS and LV.

- EV 4.3.7 Teams must be prepared to demonstrate spacing on team-built equipment. For inaccessible circuitry, fully assembled spare boards must be available.

EV 4.4 Positioning of Tractive System Parts

- EV 4.4.1 Except what is allowed according to [EV 4.4.3](#), all parts belonging to the TS including cables and wiring must be located within the rollover protection envelope, see [T 1.1.16](#). “Part” is the whole device such as the complete HVD.
- EV 4.4.2 Any part of the TS that is less than 350 mm above the ground, except what is allowed according to [EV 4.4.3](#), must be protected from impacts, see [T 3.15](#). Impact protection must follow [T 3.16](#) when having bolted attachments. TS wiring in front of the front hoop may alternatively be shielded by the front bulkhead support structure according to [T 3.14](#).
- EV 4.4.3 Outboard wheel motors are allowed only if
- an interlock is routed along the TS wiring such that the SDC, see [EV 6](#), is opened before the TS wiring or its clamping fails

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- an interlock is routed along a suspension member such that the SDC, see [EV 6](#), is opened if the suspension fails
- TS wiring is not able to reach the cockpit opening or the driver regardless of where it breaks
- wiring outside of the rollover protection envelope, see [T 1.1.16](#) is minimum length
- wiring outside of an impact structure or front bulkhead support structure, see [EV 4.4.2](#), is minimum length
- Minimum length is the shortest distance plus extra wiring caused by bending radius.

EV 4.5 Tractive System Insulation, Wiring, and Conduit

- EV 4.5.1 All live parts of the TS must be protected from being touched. This must include team members working on or inside the vehicle. This is tested with a 100 mm long, 6 mm diameter insulated test probe when the TS enclosures are in place.
- EV 4.5.2 Insulation material that is rated for the maximum TS voltage must be used. Using only insulating tape or rubber-like paint for insulation is prohibited.
- EV 4.5.3 The temperature rating for TS wiring, connections, relays, and insulation must be appropriate for the expected surrounding temperatures but at least 85 °C.
- EV 4.5.4 TS components and TS enclosures must be protected from moisture in the form of rain or puddles, see [IN 9](#).
- EV 4.5.5 It must be possible to clearly assign and prove the wire gauge, temperature rating, and insulation voltage to each used TS wire.
- EV 4.5.6 All TS wiring must be completed to professional standards with appropriately sized conductors and terminals and with adequate strain relief and protection from loosening due to vibration etc.
- EV 4.5.7 TS wiring must be located out of the way of possible snagging or damage.
- EV 4.5.8 All TS wiring that runs outside of TS enclosures must:
- be enclosed in separate orange non-conductive conduit or use an orange shielded cable. The conduit must be securely mounted to the vehicle, but not to the wire, at least at each end.
 - be securely mounted at least at each end so that it can withstand a force of 200 N without straining the cable end crimp.
- Bodywork is not sufficient to meet this enclosure requirement.
- EV 4.5.9 Any shielded TS cable must have the shield grounded.
- EV 4.5.10 Every TS connector outside TS enclosures must include a pilot contact/interlock line which is part of the SDC. TS enclosures only used to avoid interlocks are prohibited.
- EV 4.5.11 All TS connections must be designed so that they use intentional current paths through conductors such as copper or aluminium and must not rely on steel bolts to be the primary conductor.
- EV 4.5.12 All materials used in a stack-up or as a fastener of a TS connection must ensure equal pressure across the full contact area over the entire temperature range.
- EV 4.5.13 All electric connections, including bolts, nuts, and other fasteners, in the high current path, see [EV 1.2.2](#), of the TS must be secured from unintentional loosening. Fasteners must use positive locking mechanisms, see [T 10.2](#), that are suitable for high temperatures, see [EV 4.5.3](#). Components, e.g. inverters, certified for automotive use might be allowed without a positive locking feature if connections are completed as recommended by the manufacturers' datasheet and no positive locking is possible.

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- EV 4.5.14 Teams must be prepared to demonstrate positive locking. For inaccessible connections, appropriate photographs must be available.
- EV 4.5.15 Soldered connections in the TS high current path, see [EV 1.2.2](#), are only allowed if all of the following are true:
- connections on PCBs
 - the connected devices are not cells or wires
 - the devices are additionally mechanically secured against loosening

EV 4.6 Data Logger

- EV 4.6.1 A data logger is provided by the officials and must be inserted during the event. The data logger measures TS voltage and TS current.
- EV 4.6.2 The data logger must be in an easily accessible location so that it is possible to insert, remove, or replace it within 15 min in ready-to-race condition.
- EV 4.6.3 The data logger must not be placed within the TSAC.
- EV 4.6.4 All current supplying the TS must run through the data logger. The data logger must be inserted in the negative TS supply between the most negative AIR and the inverters.
- EV 4.6.5 The TS voltage sense connection of the data logger must be directly connected, see [T 1.3.1](#), to the most positive AIR on the vehicle side. If overcurrent protection is used, it must be a fuse.
- EV 4.6.6 The specification of the data logger is available in the Event Handbook.

EV 4.7 Tractive System Measuring Point (TSMP)

- EV 4.7.1 Two TSMPs must be installed directly next to the master switches, see [T 11.2](#).
- EV 4.7.2 The TSMPs must be directly connected, see [T 1.3.1](#), to the intermediate circuit capacitors even if the HVD has been removed or the TS accumulator is disconnected.
- EV 4.7.3 4 mm shrouded banana jacks rated for 600 V CAT III or better must be used.
- EV 4.7.4 The TSMPs must be marked “TS+” and “TS-” and exclusively mounted on an orange background.
- EV 4.7.5 The TSMPs must be protected by a non-conductive cover that can be opened without tools. The cover must always be mechanically linked to the vehicle.
- EV 4.7.6 Each TSMP must be secured with a current limiting resistor according to the following table. Fusing the TSMPs is prohibited. The resistor's power rating must be chosen such that they can continuously carry the current if both TSMPs are short-circuited.

Maximum <u>TS</u> Voltage	Resistor Value
$U_{max} < 200 \text{ V DC}$	5 k Ω
$200 \text{ V DC} < U_{max} \leq 400 \text{ V DC}$	10 k Ω
$400 \text{ V DC} < U_{max} \leq 600 \text{ V DC}$	15 k Ω

- EV 4.7.7 All electric connections needed to connect the TSMP to the intermediate circuit capacitors, including bolts, nuts, and other fasteners, must be secured from unintentional loosening by the use of positive locking mechanisms. Bolted connections must follow [T 10.2](#), soldered connections [EV 4.5.15](#).
- EV 4.7.8 Next to the TSMPs an LVS ground measuring point must be installed. A 4 mm black shrouded banana jack must be connected to LVS ground and must be marked “GND”.

EV 4.8 High Voltage Disconnect (HVD)

- EV 4.8.1 It must be possible to disconnect at least one pole of the TS accumulator by quickly removing an unobstructed and directly accessible element, fuse, or connector. It must be possible to remove the HVD without removing any bodywork. The HVD must be above 350 mm from the ground. Remote actuation of the HVD through a long handle, rope, or wire is not allowed.
- EV 4.8.2 Any trained person must be able to remove the HVD within 10 s when the vehicle is in ready-to-race condition.
- EV 4.8.3 A dummy connector or similar may be required to restore the system's insulation, see [EV 4.5](#). The dummy connector must be attached to the push bar, see [T 13.1](#), if not in use.
- EV 4.8.4 The HVD must be clearly marked with "HVD".
- EV 4.8.5 No tools must be necessary to remove the HVD. An interlock is required, see [EV 4.5.10](#).

EV 4.9 Discharge Circuit

- EV 4.9.1 If a discharge circuit is required to meet [EV 6.1.5](#), it must be designed to handle the maximum TS voltage permanently. After three subsequent discharges within 15 s in total, the discharge time specified in [EV 6.1.5](#) may be exceeded. Full discharging functionality must be given after a reasonable time with a deactivated discharge circuit.
- EV 4.9.2 The discharge circuit must be wired in a way that it is always active whenever the SDC is open. Furthermore, the discharge circuit must be fail-safe such that it still discharges the intermediate circuit capacitors if the HVD has been removed or the TS accumulator is disconnected.
- EV 4.9.3 Fusing of the discharge main current path is prohibited.

EV 4.10 Tractive System Active Light (TSAL)

- EV 4.10.1 The vehicles must include a single TSAL that must indicate the TS status. The TSAL must not perform any other functions. A TSAL with multiple LEDs in one housing is allowed.
- EV 4.10.2 The TSAL itself must have a red light, flashing continuously with a frequency between 2 Hz and 5 Hz and a duty cycle of 50 %, active if and only if the LVS is active and the voltage across any DC-link capacitor exceeds
- 60 V DC or 50 V AC RMS
 - half the nominal TS voltage
- whichever is lower.
- EV 4.10.3 The TSAL itself must have a green light, continuously on, active if and only if the LVS is active and ALL of the following conditions are true:
- All AIRs are opened.
 - The pre-charge relay, see [EV 5.7.2](#), is opened.
 - The voltage at the vehicle side of the AIRs inside the TSAC does not exceed 60 V DC or 50 V AC RMS.
- EV 4.10.4 The mentioned voltage detection must be performed inside the respective TS enclosure.
- EV 4.10.5 The mentioned states of the relays (opened/closed) are the actual mechanical states. The mechanical state can differ from the intentional state, i.e. if a relay is stuck. Any circuitry detecting the mechanical state must meet [EV 5.6.2](#).
- EV 4.10.6 The voltage detection circuit of the red light and the relay state and voltage detection circuit of the green light must be independent. Any plausibility check between both lights is not allowed. A TSAL state with both lights simultaneously active might occur and must not be prevented.

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EV 4.10.7 The TSAL must:

- be located lower than the highest point of the main hoop and including the mounting within the rollover protection envelope, see [T 1.1.16](#).
- be no lower than 75 mm from the highest point of the main hoop.
- not be able to contact the driver's helmet in any circumstances.

EV 4.10.8 The entire illuminated surface of the TSAL must be clearly visible:

- except for angles less than 10° on each side which are blocked by the main hoop.
- from a point 1.60 m vertically from ground level, within 3 m horizontal radius from the TSAL.
- in direct sunlight.

EV 4.10.9 The TSAL and all needed circuitry must be hard-wired electronics. Software control is not allowed.

EV 4.10.10 A green indicator light in the cockpit that is easily visible even in bright sunlight and clearly marked with "TS off" must light up if TSAL green light is on, see [EV 4.10.3](#).

EV 4.10.11 Signals influencing the TSAL and the indicator according to [EV 4.10.10](#) are SCS, see [T 11.9](#). The individual safe state of each of the TSAL lights is off. The TSAL's red light must not be illuminated for a visible check, see [T 11.9.6](#).

EV 4.10.12 The TSAL's red light voltage detection circuit, see [EV 4.10.2](#), does not need to detect an open circuit, as required by [T 11.9](#). A plausibility check must not be implemented.

EV 4.10.13 The TSAL's green light relay state detection circuit, see [EV 4.10.3](#), does not need to detect an open circuit, as required by [T 11.9](#), when the intentional state of the used (auxiliary) contact is opened. A plausibility check against the intentional relay state must be implemented in a way that the TSAL's green light stays off after the open circuit is detectable.

EV 4.10.14 The TSAL's green light voltage detection circuit, see [EV 4.10.3](#), does not need to detect an open circuit, as required by [T 11.9](#), when no voltage is present. A plausibility check against the intentional relay states must be implemented in a way that the TSAL's green light stays off after the open circuit of the TS accumulator voltage detection circuit is detectable.

EV 4.10.15 The latching required by [EV 4.10.13](#) and [EV 4.10.14](#) must not be triggered during normal operation conditions and must only be reset by power cycling the LVS.

EV 4.11 Activating the Tractive System

EV 4.11.1 The TS is active if any of the AIRs or the pre-charge relay is closed.

EV 4.11.2 The driver must be able to activate and deactivate the TS from within the cockpit without the assistance of any other person.

EV 4.11.3 The ASR must be able to activate the TS from outside the vehicle with an external TS activation button in proximity to the TSMS if and only if the ASMS is in "On" position.

EV 4.11.4 Closing the SDC by any part defined in [EV 6.1.2](#) must not (re-)activate the TS. Additional action must be required.

EV 4.11.5 The AS must not be able to (re-)activate the TS.

EV 4.11.6 The vehicle is in Ready-to-drive (R2D) mode as soon as the motors respond to the input of the APPS.

EV 4.11.7 After the TS has been activated, additional actions must be required by the driver to set the vehicle to R2D mode, e.g. pressing a dedicated start button. The transition to R2D mode must only be possible during the actuation of the mechanical brakes and a simultaneous dedicated additional action.

EV 5 Tractive System Energy Storage

EV 4.11.8 The R2D mode must be left immediately when the SDC is opened.

EV 4.12 Ready-To-Drive Sound

- EV 4.12.1 The vehicle must make a characteristic sound, continuously for at least 1 s and a maximum of 3 s while entering R2D mode.
- EV 4.12.2 The sound level must be a minimum of 80 dB(A) and a maximum of 90 dB(A), fast weighting in a radius of 2 m around the vehicle.
- EV 4.12.3 The used sound must be easily recognizable. No animal voices, song parts, or sounds that could be interpreted as offensive are allowed.
- EV 4.12.4 The vehicle must not make any other sounds similar to the R2D sound.

EV 5 Tractive System Energy Storage

EV 5.1 Definitions

- EV 5.1.1 **Cell** — a battery cell or super-capacitor.
- EV 5.1.2 **Cell Energy** — the maximum cell voltage times the nominal capacity of the used cell.
- EV 5.1.3 **TS Accumulator Segments** — an electrical sub-division consisting of a group of cells that store electric energy to be used by the TS.
- EV 5.1.4 **TS Accumulator Section** — a mechanical sub-division of the TS accumulator separated by the TSAC walls.
- EV 5.1.5 **TS Accumulator** — all cells that store the electric energy to be used by the TS as a whole.
- EV 5.1.6 **Tractive System Accumulator Container (TSAC)** — the container itself, which contains the TS accumulator.

EV 5.2 Tractive System Cells

- EV 5.2.1 All cells that store the TS energy must be enclosed in TSACs
- EV 5.2.2 Spare cells must be stored in an electrically insulated container made of fire retardant material, see [T 1.2.1](#). The cells must be securely mounted inside the container to prevent any movement. The container must be labeled as specified in [EV 5.3.5](#). If this container is placed within another container, the outer container must also be labeled in accordance with [EV 5.3.5](#).
- EV 5.2.3 Cells must be operated within the limits written in the datasheet or what is given in the respective rules, whichever is more restrictive.
- EV 5.2.4 Molten salt and thermal batteries are prohibited.
- EV 5.2.5 Fuel cells are prohibited.

EV 5.3 Tractive System Energy Storage – General Requirements

- EV 5.3.1 Each TSAC must be removable from the vehicle while still remaining rules compliant without the need to install extra components. A dummy connector or similar may be used to restore the system's insulation, see [EV 4.5](#).
- EV 5.3.2 It must be possible to open the TSAC for Technical Inspection.
- EV 5.3.3 When the TSAC is outside of the vehicle, it must be possible to remove all TS accumulator segments at any time without the need to move, lift, or tilt the TSAC. The time to remove each TS accumulator segment after opening the lid must be less than 15 min.

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- EV 5.3.4 Each TS accumulator section, see [EV 5.1.4](#), must never exceed a maximum voltage of 120V DC, a maximum energy of 6 MJ, see [EV 5.1.2](#), and a maximum mass of 12 kg.
- EV 5.3.5 The vehicle number, the university name, and the ESO phone numbers must be displayed and written in Roman Sans-Serif characters of at least 20 mm high on the lid of each TSAC. The characters must be clearly visible and placed on a high-contrast background.

EV 5.4 Tractive System Energy Storage – Electrical Configuration

- EV 5.4.1 Every TSAC must contain at least one fuse and at least two AIRs, see [EV 5.6](#) and [EV 3.2.7](#).
- EV 5.4.2 Every TSAC must have a prominent indicator, a voltmeter, or a red LED visible even in bright sunlight. If a red LED is used as an indicator, it must continuously illuminate at its full brightness whenever a voltage greater than 60 VDC or half the maximum TS voltage, whichever is lower, is present at the vehicle side of the AIRs.
- EV 5.4.3 The indicator must be clearly visible while disconnecting the TSAC from the vehicles. The indicator must be clearly marked with "Voltage Indicator"
- EV 5.4.4 The indicator must be hard-wired electronics without software control, directly and only supplied by the TS from the vehicle side of the AIRs, and always working, even if the TS accumulator is disconnected from the LVS or removed from the vehicle.
- EV 5.4.5 LVS must not be included in the TSAC except where inherently required. Exceptions include the AIRs, TS DC/DC converters, the AMS, the IMD, the TSAL's green light circuitry, and cooling fans.
- EV 5.4.6 Every wire used in a TSAC, regardless of whether it is part of the LVS or TS, must follow [EV 4.5.2](#), [EV 4.5.3](#), and [EV 4.5.5](#), the exception being wires used in the intent of [EV 5.8.11](#).
- EV 5.4.7 Maintenance plugs must allow electrical separation of all TS accumulator segments, see [EV 5.3.4](#). The separation must affect both poles of all segments including the first and last segment.
- EV 5.4.8 Maintenance plugs must:
- not require tools to separate the TS accumulator segments.
 - be non-conductive on surfaces that do not provide any electric connection.
 - be designed in a way, that it is physically impossible to electrically connect them in any way other than the design intent configuration.
 - be protected against accidental reconnection.
 - be designed such that it is clearly visible whether the connection is open or closed. Electrically controlled switches must not be used.
- EV 5.4.9 Each TS accumulator segment must be electrically insulated by the use of suitable rigid and fire retardant material, see [T 1.2.1](#), on top of the segment to prevent arc flashes caused by inter-segment contact or by parts/tools accidentally falling into the TSAC during maintenance.

EV 5.5 Tractive System Energy Storage – Mechanical Configuration

- EV 5.5.1 All TSACs must lie fully within and be attached to the primary structure. In side view the TSACs may not exceed the height of the impact structure, see [T 3.15](#).
- EV 5.5.2 The structures surrounding the TSACs must protect it from impacts, see [T 3.15](#). Impact protection must follow [T 3.16](#) when having bolted attachments. No part of the TSAC may be part of the primary structure.
- EV 5.5.3 All TSAC materials as well as all structural parts used to comply with [EV 5.5.8](#) must be fire retardant, see [T 1.2.1](#). All calculations must be conducted for an ambient temperature of 60 °C except for metallic materials and continuous fiber-reinforced laminates.

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- EV 5.5.4 The design of the TSAC and its contents, calculations and/or tests must be documented in the ASES. This includes materials used, drawings, images, fastener locations, section weight, cell, and section position.
- EV 5.5.5 TSACs must be constructed of steel or aluminium, with the following requirements:
- The bottom of the TSAC must be at least 1.25 mm thick if made from steel or 3.2 mm if made from aluminium.
 - The internal and external vertical walls, covers, and lids must be at least 0.9 mm thick if made from steel or 2.3 mm if made from aluminium.
- Alternative materials are allowed with proof of equivalency per [T 3.4](#) or for composite materials per [EV 5.5.6](#). When alternative materials are used, test samples must be presented at Technical Inspection.
- EV 5.5.6 Composite TSACs must satisfy the following requirements:
- Data obtained from the laminate perimeter shear strength test and three-point bending test, see [T 3.6](#), should be used to prove adequate strength is provided.
 - Each attachment point requires steel backing plates with a minimum thickness of 2 mm. Alternate materials may be used for backing plates if equivalency is approved.
 - The calculations and physical test results must be included in the ASES.
- EV 5.5.7 If the TSAC is made from an electrically conductive material, the insulation barrier must be adequately protected against conductive penetrations.
- EV 5.5.8 The TSAC, its mounting to the chassis, and the mounting of each cell to the TSAC must be designed to withstand the accelerations defined in [T 3.3.1](#). All TSAC attachment points must follow [EV 5.5.9](#). TSACs made of materials as stated in [EV 5.5.5](#) or [EV 5.5.6](#) may need further reinforcement to comply with this rule. [T 3.2.8](#) applies if bonded joints are used to fulfill this rule.
- EV 5.5.9 The TSAC must be mounted using at least 2 attachment points. [T 3.16.5](#) does not apply to the number of attachment points. Any brackets used must be made of either 1.6 mm thick steel or 4 mm thick aluminium and must have gussets to support bending loads.
- EV 5.5.10 If any structural part of or in the TSAC is manufactured using additive processes, physical testing must verify compliance with [EV 5.5.8](#). Test samples must be produced on the same machine with identical parameters as the final part. Calculations must account for anisotropic material properties, and all results must be documented in the ASES.
- EV 5.5.11 TS accumulator section, see [EV 5.1.4](#), must be separated by internal vertical walls which extend upwards until the lid. Openings must follow [EV 5.5.18](#).
- EV 5.5.12 Pouch cells must be fixed using one or both of the large surfaces only. Each used surface must be fixed on at least 80 %. Tabs of pouch cells must not press into the pouch. The method used for the application of pretension must be documented in the ASES. Any variation in force must be compensated for with compressible materials.
- EV 5.5.13 Any part of the segment in the high current path, see [EV 1.2.2](#), must not carry any mechanical loads.
- EV 5.5.14 Mechanical stress on each cell must not exceed 0.05 MPa shear and 1.0 MPa compression, or the maximum written in the datasheet, whichever is lower, in any state of the cell. This must be calculated using 80 % of the nominal large cell surface area. Proof must be provided in the ASES.
- EV 5.5.15 Friction-based cell mounting requires physical friction testing. Mechanically representative test cells or cell mockups can be used for friction testing.

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- EV 5.5.16 All fasteners used within or to mount the TSAC must comply with [T 10](#). Fasteners within the TSAC used for non-structural parts, e.g. PCBs, do not have to follow [T 10.1.2](#). Fasteners made of electrically non-conductive material within the TSAC used for non-structural parts do not have to follow [T 10](#).
- EV 5.5.17 The AIRs and the main fuse, see [EV 3.2.7](#), must be separated with an electrically insulating and fire retardant material, see [T 1.2.1](#), from the TS accumulator, see [EV 5.1.5](#). Air is not considered to be a suitable insulation material in this case.
- EV 5.5.18 Holes, both internal and external, in the TSAC, are only allowed for the wiring harness, ventilation, cooling, or fasteners. Holes can extend to the edge of a wall. The TSAC must still be compliant with all other rules, especially the ones concerning its structural requirements. The total cutout area must be below 25 % of the area of the respective single wall. External holes must be sealed according to [EV 4.5](#).
- EV 5.5.19 External openings in the TSAC or mounted ducts must not point toward the driver or the operator of the TS accumulator hand cart. Ducts and their mounting must be fire retardant, see [T 1.2.1](#), and properly sealed.
- EV 5.5.20 Any TS accumulators that may vent an explosive gas must have a ventilation system to prevent the vented gas from reaching an explosive concentration.
- EV 5.5.21 Every TSAC which is completely sealed must have a pressure relief valve to prevent high pressure in the TSAC.
- EV 5.5.22 A sticker according to "ISO 7010-W012" (triangle with a black lightning bolt on a yellow background) with the triangle side length of at least 100 mm and the text "Always Energized" must be applied on every TSAC. The sticker must also contain the text "High Voltage" if the voltage is more than 60 V DC or 50 V AC RMS.

EV 5.6 Accumulator Isolation Relays (AIRs)

- EV 5.6.1 At least two AIRs must be fitted inside each TSAC.
- EV 5.6.2 The AIRs must open both poles of the TS accumulator. If the AIRs are open, no TS voltage may be present outside of the TSAC and the vehicle side of the AIRs must be galvanically isolated from the TS accumulator side, see [EV 1.2.1](#).
- EV 5.6.3 The AIRs must be mechanical relays of a "normally open" type.

EV 5.7 Pre-Charge Circuit

- EV 5.7.1 A circuit that ensures that the voltage at the vehicle side of the AIRs is pre-charged to at least 95 % of the actual TS accumulator voltage before closing the second AIR must be implemented.
- EV 5.7.2 The pre-charge circuit must use a mechanical, normally open relay. All pre-charge current must pass through this relay.

EV 5.8 Accumulator Management System (AMS)

- EV 5.8.1 Each TS accumulator must be monitored by an AMS whenever the LVS is active or the TS accumulator is connected to a charger.
- EV 5.8.2 Every TSAC must contain its full AMS including its own and AMS exclusive SDC power stage, see [EV 6.1](#)
- EV 5.8.3 The AMS must continuously measure
- all cell voltages

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- the TS current
 - the temperature of thermally critical cells
 - for lithium-based cells: the temperature of at least 30 % of the cells equally distributed within the TSAC
- EV 5.8.4 The maximum cell temperature is 60 °C or the limit stated in the cell datasheet, whichever is lower.
- EV 5.8.5 The sensor to measure the cell temperature must be less than 15 mm away from the cell temperature reference point, see [EV 5.8.7](#), measured along the high current path, see [EV 1.2.2](#).
- EV 5.8.6 The sensor must be securely mounted and have less than 0.5 mm solid non-metal material, e.g. coating, adhesive, in between the cell temperature sensor and the measuring spot.
- EV 5.8.7 The cell temperature reference point is defined as:
- the point of the negative cell terminal surface that is furthest away from the positive terminal along the axis of the cell for cylindric cells.
 - the point where the negative cell tab exits the pouch for pouch cells.
- EV 5.8.8 A single temperature sensor may monitor multiple cells only if the temperature sensor complies with the above stated requirements for all cells it monitors.
- EV 5.8.9 An independent cell temperature monitoring device may be provided by the officials during Accumulator Inspection and must be installed, see [IN 3](#). The device must be placed on the warmest negative cell terminal of the TSAC and in direct contact with the terminal or less than 30 mm away from it on the busbar.
- EV 5.8.10 The AMS must open the SDC, if a critical voltage, temperature, or current value according to the cell manufacturer's datasheet or these rules persistently occurs for more than:
- 500 ms for voltage and current values
 - 1 s for temperature values
- The accuracy, noise, and sample rate of the measurement must be taken into account.
- EV 5.8.11 AMS cell voltage measurement inputs, temperature measurement inputs, and supply voltage of decentralized AMS slaves may be rated below the maximum TS voltage.
- EV 5.8.12 A red indicator light in the cockpit that is easily visible from inside and outside the cockpit even in bright sunlight and clearly marked with the lettering "AMS" must light up if and only if the AMS opens the SDC. It must stay illuminated until the error state has been manually reset, see [EV 6.1.6](#). Signals controlling this indicator are SCS, see [T 11.9](#).
- EV 5.8.13 AMS signals are System Critical Signals, see [T 11.9](#). The loss of a measurement connection, that leads to a violation of [EV 5.8.3](#), must result in an open SDC.
- EV 5.8.14 It must be possible to interrupt the measurement signal path such that at least one violation for each part of the continuous monitoring of the AMS, see [EV 5.8.3](#), can be individually triggered on all of the segments of the accumulator during Technical Inspection.
- EV 5.8.15 The AMS must be able to read and display all measured values according to [EV 5.8.3](#) at any time in a single overview e.g. by connecting a laptop to the AMS.

EV 6 Shutdown Circuit and Systems

EV 6.1 Shutdown Circuit (SDC)

EV 6.1.1

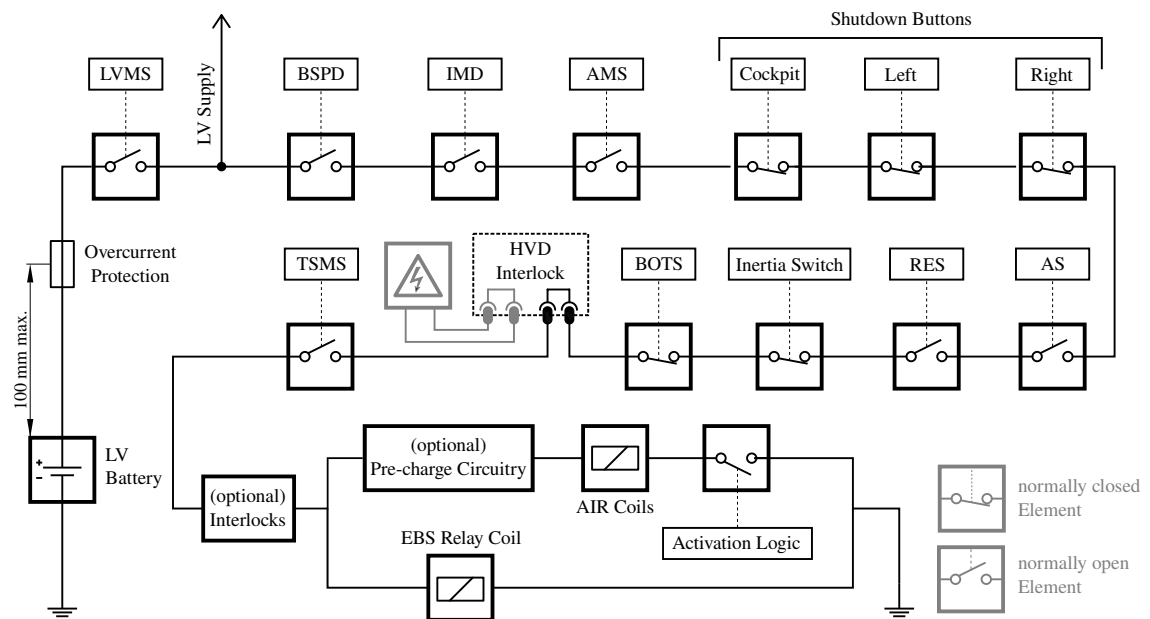


Figure 21: Explanatory example schematic of the required Shutdown Circuit (SDC)

The Shutdown Circuit (SDC) directly carries the power driving the AIRs, see EV 5.6, and the pre-charge circuitry, see EV 5.7.

- EV 6.1.2 The SDC is defined as a series connection of at least two master switches, three shutdown buttons, the BOTS, see T 6.2, the IMD, the inertia switch, see T 11.5, the BSPD, see T 11.6, all required interlocks, and the AMS. An explanatory schematic of the required SDC, excluding any possible interlock circuitry, is shown in Figure 21.
- EV 6.1.3 All parts of the SDC defined in EV 6.1.2 must be on the high-side connection of the AIR coils and the pre-charge circuitry.
- EV 6.1.4 The Tractive System Master Switch (TSMS), see EV 6.2, must be the last switch before the AIRs except for pre-charge circuitry and hardwired interlocks.
- EV 6.1.5 If the SDC is opened the TS must be shut down by opening all AIRs as well as the pre-charge relay and the voltage in the TS must drop to below 60 VDC and 50 VAC RMS in less than 5 s. All TS accumulator current flow must stop immediately. The action of opening the AIRs may be delayed by ≤ 250 ms to signal the action to the motor controllers and reduce the TS current before the AIRs are opened. The AIR supply must be abruptly switched off before reaching the minimum AIR supply voltage.
- EV 6.1.6 If the SDC is opened by the AMS or the IMD, it has to be latched open by a non-programmable logic that can only be manually reset by a person at the vehicle who is not the driver.
- EV 6.1.7 All circuits that are part of the SDC must be designed in a way, that in the de-energized/ disconnected state, they open the SDC.
- EV 6.1.8 It must be possible to demonstrate that all features of the SDC function correctly. This includes all interlocks.
- EV 6.1.9 Every system that is required to or can open the SDC must have its own, non-programmable, power stage to achieve this. The respective power stages must be designed to be able to carry

EV 7 Chargers

the SDC current, e.g. AIR inrush currents, and such that a failure cannot result in electric power being fed back into the electric SDC.

EV 6.1.10 The shutdown buttons, the BOTS, the TSMS, and all interlocks must not act through any power stage.

EV 6.1.11 All signals influencing the SDC are SCSs, see [T 11.9](#).

EV 6.2 Tractive System Master Switch (TSMS)

EV 6.2.1 An TSMS according to [T 11.2](#) must be part of the SDC, see [EV 6.1.2](#).

EV 6.2.2 The TSMS must be fitted with a “lockout/tagout” capability to prevent accidental activation of the TS. The ESO must ensure that it is locked in the off position whenever work is done on the vehicle or no ESO is present.

EV 6.2.3 The TSMS must be mounted in the middle of a completely orange circular area of ≥ 50 mm diameter placed on a high contrast background.

EV 6.2.4 The TSMS must be marked with “TS” and a symbol according to “ISO 7010-W012” (triangle with a black lightning bolt on a yellow background).

EV 6.3 Insulation Monitoring Device (IMD)

EV 6.3.1 Every vehicle must have an IMD installed in the TS system.

EV 6.3.2 The IMD must be a Bender A-ISOMETER® iso-F1 IR155-3203, -3204, -4203, or -4204, or a Bender ISOMETER® iso165C-1, iso175, or equivalent IMD approved for automotive use. Equivalency may be approved by the officials based on the following criteria: robustness to vibration, operating temperature range, IP rating, availability of a direct output, a self-test facility, and must not be powered by the system that is monitored.

EV 6.3.3 The response value of the IMD must be set to $\geq 500 \Omega/V$, related to the maximum TS voltage.

EV 6.3.4 The response value must not be changed after Electrical Inspection.

EV 6.3.5 The IMD must be connected on the vehicle side of the AIRs.

EV 6.3.6 One IMD chassis ground measurement line must be connected to the grounded TSAC or the respective grounded TS enclosure of the IMD. The other chassis ground measurement line must be connected to the main hoop. Each connection must use a separate conductor, rated for at least maximum TS voltage. An open circuit in any of these ground measurement connections must result in an opened SDC.

EV 6.3.7 In case of an insulation failure or an IMD failure, the IMD must open the SDC. This must be done without the influence of any programmable logic. See also [EV 6.1.6](#) regarding the re-activation of the TS after an insulation fault.

EV 6.3.8 A red indicator light in the cockpit that is easily visible from inside and outside the cockpit even in bright sunlight and clearly marked with the lettering “IMD” must light up if and only if the IMD opens the SDC. It must stay illuminated until the error state has been manually reset, see [EV 6.1.6](#). Signals controlling this indicator are SCS, see [T 11.9](#).

EV 7 Chargers

EV 7.1 Chargers General Requirements

EV 7.1.1 Only chargers presented at Technical Inspection are allowed. All connections of the chargers must be insulated and covered. No open connections are allowed.

EV 8 Tractive System Accumulator Container Hand Cart

- EV 7.1.2 All components interfacing with mains must be accredited to a recognized standard e.g. CE. All remaining parts must comply with all electrical requirements for the vehicle TS, e.g. [EV 4.3](#), [EV 3.1](#), and [EV 4.2](#).
- EV 7.1.3 Exposed conductive parts and the TSAC must be connected to protective earth (PE).
- EV 7.1.4 TS charging leads must be orange.
- EV 7.1.5 Each charger must include:
- an easily accessible push-type emergency stop button with a minimum diameter of 40 mm
 - an IMD indicator light as defined in [EV 6.3.8](#)
 - an AMS indicator light as defined in [EV 5.8.12](#)
 - TSMPs as described in [EV 4.7](#), connected to the TS output of the charger
 - a green indicator representing the TSAL green light, see [EV 4.10.3](#) that is easily visible and marked with “TS OFF”
- EV 7.1.6 When charging, the AMS must be active and must be able to open the SDC, see [EV 7.2](#).
- EV 7.1.7 When charging, an IMD as described in [EV 6.3](#) must be active and must be able to open the SDC, see [EV 7.2](#). The second chassis ground measurement line must be connected to the casing of the charger.
- EV 7.1.8 All indicators, switches, buttons, and connectors of the charger must be labeled.

EV 7.2 Charging Shutdown Circuit

- EV 7.2.1 The charging Shutdown Circuit (SDC) consists of at least the charger shutdown button, see [EV 7.1.5](#), the IMD, the AMS, and the charging leads’ interlocks.
- EV 7.2.2 If the SDC is opened the charging system must remain disabled and the SDC opened until it is manually reset. Closing the SDC by any part defined in [EV 7.2.1](#) must not (re-)activate the TS, see [EV 4.11.1](#), and the voltage at the TS connector has to remain below 60 V DC or 50 V AC RMS.
- EV 7.2.3 The charging shutdown systems must comply with [EV 6.1.1](#), [EV 6.1.5](#), [EV 6.1.7](#), [EV 6.1.8](#) and [EV 6.1.9](#).
- EV 7.2.4 All signals influencing the charger SDC are SCS, see [T 11.9](#).

EV 8 Tractive System Accumulator Container Hand Cart

EV 8.1 Tractive System Accumulator Container Hand Cart

- EV 8.1.1 A hand cart must be used for transporting the TSACs around the event site.
- EV 8.1.2 The hand cart must have at least four wheels.
- EV 8.1.3 The hand cart must have a brake that is always on and only released if someone pushes the handle, or similar.
- EV 8.1.4 The brake must be capable of safely stopping the fully loaded hand cart.
- EV 8.1.5 The hand cart must be easily moved when the brake is released.
- EV 8.1.6 The hand cart must be able to carry the load of the TSACs.
- EV 8.1.7 The hand cart must provide a firewall to protect the person while moving the hand cart. The firewall must have the same width as the hand cart, appropriately protect the legs and body,

EV 9 Electrical System Form (ESF)

and be at least 30 cm higher than the hand cart handle and the TSAC. The firewall must be made from a rigid, fire retardant material, see [T 1.2.1](#), and be transparent from 1.3 m above the ground.

- EV 8.1.8 The TSACs must be mechanically fixed to the hand cart to enable safe transportation.
- EV 8.1.9 The TSACs must not protrude the hand cart.
- EV 8.1.10 The TSACs must be protected from vibrations and shocks during normal operation of the cart, e.g. by the use of air tires.
- EV 8.1.11 The hand cart itself must have a label according to [EV 5.3.5](#) on its firewall maximum 1.3 m above the ground.
- EV 8.1.12 The footprint of the fully loaded hand cart must not exceed 1200 mm × 800 mm or (TSAC length + 300 mm) × 800 mm, whichever is larger.

EV 9 Electrical System Form (ESF)

EV 9.1 Electrical System Form

- EV 9.1.1 Before the event, all teams must submit structured documentation of their entire electrical system (including control and TS) called the ESF.

IN Technical Inspections

IN 1 General

IN 1.1 Technical Inspection Process

IN 1.1.1 Technical Inspection is divided into the following parts:

- Pre Inspection
- [EV ONLY] Accumulator Inspection
- [EV ONLY] Electrical Inspection
- Mechanical Inspection
- Autonomous System Inspection
- Tilt Test
- Vehicle Weighing
- [CV ONLY] Noise Test
- [EV ONLY] Rain Test
- Brake Test
- EBS Test

IN 1.2 General Rules

- IN 1.2.1 Each vehicle must pass all parts of Technical Inspection except Autonomous System Inspection and EBS Test before it may be driven under its own power.
- IN 1.2.2 Each vehicle must pass all parts of Technical Inspection including Autonomous System Inspection and EBS Test before it may run in Autonomous Mode.
- IN 1.2.3 Passing Technical Inspection is not a certification of complete rules compliance of the vehicle.
- IN 1.2.4 The Technical Inspection Sheet includes several inspection points and is provided on the competition website before the event. It must always stay with the push bar.
- IN 1.2.5 The officials may inspect other points not mentioned on the Technical Inspection Sheet to ensure compliance with the rules.
- IN 1.2.6 Teams are responsible for confirming that their vehicle and the required equipment satisfies the requirements and restrictions of the rules before presenting it for Technical Inspection.
- IN 1.2.7 Vehicles must be presented for Technical Inspection in ready-to-race condition.
- IN 1.2.8 Safety-uncritical violations without team benefit that cannot be changed at the event may result in at least 20 penalty points, depending on officials decision. If such violations give a minor benefit, at least 40 penalty points may be applied. Penalties are deducted from the team's overall score.
- IN 1.2.9 All items on the Inspection Sheet must be clearly visible for the officials without using instruments such as endoscopes or mirrors. Visible access may be provided by removing body panels or by providing removable access panels.
- IN 1.2.10 The vehicle must maintain all required specifications throughout the event.
- IN 1.2.11 Officials reserve the right to ask the team for drilling of additional inspection holes to check the chassis for compliance with the rules.
- IN 1.2.12 Officials mark or seal various different approved parts. Removal of or damage to the seals voids the Technical Inspection approval.

IN 1 General

- IN 1.2.13 Once the vehicle is approved for competition, any damage to the vehicle that requires repair(s) voids the Technical Inspection approval. After completion of the repair(s), the vehicle must be re-submitted to Technical Inspection for re-approval.
- IN 1.2.14 The officials reserve the right to prohibit the use of parts that could pose a safety risk to drivers, track marshals or the environment.

IN 1.3 Technical Inspection Sticker

- IN 1.3.1 The event's Technical Inspection stickers are placed on the nose of the vehicle.
- IN 1.3.2 If a vehicle is no longer in compliance with the rules, the officials set the vehicle's Technical Inspection status to fail, remove the respective Technical Inspection sticker(s) from the vehicle and note the reason for revoking the Technical Inspection approval in the Technical Inspection Sheet.

IN 1.4 Inspection Responsible Person

- IN 1.4.1 To accelerate the Technical Inspection process, the team must appoint one team member as Technical Inspection responsible person. For the Autonomous System Inspection, see [IN 6](#), this has to be an [ASR](#). [EV ONLY] For Electrical Inspection and Accumulator Inspection this has to be an [ESO](#).
- IN 1.4.2 This inspection responsible person must be:
- familiar with the vehicle.
 - able to show compliance of the vehicle with all points mentioned on the Technical Inspection Sheet.
 - able to perform Technical Inspection autonomously observed by the officials, when asked.
- IN 1.4.3 Should the inspection responsible person be unable to perform one of these requirements, or the vehicle and all necessary items are not ready, Technical Inspection is aborted and the team is asked to leave the Technical Inspection area.

IN 1.5 Modifications and Repairs

- IN 1.5.1 After Technical Inspection, the only modifications allowed to the vehicle are:
- Adjustment of belts, chains and clutches
 - Adjustment of the brake bias
 - Adjustment of the driver restraint system, head restraint, seat and pedal assembly
 - Substitution of the head restraint or seat insert for different drivers
 - Adjustment to engine operating parameters, e.g. fuel mixture and ignition timing
 - Adjustment of mirrors
 - Adjustment of the suspension where no part substitution, other than springs, sway bars and shims, is required
 - Adjustment of tire pressure
 - Adjustment of winglet angles, but not the position of the complete aerodynamic device in relation to the vehicle
 - Replenishment of fluids
 - Replacement of defective tires or brake pads. Replacement tires and brake pads must be identical in material/composition/size to those presented and approved at Technical Inspection.
 - Changing wheels and tires for "wet" or "damp" conditions as allowed in [D 3.2](#) and [D 7.6](#)
 - Software calibration changes
 - Recharging [LV](#) batteries
 - Recharging [TS](#) accumulators
 - (De-)Coupling of actuators and sensor repositioning as allowed in [T 14.7](#)

IN 3 [EV ONLY] Accumulator Inspection

- Installing and removing protective sensor covers, if approved during Technical Inspection
- Replacement of LV batteries. Replacement LV batteries must be identical and approved at Technical Inspection.

Any other modification must be either approved by or performed under the supervision of Technical Inspection officials.

IN 2 Pre Inspection

IN 2.1 Pre Inspection Required Items

IN 2.1.1 The following items must be presented for Pre Inspection:

- All helmets
- All driver's equipment and other safety gear
- Two unused and in date fire extinguishers
- One set of four tires on rims for wet conditions
- One set of four tires on rims for dry conditions

The tire type/rim type combination presented during Pre Inspection must be the same during the whole event (including Technical Inspection). The rims for dry tires and wet tires may be different, but for all dry tire sets and all wet tire sets the same.

IN 3 [EV ONLY] Accumulator Inspection

IN 3.1 Accumulator Inspection General Definitions

- IN 3.1.1 Spare TS accumulator segments and spare cells must be presented at Technical Inspection in a container that complies with [EV 5.3.5](#).
- IN 3.1.2 An official temperature logging device must be installed if used by the event, see [EV 5.8.9](#).
- IN 3.1.3 The accumulator charger is inspected and sealed.
- IN 3.1.4 The set of basic tools is checked.

IN 3.2 Accumulator Inspection Required Items

IN 3.2.1 The following items must be presented at Accumulator Inspection:

- All TS accumulators
- Accumulator hand cart
- Accumulator charger
- Basic Tools, see [IN 3.2.2](#)
- Tools needed for the (dis)assembly of parts
- Samples of self designed PCBs that are part of the TS and inside the TSAC, see [EV 4.3.7](#)
- Datasheets for all parts used in the accumulator
- Original delivery notes for material without serial number printed on according to [T 1.2.1](#)
- Print-outs of rule questions (if applicable)
- Print-out or digital version of ASES

IN 3.2.2 The following basic tools in good condition must be presented:

- Insulated cable shears
- Insulated screw drivers
- Multimeter with protected probe tips and two 4 mm banana plug test leads rated for 600 V CAT III or better
- Insulated tools, if bolted connections are used in the TS
- Face shield
- At least two pairs of HV insulating gloves (not expired)

IN 5 Mechanical Inspection

- Two HV insulating blankets of at least 1.0 m² each
- Safety glasses with side shields for all team members that might work on the TS or accumulator

All electrical safety items must be rated for at least the maximum TS voltage.

IN 4 [EV ONLY] Electrical Inspection

IN 4.1 Electrical Inspection General Definitions

IN 4.1.1 The insulation resistance between the TS and LVS ground is measured. Vehicles are probed with a test voltage based on their maximum TS voltage according to the following table:

Maximum <u>TS</u> Voltage	Test Voltage
≤250 V	250 V
>250 V to ≤500 V	500 V
>500 V	1000 V

To pass this test, the measured insulation resistance must be at least 500 Ω/V related to the maximum TS voltage of the vehicle.

IN 4.1.2 The IMD is tested by connecting a resistor between the TSMP, see [EV 4.7](#) and LVS ground connector, see [EV 4.7.8](#). The test is passed if the IMD shuts down the TS within 30 s at a fault resistance of 50 % below the response value which corresponds to 250 Ω/V.

IN 4.1.3 The BSPD is tested by sending an appropriate signal that represents the current, to achieve ≤ 5 kW whilst pressing the brake pedal. This test must prove the functionality of the complete BSPD except for any commercially available current sensors. Ends of a current transducer's auxiliary winding must be insulated.

IN 4.2 Electrical Inspection Required Items

IN 4.2.1 The following items must be presented at Electrical Inspection:

- Vehicle with mounted TS accumulator
- Jacks and push bar
- Samples of self designed PCBs that are part of the TS and are outside of the TSAC, see [EV 4.3.7](#)
- Tools needed for the BSPD check, see [IN 4.1.3](#)
- Datasheets for all parts used in the TS
- Original delivery notes for material without serial number printed on according to [T 1.2.1](#)
- Tools needed for the (dis)assembly of parts for Electrical Inspection
- Print-outs of rule questions (if applicable)
- The connectors needed
 - to safely close the SDC while the HVD is removed and
 - to safely supply the TS using the same shrouded receptacles as used for TSMP when the TS accumulator is unconnected
- LV Battery (if applicable), and possible replacement batteries

IN 5 Mechanical Inspection

IN 5.1 Mechanical Inspection General Definition

IN 5.1.1 The pedal box must be accessible for visual and manual inspection, using only tools carried with the push bar.

IN 5.2 Mechanical Inspection Required Items

IN 5.2.1 The following items must be presented at Mechanical Inspection:

- Jacks and push bar
- The tallest driver of the team
- Copies of any safety structure equivalency forms
- Copies of any impact attenuator data requirement
- Print-outs of rule questions (if applicable)
- Physically tested IA assembly, including (representative) test fixture (if applicable)
- Teams with a monocoque: laminate test specimen(s)
- Teams using alloyed steel: test specimen(s)
- Only tools needed for the (dis)assembly of parts for Mechanical Inspection
- [HY ONLY] Copies of the HSE
- [HY ONLY] HSC and any tools and/or materials needed to determine the weight limit as per [CV 5.2.3](#)

IN 6 Autonomous System Inspection

IN 6.1 Autonomous System Inspection Required Items

IN 6.1.1 The following items are required:

- Datasheets for all perception sensors
- Documents which prove that all perception sensors meet local legislation
- RES remote control
- ASF
- Tools needed for the (dis)assembly of parts for Autonomous System Inspection
- Print-outs of rule questions (if applicable)

IN 7 Tilt Test

IN 7.1 Tilt Test Procedure

IN 7.1.1 The Tilt Test is conducted with the tallest driver fully strapped in normal driving position.

IN 7.1.2 The Tilt Test is conducted with all vehicle fluids at their maximum fill level.

IN 7.1.3 The vehicle is placed upon the tilt table and to an angle of 60°. There must be no fluid leaks and all wheels must remain in contact with the tilt table surface.

IN 8 Vehicle Weighing

IN 8.1 Vehicle Weighing Procedure

IN 8.1.1 All vehicles must be weighed in ready-to-race condition.

IN 8.1.2 All fluids must be at their maximum fill level for weighing.

IN 9 [EV ONLY] Rain Test

IN 9.1 Rain Test General Definitions

IN 9.1.1 Vehicles must have passed Electrical Inspection, see [IN 4](#), to attempt the Rain Test.

IN 9.2 Rain Test Procedure

- IN 9.2.1 The vehicle must be in ready-to-race condition. All components and constructions used to protect the vehicle from water during the Rain Test must be used during the entire event. Any fans (if applicable) must be switched on.
- IN 9.2.2 The TS must be active during the Rain Test.
- IN 9.2.3 The vehicle must be jacked up using the jacks, see [T 13.2](#), and all driven wheels must be removed.
- IN 9.2.4 The vehicle must not be in R2D mode, see [EV 4.11](#).
- IN 9.2.5 The test is conducted without a driver.
- IN 9.2.6 Water is sprayed at the vehicle from any possible direction. The water spray is similar to a vehicle driving in rain and not a direct high-pressure stream of water.
- IN 9.2.7 The test is passed if the IMD is not triggered and the TS is still active, see [EV 4.11.1](#), while water is sprayed at the vehicle for 120 s and 120 s after the water spray has stopped.

IN 10 [CV ONLY] Noise Test

IN 10.1 Noise Test Procedure

- IN 10.1.1 The sound level is measured during a static test.
- IN 10.1.2 The vehicle must be compliant at all engine speeds up to the maximum test speed, see [CV 3.2.1](#).
- IN 10.1.3 Teams must bring a laptop to indicate the engine speed measured by the engine control unit.
- IN 10.1.4 Measurements are made with a free-field microphone placed free from obstructions at the exhaust outlet level, 0.5 m from the end of the exhaust outlet, at an angle of 45° with the outlet in the horizontal plane.
- IN 10.1.5 Where more than one exhaust outlet is present, the test is repeated for each exhaust and the highest reading is used.
- IN 10.1.6 If the exhaust has any form of active tuning or throttling device or system, it must be compliant with the rules in all positions. Manually adjustable tuning devices must require tools to change them and must not be moved or modified after the Noise Test is passed. The position of the device must be visible to the officials and manually operable by the officials during the Noise Test.
- IN 10.1.7 The test is run with the gearbox in neutral. During this test the vehicle must be jacked up using the jacks, see [T 13.2](#).
- IN 10.1.8 After passing the Noise Test the function of the master switch, the cockpit-mounted shutdown button and the inertia switch is tested.
- IN 10.1.9 After passing [IN 10.1.8](#) the air tightness of the intake system is tested by closing off the inlet after which the engine must stall.

IN 11 Brake Test

IN 11.1 Brake Test Procedure

- IN 11.1.1 Lock all four wheels and stop the vehicle in a substantially straight line at the end of an acceleration run specified by the officials. All four wheels must lock before the vehicle stops; wheels may lock sequentially.
- IN 11.1.2 [CV ONLY] The vehicle must come to standstill without stalling the engine.
- IN 11.1.3 [EV ONLY] After accelerating, the TS must be switched off by the driver, using the cockpit-mounted shutdown button, and the driver must brake using only the mechanical brakes.
- IN 11.1.4 After the Brake Test, the vehicle must be able to continue driving under its own power without external assistance.
- IN 11.1.5 The brake light, [EV ONLY] and TSAL, illumination is checked and the officials verify if the illumination is satisfactory for external observation.
- IN 11.1.6 [EV ONLY] The R2D sound is checked and the officials verify if the sound level is satisfactory.
- IN 11.1.7 During the Brake Test, tire pressure may be max. 1.0 bar above cold pressure before any dynamic discipline and must never exceed the manufacturer's limit.

IN 11.2 EBS Test

- IN 11.2.1 The EBS performance is tested dynamically and must demonstrate the performance described in [T 15.4](#).
- IN 11.2.2 The test is performed in a straight line marked with cones similar to Acceleration.
- IN 11.2.3 During the Brake Test, the vehicle must accelerate in autonomous mode up to at least 40 km/h within 20 m. From the point where the RES is triggered, the vehicle must come to a safe stop within a maximum distance that corresponds to the [Emergency Brake System Performance](#) requirements according to [T 15.4](#).
- IN 11.2.4 In case of wet track conditions, the stopping distance is scaled by the officials dependent on the friction level of the track.
- IN 11.2.5 The EBS Test is conducted after all other elements of [IN 11](#) have been passed.

IN 12 Post Inspection

IN 12.1 Post Inspection Procedure

- IN 12.1.1 The officials reserve the right to impound any vehicle at any time during or after any of the dynamic disciplines to check for compliance with the rules. If necessary, the vehicle remains with the officials until the violation of the rule has been discussed with the team.
- IN 12.1.2 After Endurance and Trackdrive, the vehicle must be placed in parc fermé where no team member may access the vehicle.
- IN 12.1.3 During and after any dynamic discipline, the vehicle must be in compliance with the rules.
- IN 12.1.4 For each violation of the rules, the team receives a separate penalty as follows:
 - Group A: Violation of the rules without advantage for the team
 - Group B: Violation of the rules with advantage for the team (e.g. aerodynamics)

Acceleration Skidpad Autocross Endurance Trackdrive

IN 12 Post Inspection

Group A	0.3 s	0.2 s	2 s	30 s	30 s
Group B	1 s	0.6 s	5 s	2 min	2 min

The penalty applies to all runs since the vehicle entered the dynamic area the last time.

- IN 12.1.5 Violation of the rules concerning safety or the environment (e.g. [BOTS](#), safety harness issues, ground clearance, fluid leaks other than plain water, noise) results in [DQ](#) since entering the dynamic area.
- IN 12.1.6 Losing a part of the vehicle on the track results in [DQ](#) for that particular run.
- IN 12.1.7 A change in vehicle weight of more than ± 5 kg compared to the official Technical Inspection weight, see [IN 8](#), results in 20 penalty points for each kg the tolerance is exceeded by. E.g. a weight difference of ± 6.2 kg results in 40 penalty points.
- IN 12.1.8 If the vehicle previously attempted any dynamic discipline in autonomous mode, see [T 1.4.2](#), the vehicle must be able to:
- enter “AS Ready”, see [T 14.8.3](#), during Post Inspection after any dynamic discipline that was attempted in manual mode, see [T 1.4.1](#).
 - fully execute the Inspection Mission, see [T 14.10.2](#), during Post Inspection after any dynamic discipline that was attempted in autonomous mode, see [T 1.4.2](#).
- To achieve this, the vehicle may be modified according to [IN 1.5.1](#). Violation results in [DQ](#).
- IN 12.1.9 [EV ONLY] The vehicle must be able to enter [R2D](#) mode, see [EV 4.11](#), for all tests requiring this mode during Post Inspection. This capability must be maintained under all circumstances for at least 15 min and without any modifications to the vehicle. Violation results in [DQ](#).
- IN 12.1.10 [EV ONLY] Directly after Endurance and leaving parc fermé, every team must come back to the charging area to disassemble the temperature logging device (if used by the event) from the [TS](#) accumulator, see [EV 5.8.9](#).
- IN 12.1.11 Directly after the last dynamic discipline and leaving parc fermé, the data logger (if installed), see [EV 4.6](#) or [T 14.2](#), is disassembled from the vehicle.

S Static Disciplines

S 1 General Rules

S 1.1 Vehicle Condition

- S 1.1.1 Vehicles must be presented for judging in finished condition, fully assembled, complete and ready-to-race.
- S 1.1.2 The judges will not evaluate any vehicle that is presented in what they consider to be an unfinished state.
- S 1.1.3 Vehicles may be presented for judging without having passed Technical Inspection, even if final tuning and setup is in progress.
- S 1.1.4 Covers and/or parts may be removed during the judging to facilitate access and presentation of components or concepts.

S 2 Business Plan Presentation (BPP)

S 2.1 Business Plan Presentation Objective

- S 2.1.1 The objective of the BPP is to assess the team's ability to develop, present and defend a comprehensive business model in an investment round scenario. The business model must offer a product or a service integrating the team's specific prototype vehicle or a specific component of it. The business model must be for-profit and provide a rewarding business opportunity. Therefore, non-profit organizations are not permitted.
- S 2.1.2 The following are not eligible as components of the team's current prototype vehicle for the creation of the business model:
 - Consumables and fasteners
 - Offcuts, scrap, or other by-products arising from the manufacturing process
 - Tooling or tools used in the manufacturing process
 - Off-vehicle infrastructure or equipment not permanently integrated into the vehicle
 - Test, measurement, and calibration equipment not permanently integrated into the vehicle
 - Personal protective equipment and driver apparel
 - Standalone services, data, or cloud solutions that do not rely on a team-developed on-vehicle component or software
 - Using, rebranding, or reselling bought parts without the team's involvement in design, production, or further processing
 - Items that are not part of the team's current prototype vehicle
- S 2.1.3 During the presentation, teams must explicitly show how they derived their proposed product or service from the team's current prototype vehicle.
- S 2.1.4 The judges should be treated as if they were potential investors or partners for the presented business model. A team may assign the judges specific roles for their BPP, i.e. as representatives of a specific company as strategic investors. Nonetheless, these roles have to allow the judges to pose as potential investors or partners.
- S 2.1.5 The quality of the actual prototype is not considered as part of the BPP judging, therefore [S 1.1](#) does not apply.

S 2 Business Plan Presentation (BPP)

S 2.2 Business Plan Presentation Procedure

- S 2.2.1 Presentations are limited to a maximum of 10 min. Anything presented after the 10 min time limit is not considered for the scoring. The judges may stop any presentation exceeding 10 min.
- S 2.2.2 The presentation will not be interrupted by questions. Immediately following the presentation there is a question and answer session.
- S 2.2.3 One or more team members may present the business plan.
- S 2.2.4 All team members participating in the BPP must be introduced to the judges at the beginning of the presentation. Only team members who were introduced at the beginning and who presented a part of the 10 min presentation may answer the judges' questions.
- S 2.2.5 Handouts, prototypes, or other supporting material may be introduced at any time from the start of the presentation until the end of the Q&A. Such material is only evaluated if explicitly referenced within the presentation or Q&A. All items must be removed by the team after their presentation slot.
- S 2.2.6 Before the event a specific deep dive topic, which has to be part of the 10 min presentation, is published on the competition website.

S 2.3 Executive & Financial Summary

- S 2.3.1 To prepare the potential investors or partners for your team's upcoming presentation, it is required that a Business Plan Executive & Financial Summary (BPEFS) is submitted before the event.
- S 2.3.2 Further requirements regarding the BPEFS are published on the competition website on the date specified in the Event Handbook.
- S 2.3.3 The BPEFS must be submitted online in PDF format via the competition website, not later than the deadline specified in the Event Handbook.

S 2.4 Business Plan Presentation Scoring

- S 2.4.1 The BPP is evaluated on the categories specified in Table 4.

Category	Points
Executive & Financial Summary	10
Content	20
Finances	15
Deep Dive Topic	10
Demonstration and Structure	15
Delivery	10
Q&A – Content	10
Q&A – Delivery and Demonstration	5
General Impression	5
Total	100

Table 4: BPP evaluation categories

Note: All aspects of the Executive and Financial Summary are only evaluated in the "Executive & Financial Summary" category. Presented content, delivery and demonstration during the Q&A is only evaluated in the "Q&A" categories.

S 3 Cost and Manufacturing

- S 2.4.2 If the business model presented fails to meet any requirement from [S 2.1.1](#) or [S 2.1.2](#), 15 penalty points are deducted from the team's BPP scoring.
- S 2.4.3 The judging at the event starts with an initial judging, where all teams are judged by different judging groups.
- S 2.4.4 Some teams may be chosen to participate in the BPP finals to determine the BPP winner. The BPP finals are held separately from the initial judging and teams are informed about their participation during the event. The requirements of [S 2.2.5](#) also apply to the BPP finals. During the finals, physical supporting material may be displayed or demonstrated on stage but no supporting material must be handed out to the judges.
- S 2.4.5 The scoring of the BPP is based on the average of the scores given by each of the judges.
- S 2.4.6 The scoring for the non-finalists is calculated as follows:

$$\text{BPP_SCORE} = 65 \left(\frac{P_{\text{team}}}{P_{\text{max}}} \right)$$

P_{team} is the score awarded to the team

P_{max} is the highest score awarded to any team not participating in the finals

- S 2.4.7 The scoring of the BPP finalists varies from 75 to 71 points and is scored immediately after the BPP finals by all judges.

S 3 Cost and Manufacturing

S 3.1 Cost and Manufacturing Objective

- S 3.1.1 The objective of Cost and Manufacturing is to evaluate the team's understanding of the manufacturing processes, costs and greenhouse gas emissions associated with the construction of a prototype vehicle. This includes trade off decisions between content and cost, make or buy decisions and understanding the differences between prototype and mass production.

S 3.2 Cost and Manufacturing Procedure

- S 3.2.1 Prior to the event, three CRD, see [S 3.3](#), must be submitted to the competition website by the deadline specified in the Event Handbook.
- S 3.2.2 During the event, a discussion with the judges takes place, next to the team's vehicle. The discussion is split into three parts:
- S 3.2.3 Part 1 "CCBOM Discussion":
- A discussion to evaluate the team's ability to prepare an accurate engineering and manufacturing CCBOM for the required system(s). The team must prove the following:
- The specification of the system(s) in the CRD accurately reflects the vehicle brought to the event.
 - The costs and emissions within the CCBOM are correct and realistic.
 - The manufacturing feasibility of the system(s).
- S 3.2.4 Part 2 "Cost Understanding":
- A discussion to evaluate the general Cost and Manufacturing knowledge of the team.
- S 3.2.5 Part 3 "Real Case":
- A specific task to evaluate the Cost and Manufacturing knowledge in a certain field. The real case task is published on the competition website before the event.

S 3 Cost and Manufacturing

- S 3.2.6 Teams are allowed to bring electronic, handwritten or printed handouts, flip charts or similar, but the space available may be limited.

S 3.3 Cost Report Documents (CRD)

- S 3.3.1 The CRD consist of the following documents:

- The CCBOM, created and submitted online on the competition website
- The Supporting Material File, submitted as a PDF file to the competition website
- The Cost and Emissions Explanation File, submitted as a PDF file to the competition website

- S 3.3.2 A changelog of the changes on the vehicle since the submission deadline of the CRD can be presented to the judges at the CCBOM discussion as a separate document. Missing parts of the CCBOM cannot be part of the changelog.

- S 3.3.3 All CRD must be brought either as hard copy or as digital version.

- S 3.3.4 The website tool provides a printable pdf version of the CCBOM.

- S 3.3.5 The team must ensure that at least two judges are able to review the CRD independently.

- S 3.3.6 The presented CRD must be identical to the submitted versions, otherwise 0 points are scored for the "CCBOM Discussion" part.

S 3.4 Costed Carbonized Bill of Material (CCBOM)

- S 3.4.1 The CCBOM is a sorted list of all parts belonging to the required system(s).

- S 3.4.2 The CCBOM must:

- include the actual associated manufacturing processes for each part as used for the prototype vehicle. This includes the assembly of parts and sub-assemblies as well as the assembly on the vehicle.
- include tooling (e.g. welding jigs, molds, patterns and dies).
- include the actual costs of materials, fabrication, bought parts and assembly for each part as used for the prototype vehicle.
- include the system's carbon footprint for each part, manufacturing and assembly process of the prototype vehicle in CO₂e in kg.

- S 3.4.3 The CCBOM is structured as follows:

- The CCBOM is broken down into the required "system(s)", see [S 3.4.5](#).
- Each "system" is broken down into "assemblies", which are defined by the website tool.
- Each "assembly" can optionally be broken down into "sub-assemblies", which must be defined by the team.
- Each "assembly" or "sub-assembly" is broken down into "parts", which must be defined by the team.
- Each "part" is broken down into
 - "materials" used (e.g. aluminium)
 - "processes" that describe the necessary steps for manufacturing
 - "fasteners" used to assemble the part (if applicable)
 - "tooling" required for manufacturing (if applicable)
 - "overhead" for additional associated costs and emissions of manufacturing and assembly of a part that are not covered by the points above

- S 3.4.4 The CCBOM must list all parts belonging to the required system(s) that are fitted to the prototype vehicle at any time during the event.

- S 3.4.5 The "systems" are:

- Autonomous System

S 3 Cost and Manufacturing

- Brake System
 - Chassis and Body
 - Drivetrain
 - Engine and Tractive System
 - Grounded Low Voltage System
 - Miscellaneous, Fit and Finish
 - Steering System
 - Suspension System
 - Wheels, Wheel Bearings and Tires
- S 3.4.6 For 2027, a CCBOM for the “Grounded Low Voltage System” and “Chassis and Body” must be submitted.
- S 3.4.7 When adding “parts” to the CCBOM, the comments section should be reviewed thoroughly. The chosen part name must clearly describe what is included. For example, if a spring is included, the “part” should not be called “damper”.
- S 3.4.8 Each constituent “part” of an “assembly” must be classified as “bought” or “made”.
- S 3.4.9 A modified bought part is considered a made part, where the bought part acts as the material. Associated “processes”, “tooling”, “fasteners” and “overhead” (if applicable) for the modifications must be included. However, the “carbon footprint” of the bought part itself should not be considered.
- S 3.4.10 Only parts that can be bought from a catalog may be designated as “bought parts”.
- S 3.4.11 For “bought parts”, only fasteners have to be included in the emissions calculation (if applicable). Fasteners themselves are not considered bought parts.
- S 3.4.12 “Fasteners” are additional items necessary to assemble a “part” or “(sub-)assembly”.
- Examples for “fasteners”:
DIN-931 hexagon screws, glue, zip ties, tape used for mounting or fixing, ...
 - Examples that are considered “parts” and not “fasteners”:
Tape used for edge protection, safety wire, ...
- S 3.4.13 A self-manufactured “fastener” is considered a “part”.
- S 3.4.14 “Tooling” is the necessary tools used for transforming the “material” into the desired shape. If production tooling is associated with processes that are specific to the part geometry, it must be included. For example the dies to stamp out a chassis bracket are tooling.
- S 3.4.15 Hand or power tools must not be included.
- S 3.4.16 “Processes” are the operations necessary to produce the “part” out of the “material” as well as the steps needed for assembly.
- S 3.4.17 Only metric units must be used within the CCBOM.
- S 3.4.18 The cost calculations must be completed as realistically as possible and exclude research, development and capital expenditures for real estate (e.g. plant or development hours of the team).
- S 3.4.19 All costs must be displayed in EUR. For calculating the prices in EUR from other currencies, the team must provide the exchange rates used.
- S 3.4.20 There is no maximum cost. Receipts are not required for any items.
- S 3.4.21 The calculations for machine hourly rates, wages (including overheads) etc. must be shown.
- S 3.4.22 The “carbon footprint” is calculated in CO₂e based on a “Cradle-to-Gate” Life Cycle Assessment (LCA) of the selected “system(s)”.

S 4 Engineering Design

- S 3.4.23 The “Cradle-to-Gate” analysis includes extraction of raw materials needed, manufacturing of “made parts”, “sub-assemblies”, “assemblies” and the “system” as well as the mounting into the final prototype vehicle.
- S 3.4.24 The “carbon footprint” for the transportation between the steps must be included.
- S 3.4.25 The “carbon footprint” must exclude research and development.

S 3.5 Supporting Material File

- S 3.5.1 The Supporting Material File is a document containing additional information which allows the judges to understand the CCBOM. It may include drawings, exploded view drawings and/or pictures of the vehicle and the parts included in the CCBOM.

S 3.6 Cost and Emissions Explanation File

- S 3.6.1 The Cost and Emissions Explanation File is a document which allows the judges to understand the costs and greenhouse gas emissions within the CCBOM.
- S 3.6.2 The cost model used must be pointed out, especially which specific costs are included, e.g. machine operation costs.
- S 3.6.3 The methodology used for determining the Global Warming Potential (GWP) in CO₂e also needs to be explained. The specific carbon emission factors associated with, e.g., one material type need to be included.

S 3.7 Cost and Manufacturing Scoring

- S 3.7.1 The team must present their vehicle according to [S 1.1](#) to score points in Cost and Manufacturing.
- S 3.7.2 Cost and Manufacturing is evaluated on the categories specified in the following table:

Category	Points
Part 1 “ <u>CCBOM</u> Discussion”	50
Part 2 “Cost Understanding”	25
Part 3 “Real Case”	25
Total	100

- S 3.7.3 If items are missing from the CCBOM, points are deducted until 0 points are scored for “CCBOM Discussion”.
- S 3.7.4 Some teams may be chosen to participate in the Cost and Manufacturing finals to determine the Cost and Manufacturing winner. The Cost and Manufacturing finals are held separately from the initial judging and teams are informed about their participation during the event.
- S 3.7.5 In case of finals the highest achievable scoring for the non-finalists is 80 points.

S 4 Engineering Design

S 4.1 Engineering Design Objective

- S 4.1.1 The concept of the Engineering Design is to evaluate the student’s engineering process and effort that went into the design of a vehicle, meeting the intent of the competition.

S 4 Engineering Design

- S 4.1.2 Proprietary components and systems that are incorporated into the vehicle design as finished items are not evaluated as a student designed unit, but are only assessed on the team's selection and application of that unit.
- S 4.1.3 An evaluation concerning the capability of the vehicle to drive autonomously is also part of this discipline. Therefore, all systems that are required to drive autonomously are investigated. This also includes a discussion about the hardware and the software used in the AS.

S 4.2 Technical Vehicle System Documentation (TVSD)

- S 4.2.1 The TVSD should provide the judges with a structured overview of the vehicle and the team's design approach. The level of detail must allow the evaluation of design decisions, engineering methods, and validation procedures according to the engineering categories listed in Table 5. Any information required to scope, explain or highlight design features, concepts, methods or objectives that demonstrate the value and performance of the vehicle should be included at the team's discretion.
- S 4.2.2 A completed TVSD must be submitted online via the competition website, not later than the deadline specified in the Event Handbook.
- S 4.2.3 The TVSD must include three technical drawings (no renderings) showing the vehicle from the front, top and side. Each drawing must contain the overall vehicle dimensions (length, width, and height) clearly annotated at least once.
- S 4.2.4 The TVSD is used to sort the teams into appropriate Engineering Design queues and to support the evaluation during Engineering Design.
- S 4.2.5 Any evidence or supplementary documentation referenced in the TVSD must be available at the event and presented to the judges upon request.

S 4.3 Engineering Design Procedure

- S 4.3.1 Engineering Design starts with the submission of the TVSD and its review.
- S 4.3.2 At the event, teams present their knowledge and their vehicle to the judges, which evaluate the team's performance following the design objectives stated in chapter [S 4.1](#).
- S 4.3.3 Some teams may be chosen to participate in the Engineering Design finals to determine the Engineering Design winner. The Engineering Design finals are held separately from the initial judging and teams are informed about their participation during the event.
- S 4.3.4 Teams may bring any photographs, drawings, charts, spare parts or other material that they believe are supportive, but the space provided for Engineering Design judging may be limited.
- S 4.3.5 [EV ONLY] Only sealed TSACs which passed the Accumulator Inspection may be presented or mounted in the vehicle. They must not be opened.
- S 4.3.6 [EV ONLY] Only discharged, below 5 % State of Charge, accumulator cells each having all terminals electrically insulated may be presented.
- S 4.3.7 [DC ONLY] There is no extra Engineering Design presentation for the DC, however scoring is adapted, see Table 6, and there are more judges during the normal presentation focusing on the AS.
- S 4.3.8 [DC ONLY] There are separate Engineering Design finals for the DC.

S 4.4 Engineering Design Judging Criteria

- S 4.4.1 The judges evaluate the engineering effort based upon the team's TVSD, responses to questions and an inspection of the vehicle.

S 4 Engineering Design

- S 4.4.2 The judges inspect the vehicle to determine if the design concepts are adequate and appropriate for the application (relative to the objectives set forth in the rules).
- S 4.4.3 The judges may deduct points if the team cannot adequately explain the engineering and construction of the vehicle.

S 4.5 Engineering Design Scoring

- S 4.5.1 The team must present their vehicle according to [S 1.1](#) to score points in Engineering Design.
- S 4.5.2 The overall Engineering Design maximum scoring is 150 points.
- S 4.5.3 The maximum scores listed in Table 5 apply for Engineering Design.

Category	Points
Overall Vehicle Concept	25
Software	10
Vehicle Performance	30
Mechanical / Structural	15
Tractive System / Powertrain	25
LV-Electrics / Electronics	15
Autonomous Functionality	15
Driver Interface	10
Technical Vehicle System Documentation (<u>TVSD</u>)	5

Table 5: Maximum scores in Engineering Design

Category	Points
Software / Compute	35
Vehicle Performance	30
Autonomous Functionality	85

Table 6: [DC ONLY] Maximum scores in Engineering Design

D Dynamic Disciplines

D 1 Dynamic Disciplines General

D 1.1 Definitions

- D 1.1.1 **Starting** — crossing the light barrier at the starting line starts a lap.
- D 1.1.2 **Finishing** — crossing the light barrier at the finish line ends a lap. Can be the same as the starting line.
- D 1.1.3 **Lap** — a complete trip on a defined track, beginning at the start line and ending at the finish line.
- D 1.1.4 **Run** — a single attempt to compete in a dynamic discipline. Runs consist of one or more lap(s). A run is started when receiving a GREEN FLAG or the go-signal by the officials.
- D 1.1.5 **Did Not Finish (DNF)** — starting a run without finishing it. For the purpose of scoring, a DNF is treated as a DQ.
- D 1.1.6 **disqualified (DQ)** — being removed from the scoring due to a rule violation.
- D 1.1.7 **Valid Run** — a run that is neither DNF nor DQ.
- D 1.1.8 **Running Order** — the sequence in which teams take part in a dynamic discipline. If not defined otherwise, the running order is queue based and teams on their first run receive priority.
- D 1.1.9 To score points in any dynamic discipline, a team must have at least one valid run in that discipline.
- D 1.1.10 **Turning Diameter** — the diameter of the smallest circle within which a vehicle can be turned round completely.

D 1.2 Driver Limitations

- D 1.2.1 In total, a minimum of three and a maximum of four drivers are allowed for each team.
- D 1.2.2 An individual driver must not drive in more than two dynamic disciplines.
- D 1.2.3 Endurance and Efficiency are considered a single discipline.
- D 1.2.4 An individual driver must not drive more than two runs in each dynamic discipline.
- D 1.2.5 No more than two drivers are allowed to drive in each dynamic discipline.

D 1.3 Number of Runs in the Dynamic Disciplines

- D 1.3.1 Table 7 defines the number of runs for each dynamic discipline.

D 2 Driving Rules

Dynamic Discipline	Runs in Manual Mode	Runs in Autonomous Mode
Acceleration	2	-
<u>DV</u> Acceleration	-	2
Autocross	4	-
[DC ONLY] <u>DV</u> Autocross	-	2
Endurance	1	-
Skidpad	2	-
<u>DV</u> Skidpad	-	2
[DC ONLY] Trackdrive	-	1

Table 7: Number of runs for each dynamic discipline

D 1.4 Dynamic Area and Dynamic Vests

- D 1.4.1 Technical Inspection and all dynamic disciplines are held in the dynamic area.
- D 1.4.2 Four dynamic vests are handed out to each team by the officials and must be worn in the dynamic area.
- D 1.4.3 Only four members per team, including the driver, may enter the dynamic area at one time.
- D 1.4.4 Drivers must not wear the dynamic vest when sitting in the vehicle. The driver's vest must be attached to the push bar.
- D 1.4.5 The number of tools that may be used in this area is restricted to those which can be safely carried by the four team members in one trip.

D 1.5 Track Walks

- D 1.5.1 There are track walks before Autocross, Endurance and [DC ONLY] Trackdrive.
- D 1.5.2 During the track walk only analog measurement devices may be used.

D 2 Driving Rules

D 2.1 Flags

- D 2.1.1 Flag signals are commands that must be obeyed immediately and without question.
- D 2.1.2 There are no flag signs for vehicles in autonomous mode.
- D 2.1.3 **BLACK FLAG** — the driver must pull into the driver change area for discussion with the officials concerning an incident. A time penalty may be assessed.
- D 2.1.4 **BLACK FLAG WITH ORANGE DOT / MECHANICAL BLACK FLAG** — the driver must pull into the driver change area for a Mechanical Inspection of the vehicle, something has been observed that requires a closer inspection.
- D 2.1.5 **BLUE FLAG** — the driver must pull into the designated passing zone to be passed by a faster competitor. The driver must obey the track marshals' signals at the end of the passing zone.
- D 2.1.6 **CHECKERED FLAG** — the session has been completed. The driver must exit the track at the first opportunity.
- D 2.1.7 **GREEN FLAG** —

D 2 Driving Rules

- The session has started, the driver may enter the track under direction of the track marshals. In case of stalling, the vehicle can be restarted, but the driver has to await another GREEN FLAG as the opening in traffic may have closed.
 - The driver is clear to re-enter the track after using the slow lane to let a faster vehicle pass.
 - The driver may pick up speed again after a YELLOW FLAG has been displayed.
- D 2.1.8 **RED FLAG** — the driver must come to an immediate safe and controlled stop on the track and must follow track marshals' directions.
- D 2.1.9 **YELLOW FLAG** — danger, the driver must slow down, something has happened beyond the flag station, no overtaking unless directed by the track marshals.
- D 2.1.10 **RED AND YELLOW STRIPED FLAG** — the track is slippery or something is on the racing surface that should not be there. The driver must be prepared for evasive maneuvers to avoid the situation.

D 2.2 Teleoperated Driving

- D 2.2.1 After the vehicle has crossed the starting line it is prohibited to change parameters, send commands or perform any software changes by wireless communication until the vehicle has crossed the finish line. Receiving information from the vehicle via one-way-telemetry is allowed.
- D 2.2.2 The only device that is allowed to send commands by wireless communication at all times is the RES described in [T 14.3](#).

D 2.3 Driving Under Power

- D 2.3.1 During driving, the mechanical integrity of the vehicle must be maintained.
- D 2.3.2 Vehicles must not be driven in reverse.
- D 2.3.3 The vehicle must be capable of starting and restarting without external assistance/batteries at all times.
- D 2.3.4 Push starts are prohibited.
- D 2.3.5 Burnouts before and during the dynamic disciplines are prohibited.
- D 2.3.6 When driving in autonomous mode, an ASR has to be present at the race control with the RES. Additionally, a single monitoring device (laptop, tablet, ...) may be brought (no complicated antenna construction or similar!).

D 2.4 Cones & Markings for Autonomous Running

- D 2.4.1 Details of the cones used and more detailed track layout figures can be found in the Event Handbook.

D 2.5 Start-up Procedure for Autonomous Running

- D 2.5.1 No additional equipment (e.g. laptop, jack-up device, pressure tank, etc.) is allowed to start up the vehicle at the staging/starting line.
- D 2.5.2 If the vehicle does not enter "AS Ready" state within 1 min after being staged, the team may be sent back to the preparation area by the officials. In this case any priority is lost.
- D 2.5.3 The vehicle may only be staged with the steering system in straight position.
- D 2.5.4 The ASMS may only be switched on by the ASR after approval from an official at the starting line.

D 3 Weather Conditions

D 2.6 Vehicle Break Downs and Usage of RES during Autonomous Running

- D 2.6.1 If a vehicle comes to standstill for any reason, it may have up to 30 s to attempt to continue to drive. If the vehicle doesn't restart within 30 s, it is disqualified.
- D 2.6.2 If the average speed of the first three laps in Trackdrive (after completing the third lap) is below 2.5 m/s or the average speed of any of the following laps is below 3.5 m/s, the vehicle is disqualified.
- D 2.6.3 If a traceable signal loss of the RES appears and doubtless proof can be brought by the team that it was not self-inflicted, a re-run may be granted for the respective autonomous run, if the officials are notified before the affected dynamic discipline closes. The data logger data must be uploaded before notifying the officials.
- D 2.6.4 At the direction of the officials, team members may be instructed to retrieve broken-down vehicles. This recovery may only be done under the control of the officials.

D 2.7 Procedure After Completing a Driverless (DV) Dynamic Discipline

- D 2.7.1 After approval from the officials the ASR has to deactivate the vehicle using the RES.
- D 2.7.2 The vehicle must be collected by the ASR and an additional team member immediately after approval from the officials.

D 2.8 Handling of Re-Runs

- D 2.8.1 A re-run may be granted in following cases:
- Usage of YELLOW FLAG in Autocross.
 - Usage of RED FLAG.
 - Usage of RES to ensure safe conditions on the track or traceable signal loss of the RES according to [D 2.6](#).
 - After official's decision due to technical or organizational issues.
- D 2.8.2 The team is informed about a granted re-run either by a sign at the exit or by direct communication through an official.
- D 2.8.3 It is at the discretion of the team whether they accept the re-run. If the re-run is accepted, it must be executed immediately and the corresponding initial run is removed from the scoring.

D 3 Weather Conditions

D 3.1 Operating Conditions

- D 3.1.1 The following track conditions are recognized:
- Dry
 - Damp
 - Wet
- D 3.1.2 The operating conditions are decided by the officials and may change at any time.
- D 3.1.3 The current operating condition is prominently displayed at the dynamic area.

D 3.2 Tires Allowed

- D 3.2.1 Teams must run the tires allowed for each operating condition:

D 4 Skidpad

Operating Condition	Tires allowed
Dry	Dry
Damp	Dry or Wet
Wet	Wet

- D 3.2.2 When the operating condition is damp, teams may change between dry tires and wet tires:
- Any time during Acceleration, Skidpad, and Autocross.
 - Any time before taking the GREEN FLAG to start Endurance and according to [D 7.6](#).
- D 3.2.3 If a discipline had varied operating conditions, the minimum performance levels to score points may be adjusted if deemed appropriate by the officials.
- D 3.2.4 Only one set of tires per type (dry/wet) may be used during all the dynamic disciplines. The Brake Test, practice area, [DC ONLY] Driverless Autocross, [DC ONLY] Trackdrive and the static disciplines are excluded from this regulation.

D 4 Skidpad

D 4.1 Skidpad Track Layout

- D 4.1.1 The Skidpad track consists of two pairs of concentric circles in a figure of eight pattern.
- D 4.1.2 The centers of these circles are 18.25 m apart. The inner circles are 15.25 m in diameter and the outer circles are 21.25 m in diameter.
- D 4.1.3 17 cones are placed around the inside of each inner circle. 13 cones are positioned around the outside of each outer circle, in the pattern shown in the Skidpad layout diagram.
- D 4.1.4 Each circle is marked with a line, outside the inner circle and inside the outer circle.
- D 4.1.5 The driving path is the 3 m wide path between the inner and outer circles. The vehicles enter and exit through gates on a 3 m wide path that is tangent to the circles where they meet.
- D 4.1.6 The line between the centers of the circles defines the start/finish line. A lap is defined as traveling around one of the circles, starting and ending at the start/finish line.

D 5 Acceleration

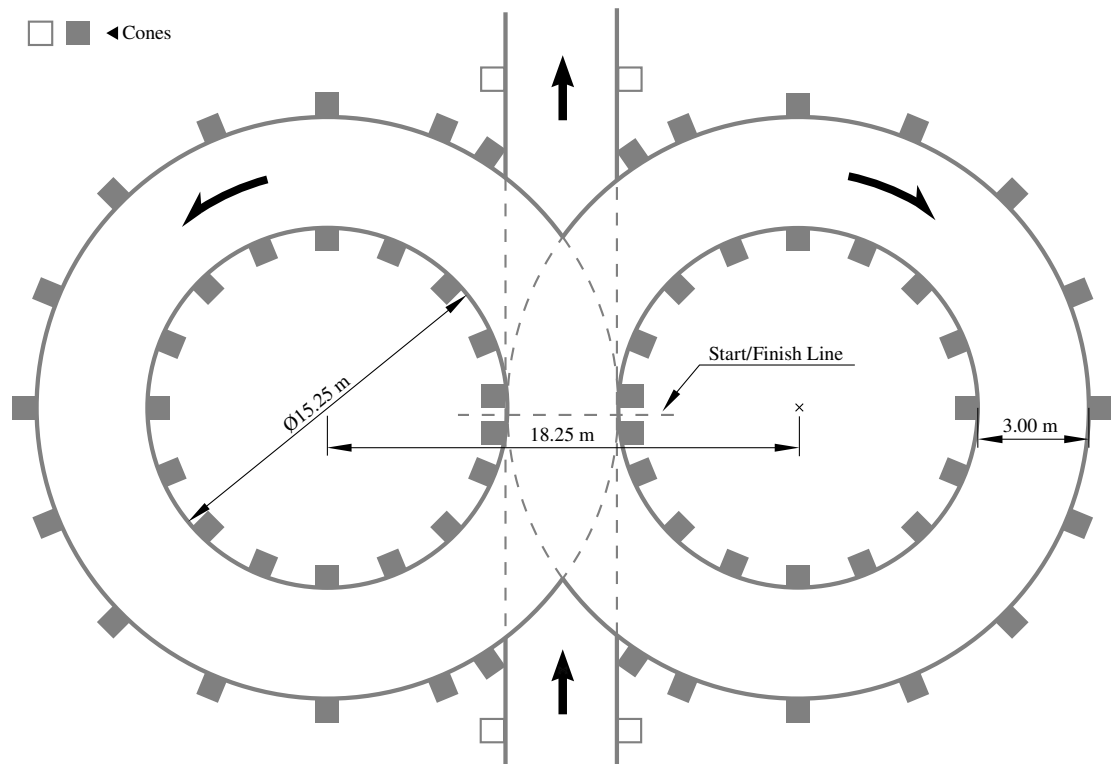


Figure 22: Skidpad Track Layout

D 4.2 Skidpad Procedure

- D 4.2.1 A run consists of a sequence of laps in exactly the following order. The vehicle must enter perpendicular to the figure of eight and must take one full lap on the right circle to establish the turn. The next lap must be on the right circle and is timed. Immediately following the second lap, the vehicle must enter the left circle for the third lap. The fourth lap must be on the left circle and is timed. Immediately upon finishing the fourth lap, the vehicle must exit the track perpendicular to the figure of eight and moving in the same direction as entered.
- D 4.2.2 The run time is the average time of the timed left and the timed right circle plus penalties which are added after the averaging.
- D 4.2.3 [Manual Mode ONLY] The driver has the option to make the second run immediately after their first run.
- D 4.2.4 [Autonomous Mode ONLY] The foremost part of the vehicle is staged 15 m in front of the starting line.
- D 4.2.5 [Autonomous Mode ONLY] After finishing the run, the foremost part of the vehicle must come to a full stop within 25 m behind the finish line, all four wheels must remain within the track boundaries and the vehicle must enter the finish-state described in [T 14.8](#).

D 5 Acceleration

D 5.1 Acceleration Track Layout

- D 5.1.1 The Acceleration track is a straight line with a length of 75 m from starting line to finish line. The track is at least 3 m wide. Cones are placed along the track at intervals of about 5 m.

D 5.2 Acceleration Procedure

- D 5.2.1 A run consists of a single lap.

D 7 Endurance and Efficiency

- D 5.2.2 [Manual Mode ONLY] The driver has the option to make the second run immediately after their first run.
- D 5.2.3 The foremost part of the vehicle is staged 0.30 m behind the starting line.
- D 5.2.4 [Autonomous Mode ONLY] After finishing the run, the foremost part of the vehicle must come to a full stop within 75 m behind the finish line, all four wheels must remain within the track boundaries and the vehicle must enter the finish-state described in [T 14.8](#).

D 6 Autocross

D 6.1 Autocross Track Layout

- D 6.1.1 The Autocross track layout is a handling track built to the following guidelines:
- Straights: No longer than 80 m
 - Slaloms: Cones in a straight line with 7.5 m to 12 m spacing
 - Miscellaneous: Chicanes, multiple turns, decreasing radius turns, hairpin turns, etc.
 - The minimum track width is 3 m
 - The minimum required turning diameter according to [D 1.1.10](#) is 9 m
- D 6.1.2 The length of the Autocross track is less than 1.5 km.
- D 6.1.3 [DC ONLY] The Autocross track layout is instead built to the guidelines defined in [D 8.1](#).

D 6.2 Autocross Procedure

- D 6.2.1 A run consists of a single lap.
- D 6.2.2 [Manual Mode ONLY] Each driver has the option to make the second run immediately after their first run.
- D 6.2.3 [Autonomous Mode ONLY] The vehicle is staged such that the front wheels are 6 m in front of the starting line on the track.
- D 6.2.4 [Autonomous Mode ONLY] After finishing the run, the foremost part of the vehicle must come to a full stop within 30 m behind the finish line, all four wheels must remain within the track boundaries and the vehicle must enter the finish-state described in [T 14.8](#).

D 7 Endurance and Efficiency

D 7.1 Endurance Track Layout

- D 7.1.1 The Endurance track layout is a closed lap circuit built to the following guidelines:
- Straights: No longer than 80 m
 - Slaloms: Cones in a straight line with 9 m to 15 m spacing
 - Miscellaneous: Chicanes, multiple turns, decreasing radius turns, hairpin turns, etc.
 - The minimum track width is 3 m
 - The minimum required turning diameter according to [D 1.1.10](#) is 9 m
- D 7.1.2 The length of one lap of the Endurance track is approximately 1 km.
- D 7.1.3 The length of the complete Endurance is approximately 22 km.

D 7.2 Endurance Procedure

- D 7.2.1 Before entering the discipline, each CV's fuel tank must be filled to the fuel level line, see [CV 2.6.4](#), at the fueling station. During fueling, once filled to the scribe line, no shaking or tilting of the tank, the fuel system or the entire vehicle is allowed.

D 7 Endurance and Efficiency

- D 7.2.2 A run consists of multiple laps driven by two different drivers.
- D 7.2.3 The first driver drives for approximately 11 km and is then signaled into the driver change area.
- D 7.2.4 After the driver change specified in [D 7.5](#), the second driver drives for an additional approximately 11 km and is signaled to exit the track after crossing the finish line.
- D 7.2.5 Each lap of Endurance is individually timed. The corrected elapsed time is determined by subtracting the extra-long lap for the driver change from the total time and adding any penalty times.
- D 7.2.6 [CV ONLY] The second driver proceeds directly to the fueling station. The tank is filled to the refill mark and the amount of fuel is recorded.

D 7.3 Endurance Running Order

- D 7.3.1 The running order for Endurance is established in a way that vehicles with similar speeds are on the track together.
- D 7.3.2 The running order is published according to the Event Handbook.
- D 7.3.3 The procedure for handling the running order is defined in the Event Handbook.

D 7.4 Passing

- D 7.4.1 During Endurance, overtaking is only permissible in the designated passing zones and under the control of the track marshals.
- D 7.4.2 Passing zones have two parallel lanes, a slow lane only used by the vehicles that are being overtaken and a fast lane for the vehicles that are overtaking.
- D 7.4.3 Passing zones may be situated on either the left or right of the fast lane.
- D 7.4.4 The passing zone procedure is as follows:
- A slower leading vehicle is shown the BLUE FLAG and must drive into the slow lane and decelerate.
 - The following faster vehicle continues in the fast lane to pass the slow vehicle.
 - The vehicle that has been overtaken may re-enter the track when the track marshal who is in charge of that passing zone is showing the GREEN FLAG.
- D 7.4.5 The passing rules do not apply to vehicles that are stopped on track or vehicles that have spun out and are not moving. When passing a stationary vehicle, it is critical to slow down, drive cautiously and to follow the instructions from the track marshals.

D 7.5 Endurance Driver Change Procedure

- D 7.5.1 Only three team members including the drivers may enter the driver change area. They may only bring the tools required to adjust the vehicle to accommodate the second driver and/or change tires.
- D 7.5.2 [EV ONLY] One team member must be an [ESO](#).
- D 7.5.3 During the driver change, the team may:
- perform changes to accommodate the second driver
 - operate the master switch(es)
 - change tires as per [D 7.6](#).

No other work may be performed during the driver change.

D 7 Endurance and Efficiency

- D 7.5.4 Each team is given 3 min to change their driver.
- D 7.5.5 The driver change time starts once the vehicle is stopped in the driver change area and the first driver has turned off the [CV ONLY] engine / [EV ONLY] TS.
- D 7.5.6 [EV ONLY] The TSMS must be switched off by the ESO and the TSAL must have changed to green color, before anyone is allowed to touch the vehicle or to climb out of the vehicle.
- D 7.5.7 The first driver climbs out of the vehicle and any necessary adjustments are made to the vehicle to fit the second driver (seat cushions, head restraint, pedal position, etc.). The second driver is then secured in the vehicle.
- D 7.5.8 When the second driver is fully secured in the vehicle, the vehicle has restarted and is R2D again, the driver change time is stopped.
- D 7.5.9 If the driver change takes longer than 3 min, the extra time is included in the final time.

D 7.6 Endurance Tire Changes

- D 7.6.1 All tire changes after a vehicle has received the GREEN FLAG to start Endurance must take place in the driver change area.
- D 7.6.2 If the operating condition changes to wet during Endurance, the track is RED FLAGGED or vehicles are BLACK FLAGGED and brought into the driver change area.
- D 7.6.3 If a team wants to change tires, the officials must be informed beforehand.
- D 7.6.4 Tire changes can be carried out during the driver change or during an extra stop.
- D 7.6.5 If the tire change is carried out during the driver change, in deviation from [D 7.5](#), four team members may enter the driver change area.
- D 7.6.6 The allowed tire changes and associated conditions are given in Table 8.

Existing Operating Condition	Currently running on	Condition changes to Dry	Condition changes to Damp	Condition changes to Wet
Dry	Dry Tires	–	A	B
Damp	Dry Tires	–	A	B
Damp	Wet Tires	C	–	–
Wet	Wet Tires	C	C	–

Requirement		Time Allowed
A	may change from dry to wet	Note 1
B	must change from dry to wet	Note 1
C	may change from wet to dry	Note 2

Table 8: Endurance tire change

Note 1: Any time in excess of 10 min without driver change, or 13 min with driver change, is added to the team's total time for Endurance.

Note 2: The time used to change to dry tires is added to the team's total time for Endurance.

- D 7.6.7 Teams that have incurred a puncture during Endurance due to external factors (e.g. debris on track) may change the tire with no time penalty, if the vehicle reaches the driver change area under its own power without external assistance.

D 8 [DC ONLY] Trackdrive

- D 7.6.8 The wheel is inspected by the officials. If the deflation was not caused by external factors, the vehicle is disqualified.
- D 7.6.9 Deflation caused by running OC or impacting barriers or other objects due to driver error is not regarded as an external factor.

D 7.7 Endurance Specific Regulations

- D 7.7.1 Teams are prohibited from working on or fueling their vehicles during the run.
- D 7.7.2 Wheel-to-wheel racing is prohibited.
- D 7.7.3 If a vehicle stalls out on the track, it is allowed one lap by the vehicle that is following it (approximately one minute) to restart.
- D 7.7.4 If a vehicle has a restart problem at the driver change or after a RED FLAG, it is allowed 2 min to restart the engine or to enter R2D. The 2 min start from the time the driver first tries to restart the engine or to enter R2D. The time is added to the Endurance time.
- D 7.7.5 If restarts are not accomplished within the above times, the vehicle is disqualified for the run.
- D 7.7.6 If a vehicle breaks down it is removed from the track and is not allowed to re-enter the track.
- D 7.7.7 Team members may be instructed by the officials to retrieve broken down vehicles. This recovery may only be carried out under the control of the officials.

D 7.8 Endurance Restart Procedure After Red Flag

- D 7.8.1 When the cause of the RED FLAG has been rectified, all vehicles on the track are guided to the driver change area under YELLOW FLAG in order to receive further instructions from the officials.
- D 7.8.2 The lap in which the RED FLAG occurred is completely ignored for Endurance and Efficiency scoring and repeated after the restart.

D 7.9 Efficiency Procedure

- D 7.9.1 Efficiency is measured during Endurance.
- D 7.9.2 Only vehicles which scored minimum $P_{\min}$ points in Endurance are considered for Efficiency scoring.
- D 7.9.3 [CV ONLY] Fuel pumps must be on and fuel valves must be opened during refueling.
- D 7.9.4 [CV ONLY] The measured fuel mass of vehicles using E85 fuel is divided by 1.45 to be comparable to the vehicles using 98 RON.
- D 7.9.5 [EV ONLY] The Endurance energy is calculated as the time integrated value of the measured voltage multiplied by the measured current logged by the data logger, see [EV 4.6](#). Regenerated energy is multiplied by 0.9 and subtracted from the used energy.

D 8 [DC ONLY] Trackdrive

D 8.1 Trackdrive Track Layout

- D 8.1.1 The Trackdrive layout is a closed loop circuit built to the following guidelines:
- Straights: No longer than 80 m
 - Miscellaneous: Chicanes, multiple turns, decreasing radius turns, hairpin turns, etc.
 - The minimum track width is 3 m
 - The minimum required turning diameter according to [D 1.1.10](#) is 9 m

D 9 Dynamic Disciplines Scoring

D 8.1.2 The length of one lap is approximately 200 m to 500 m.

D 8.2 Trackdrive Running Order

D 8.2.1 The running order is defined by the officials, based on previous dynamic discipline results.

D 8.2.2 The running order is published according to the Event Handbook.

D 8.2.3 The procedure for handling the running order is defined in the Event Handbook.

D 8.3 Trackdrive Procedure

D 8.3.1 A run consists of ten laps.

D 8.3.2 The vehicle is staged such that the front wheels are 6 m in front of the starting line on the track.

D 8.3.3 After ten laps, the foremost part of the vehicle must come to a full stop within 30 m behind the finish line, all four wheels must remain within the track boundaries and the vehicle must enter the finish-state described in [T 14.8](#).

D 8.3.4 There is no last lap signal i.e. the vehicle should count laps itself.

D 8.3.5 Each lap of Trackdrive is individually timed. The corrected elapsed time is determined by adding any penalty times.

D 9 Dynamic Disciplines Scoring

D 9.1 Default Scoring

D 9.1.1 Points are given based on the following formula:

$$\text{SCORE} = (P_{\max} - P_{\min}) \left(\frac{T_{\max} - T_{\text{team}}}{T_{\max} - T_{\min}} \right)^2 + P_{\min}$$

with

$P_{\max}$ is the maximum points for the discipline according to Table 1

$P_{\min}$ is the minimum points for the discipline according to Table 9

T_{team} is the team's best time including penalties. T_{team} is capped to $T_{\max}$.

$T_{\max}$ is defined in Table 9.

$T_{\min}$ is the time of the fastest vehicle including penalties.

Dynamic Discipline	$T_{\max}$	$P_{\min}$
Skidpad	1.35 $T_{\min}$	0.05 $P_{\max}$
Acceleration	1.7 $T_{\min}$	0.05 $P_{\max}$
Autocross	1.4 $T_{\min}$	0.1 $P_{\max}$
Endurance	1.5 $T_{\min}$	0.1 $P_{\max}$
[DC ONLY] Skidpad	1.7 $T_{\min}$	0.05 $P_{\max}$
[DC ONLY] Acceleration	2.25 $T_{\min}$	0.05 $P_{\max}$

Table 9: $T_{\max}$ and $P_{\min}$ for each dynamic discipline

D 9.2 [Autonomous Mode ONLY] Scoring for Skidpad and Acceleration

D 9.2.1 Runs with a run time without penalties >25 s are disqualified.

D 9.2.2 Points are given based on the following formula:

D 9 Dynamic Disciplines Scoring

$$DV_SCORE = P_{\max} \frac{N_{\text{all}} + 1 - R_{DV, \text{team}}}{N_{\text{all}}}$$

with

$P_{\max}$ is the maximum points for the discipline according to Table 1

$R_{DV, \text{team}}$ is the ranking of team's best autonomous mode time including penalties within the best autonomous mode times including penalties of all other teams

N_{all} is the number of teams who have at least one valid manual mode or autonomous mode run

D 9.3 [DC ONLY] Driverless Cup Scoring

D 9.3.1 Points for Skidpad and Acceleration are given according to [D 9.1.1](#).

D 9.3.2 Points for Autocross are given based on the following formula:

$$\text{AUTOCROSS_SCORE} = 0.9P_{\max} \left(\frac{T_{\max} - T_{\text{team, total}}}{T_{\max} - T_{\min}} \right) + 0.1P_{\max}$$
$$T_{\text{team, total}} = \min(T_{\text{team,1}}, \text{avg}(T_{\text{team,1}}, T_{\text{team,2}}))$$

$P_{\max}$ is the maximum points for the discipline according to Table 1.

$T_{\text{team,1}}$ is the team's autonomous mode time including penalties of run 1.

$T_{\text{team,2}}$ is the team's autonomous mode time including penalties of run 2.

$T_{\max}$ is the time for driving the lap with 6 m/s.

$T_{\min}$ is the fastest autonomous mode $T_{\text{team, total}}$ of all teams.

For runs that are DNF or DQ or for runs with a $T_{\text{team,i}}$ above $T_{\max}$, $T_{\text{team,i}}$ is set to $T_{\max}$.

D 9.3.3 Points for Trackdrive are given based on the following formula:

$$\text{TRACKDRIVE_SCORE} = 0.75P_{\max} \left(\frac{T_{\max}}{T_{\text{team}}} - 1 \right)$$

$P_{\max}$ is the maximum points for the discipline according to Table 1.

T_{team} is the team's corrected elapsed time. T_{team} is capped at $T_{\max}$.

$T_{\max}$ is 2 times of the corrected elapsed time of the fastest vehicle.

D 9.3.4 In deviation of [D 1.1.9](#), an additional 2.5 % of the maximum points for the discipline according to Table 1 are given for every completed lap.

D 9.4 Efficiency Scoring

D 9.4.1 Points for Efficiency are given based on the following formula:

$$\text{EFFICIENCY_SCORE} = P_{\max} \left(\frac{EF_{\max} - EF_{\text{team}}}{EF_{\max} - EF_{\min}} \right)^2$$

$P_{\max}$ is the maximum points for the Efficiency according to Table 1

EF_{team} the team's Efficiency Factor

$EF_{\min}$ the lowest Efficiency Factor of all teams which are considered for Efficiency

$EF_{\max}$ is defined as $2 \cdot EF_{\min}$

For runs with an EF_{team} above $EF_{\max}$, EF_{team} is set to $EF_{\max}$.

D 9.4.2 The Efficiency Factor is calculated based on the following formula:

$$EF = T^2 \cdot E$$

D 10 Dynamic Disciplines Penalties

T uncorrected elapsed driving time

E [CV ONLY] corrected used fuel mass / [EV ONLY] used energy

D 10 Dynamic Disciplines Penalties

D 10.1 General Penalties

- D 10.1.1 Penalties are not assessed for accident avoidance or other reasons deemed sufficient by the officials.
- D 10.1.2 A cone is Down or Out (DOO) if the cone has been knocked over or the entire base of the cone lies outside the box marked around the cone in its undisturbed position.
- D 10.1.3 Cones that are DOO are not replaced/reset during autonomous running. There is no re-run due to cones in the driving path or disorientation due to missing cones.
- D 10.1.4 The DOO penalty, see [D 10.1.7](#), is added for each DOO including entry and exit gate cones before the start and after the finish line, that occur on that particular run.
- D 10.1.5 Off-Course (OC)
- An OC occurs when the vehicle has all four wheels outside the track boundary as indicated by edge marking.
 - Missing one or more gates of a given slalom at Autocross or Endurance is counted as one OC per occurrence.
 - When an OC occurs, the driver must re-enter the track at the next possible point.
 - When re-entering the driver needs to wait for a gap and follow the instructions of the track marshals.
- D 10.1.6 An Unsafe Stop (USS) is defined as not stopping within the specified area and/or not maintaining the expected orientation while stopping and/or not entering the finish-state described in [T 14.8](#) within 30 s after stopping during autonomous running.
- D 10.1.7
- | | Acceleration | Skidpad | Autocross | Endurance | Trackdrive |
|------------|--------------|---------|-----------|-----------|------------|
| <u>DOO</u> | 2 s | 0.2 s | 2 s | 2 s | 2 s |
| <u>OC</u> | DQ | DQ | 10 s | 10 s | 10 s |
| <u>USS</u> | DQ | DQ | DQ | n/a | -50 points |
- D 10.1.8 Failure to obey a flag: 1 min time penalty.
- D 10.1.9 Reckless or aggressive driving or "Over Driving": BLACK FLAG, see [D 2.1.3](#).
- D 10.1.10 Vehicle to vehicle contact: 2 min up to DQ depending on the nature of the incident.
- D 10.1.11 If a vehicle is not able to start the run, within 30 s after receiving a GREEN FLAG or go-signal, the vehicle is DQ for that run.

D 10.2 Endurance Penalties

- D 10.2.1 Teams that are running out-of-order as defined in the Event Handbook, are penalized by 2 min and may then run at the discretion of the officials.
- D 10.2.2 Mechanical Problem: The time taken for Mechanical Inspection under a BLACK FLAG WITH ORANGE DOT, see [D 2.1.4](#), is considered officials' time and is not included in the team's total time. If the inspection reveals a mechanical integrity problem the vehicle is disqualified.
- D 10.2.3 If it is impossible to impose a penalty by a stop under a BLACK FLAG, e.g. not enough laps left, the officials may add an appropriate time penalty to the team's elapsed time.

D 10 Dynamic Disciplines Penalties

- D 10.2.4 A team may be disqualified if their vehicle is too slow or being driven without proper control. A vehicle is considered to be too slow if the average speed, based on lap times, for two consecutive laps is below 8 m/s in dry conditions or below 5 m/s in wet or damp conditions.
- D 10.2.5 Any violation to the procedure of driver change, see [D 7.5.3](#), leads to a time penalty of up to 2 min.
- D 10.2.6 All rules and penalties from [IN 12](#) are also applicable during driver change.

D 10.3 [CV ONLY] Efficiency Penalties

- D 10.3.1 Should the fuel level change after refueling due to effects such as described in [CV 2.7.1](#), the difference in fuel level is measured and twice the amount is added to the official fuel consumption figure.

D 10.4 [EV ONLY] Power and Voltage Violation Penalties

- D 10.4.1 A violation is defined as using more than the maximum power or IS current, see [EV 2.2](#), or exceeding the specified voltage, see [EV 4.1.1](#), after a moving average over 500 ms is applied to the respective data logger signal, see [EV 4.6](#).
- D 10.4.2 Each violation results in the team being disqualified for the fastest valid run of the discipline. If a violation has occurred during two runs, the two fastest valid runs are disqualified etc.
- D 10.4.3 The respective data and the resulting decision of violations may be made public.
- D 10.4.4 Non-availability of data logger data is treated as a violation.